DIGITAL SIGNAL PROCESSING

ΠΑΝΕΠΙΣΤΗΜΙΟ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF INFORMATION AND COMPUTER ENGINEERING

EXERCISES FOR DELIVERY 2022 - 2023

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PROGRAM OF STUDY : UNIWA
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STUDENT PHOTO:

DIGITAL SIGNAL PROCESSING

CONTENTS

EXERCISE 1

Plot the discrete time signal

$$x[n] = \begin{cases} \text{sum(AM)}, & -2 \leq n \leq 4 \\ 0, & 4 < n \leq 10 \\ \sqrt{2n}, & 10 < n \leq 20 \end{cases}$$

Where AM is the student's registration number and *sum* (*AM*) is the single-digit sum of its digits.

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For the interval $-2 \leq n \leq 4$the signal x [n] is as follows:

AM = 19390005
sum(AM) = 1 + 9 + 3 + 9 + 0 + 0 + 0 + 5 →
sum(AM) = 27 →
sum(AM) = 2 + 7 →
sum(AM) = 9

Therefore, the discrete-time signal to be plotted is as follows:

$$x[n] = \begin{cases} 9, & -2 \leq n \leq 4 \\ 0, & 4 < n \leq 10 \\ \sqrt{2n}, & 10 < n \leq 20 \end{cases}$$

Code in Matlab

```
n1 = -2:4;
n2 = 5:10;
n3 = 11:20;
n = [n1 n2 n3]

AM = [1 9 3 9 0 0 0 5]
len = length (AM);
s = 0;
decs = 0;
for i = 1:len
s = s + AM( i );
    if s >= 10
decs = decs + 1;
s = mod(s, 10);
    end
end
s = s + decs
x1 = s * ones (size (n1))
x2 = zeros(size(n2));
x3 = sqrt(2 * n3);
x = [x1 x2 x3];

stem (n, x)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'x[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( ' The signal x[n]' )
```

Illustration of used of commands

Commands	Results	Comments
>> n1 = -2:4; >> n2 = 5:10; >> n3 = 11:20; >> n = [n1 n2 n3]	n = -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20	Defining the time intervals of each member of the signal x [n] with step 1 and joining them into a vector n .
>> AM = [1 9 3 9 0 0 0 5] >> len = length(AM); >> s = 0; >> decs = 0; >> for i = 1:len >> s = s + AM(i); >> if s >= 10 >> decs = decs + 1; >> s = mod(s, 10); >> end >> end >> s = s + decs >> x1 = s * ones (size (n1))	AM = 1 9 3 9 0 0 0 5 s = 9	Calculation of the single digit sum of the Registry Number. The loop accesses the digits of AM stored in the array 'AM' and adds their sum to ' s '. The check condition helps to remember the tens of the sum in the variable ' decs ' so that the single digit sum (tens + ones) is calculated at the end.
>> x1 = s * ones (size (n1);	x1 = 9 9 9 9 9 9 9	Definition of the 1st member of the signal x [n] with the help of the unit vector ones .
>> x2 = zeros(size(n2));	x2 = 0 0 0 0 0 0	Definition of the 2nd member (0) of the signal x [n] with the help of the zero vector zeros .
>> x3 = sqrt(2 * n3);	x3 = 4.6904 4.8990 5.0990 5.2915 5.4772 5.6569 5.8310 6.0000 6.1644 6.3246	Definition of the 3rd term ($\sqrt{2 \times n3}$) of the signal x [n] with the help of the square root calculation function sqrt .
>> x = [x1 x2 x3]	x = Columns 1 through 13 9.0000 9.0000 9.0000 9.0000 9.0000 9.0000 9.0000 0 0 0 0 0 0 Columns 14 through 23 4.6904 4.8990 5.0990 5.2915 5.4772 5.6569 5.8310 6.0000 6.1644 6.3246	Union of the three members of the signal x [n] into a vector x .
>> stem (n, x)	Figure 1.1 (Page 6)	Plotting the discrete-time signal x [n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 1.1 (Page 6)	Add the label " n " to the horizontal x-axis, with a label size (FontSize) of 12 and a weight (FontWeight) of bold .

>> ylabel ('x[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 1.1 (Page 6)	Addition her label 'x[n]' in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('Signal x [n]')	Figure 1.1 (Page 6)	Add a title to the plot named "The signal x [n]".

Commenting on the resolution

The solution is based on the incremental generation of the discrete-time signal x [n], constructing the signal members one by one. The n time intervals that will be represented on the horizontal axis (n 1, n 2, n 3) are created separately, just as the values x [n] that will be represented on the vertical axis are created separately. The reason for creating the vectors n and x is to represent the entire signal x [n] with the stem function . The vector n includes all the value fields that will be represented on the horizontal axis, while the vector x includes all the values x [n] that will be represented on the vertical axis. For the signals x 1 and x 2 it was considered appropriate to use the unit vector ones and the zero vector zeros respectively, as the signals are constant, that is, their value does not change in the range of values represented on the horizontal axis.

Commenting on the results

For n ϵ [-2, 20], the vectors n and x have dimensions 1 x 23. The vector n contains the values to be represented on the horizontal axis, i.e., n ϵ [-2, 20], while the vector x contains the values to be represented on the vertical axis, that is, x [n] for n ϵ [-2, 20].
The vector x 1 contains x [n 1] in the domain $-2 \leq n1 \leq 4$, where is constant 9, since the signal is the constant function 9. The same is true for the vector x 2 that contains x [n 2] in the domain $5 \leq n2 \leq 10$, where constant 0, since the signal is the constant function 0. Finally, the vector x 3 includes the x [n 3] in the domain $11 \leq n3 \leq 20$, where is the result of the expression $\sqrt{2 \times n3}$.

Graphic representations

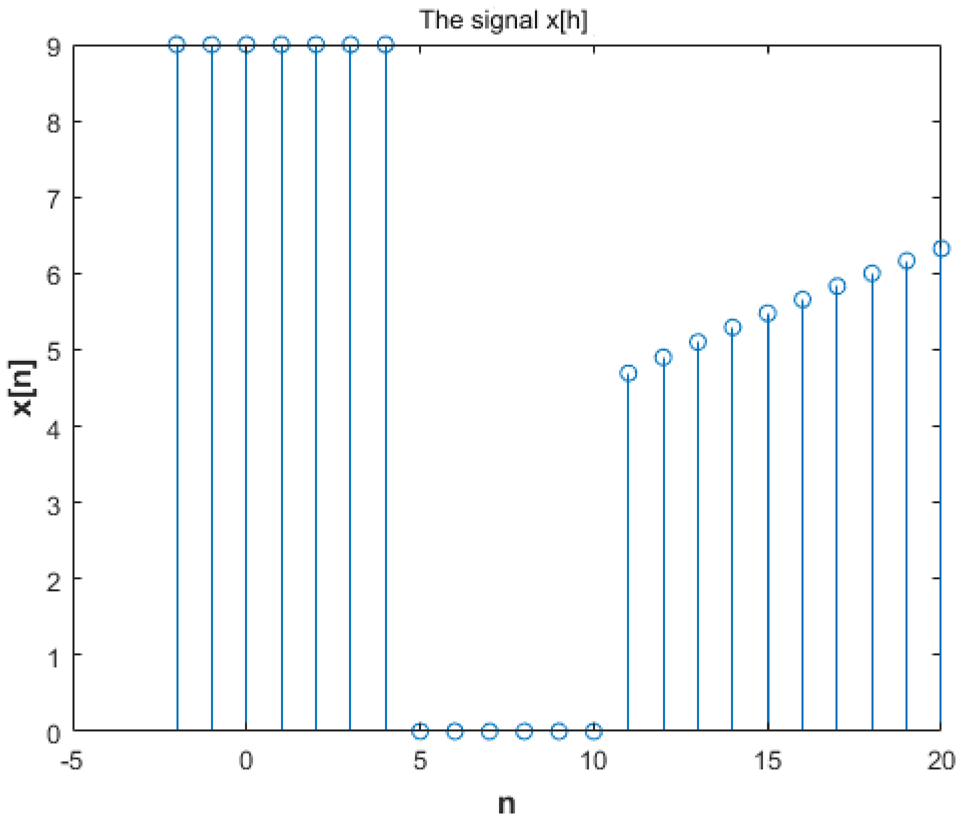

Figure 1.1 The discrete-time signal x [n]

EXERCISE 2

The discrete time signals are plotted

a) y[n] = u(n − (3 + AM mod 5)) − 6 ∗ δ (n − AM mod 4)
b) z[n] = u (n + 2 + AM mod 5) − u (n − 2 − AM mod 4)

for n = -50-(AM mod 4):50+(AM mod 4)

Where AM is the student registration number and u[n] and δ[n] are the unit step and unit percussion sequence respectively.

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The time interval of the discrete-time signals y [n] and z [n] with step 1 is :

AM = 19390005
n = -50 – (19390005 mod 4):50 + (19390005 mod 4) →
n = -50 – 1:50 + 1 → n = -51:51

a) y[n] = u(n − (3 + AM mod 5)) − 6 ∗ δ (n − AM mod 4)

The discrete-time signal y [n] is:

AM = 19390005
y [n] = u (n – (3 + 19390005 mod 5)) – 6 * δ(n – 19390005 mod 4) →
y [n] = u (n – (3 + 0)) – 6 * δ(n – 1) →
y [n] = u (n – 3) – 6 * δ(n – 1)

Code in Matlab

Main program

```
AM = 19390005

n1 = -50 - mod(AM, 4);
n2 = 50 + mod(AM, 4);
n = n 1:n 2

d = 6 * gauspuls (n - mod (AM, 4))
u = stepseq ((3 + mod (AM, 5)), n1, n2)
y = u - d

stem (n, y)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y[n] = u( n - 3) - 6 d (n - 1)' )
```

Code of the stepseq function

```
function [x, n] = stepseq (n0, n1, n2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end

n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> AM = 19390005	AM = 19390005	The student's registration number.
>> n1 = -50 - mod(AM, 4); >> n2 = 50 + mod(AM, 4); >> n = n1:n2	n = Columns 1 through 22 -51 -50 -49 -48 -47 -46 -45 -44 -43 -42 -41 -40 -39 -38 -37 -36 -35 -34 -33 -32 -31 -30	Define the time interval of the signal y [n] with step 1.

	Columns 23 through 44 -29 -28 -27 -26 -25 -24 -23 -22 -21 -20 -19 -18 -17 -16 -15 -14 -13 -12 -11 -10 -9 -8 Columns 45 through 66 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 Columns 67 through 88 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 Columns 89 through 103 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51	
>> d = 6 * gauspuls (n - mod (AM, 4))	d = Columns 1 through 22 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 23 through 44 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 45 through 66 0 0 0 0 0 0 0 0 6 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 67 through 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 89 through 103 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	Definition of the percussive sequence 6δ(n – A M mod 4) for every n that belongs to the interval -51 ≤ n ≤ 51. The definition is made with the help of the gauspuls function .
>> u = stepseq ((3 + mod (AM, 5)), n1, n2)	n1_u1 = Columns 1 through 22 -51 -50 -49 -48 -47 -46 -45 -44 -43 -42 -41 -40 -39 -38 -37 -36 -35 -34 -33 -32 -31 -30 Columns 23 through 44	Define the time interval of the step sequence u (n – (3 + AM mod 5) by calling the function " stepseq ", where the signal is equal to 0 for every n belonging to the interval for -51 ≤ n ≤ 2

	-29 -28 -27 -26 -25 -24 -23 -22 -21 -20 -19 -18 -17 -16 -15 -14 -13 -12 -11 -10 -9 -8 Columns 45 through 54 -7 -6 -5 -4 -3 -2 -1 0 1 2	
>> y = u – d	y = Columns 1 through 22 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 23 through 44 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 45 through 66 0 0 0 0 0 0 0 0 -6 0 1 1 1 1 1 1 1 1 1 1 1 1 Columns 67 through 88 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 89 through 103 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	Definition of the signal y [n] by the difference of the signals d and u .
>> stem (n, y)	Figure 2.1 (Page 12)	Plotting the discrete-time signal y[n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 2.1 (Page 12)	Add the label " n " to the horizontal x-axis, with a label size (FontSize) of 12 and a weight (FontWeight) of bold (bold).
>> ylabel ('x[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 2.1 (Page 12)	Add the label "y[n]" to the vertical y- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> title ('y[n] = u(n – 3) – 6 d (n – 1')	Figure 2.1 (Page 12)	Add a title to the graph named " y [n] = u (n – 3) – 6d(n – 1)".

Commenting on the resolution

The solution is based on the gradual creation of the discrete signal y [n], constructing one by one the two signals that in the end with their difference will represent y [n] . To begin with, gauspuls generates the shock sequence δ[n] shifted by 1 time to the right on the horizontal axis and multiplied by 6. gauspuls directly generates the δ(n – 1) signal . Separately, the step sequence u (n – 3) shifted by 3 time instants to the right on the horizontal axis is also constructed. The intervals are defined, where the signal will be equal to 1 and equal to 0 respectively. Hence the use of the zero vector zeros for the interval, where the step sequence is equal to 0 and the use of the unit vector ones , where the step sequence is equal to 1. Finally, the difference of the d and u signals is stored in the y signal , to represent the discrete-time signal y [n] with the command stem .

Commenting on the results

For $n \in [-51, 51]$, the vector y has dimensions 1 x 103. The vector n contains the values that will be represented on the horizontal axis, i.e., the $n \in [-51, 51]$, while the vector y contains the values to be represented on the vertical axis, that is, y [n] for $n \in [-51, 51]$.
The vector d includes the elements of the shock sequence $6\delta(n - 1)$ in the domain $-51 \leq n \leq 51$, where it is 6 for $n = 1$ and 0 for $n \neq 1$. The same is true for the step sequence $u(n - 3)$ in the domain $-51 \leq n \leq 51$,where 0 for $n \geq 3$ and 1 for $n < 3$. Finally, the discrete-time signal y [n] defined by the difference of the signals u and d , is 6 for $n = 1$, 1 for $n \geq 3$ and 0 for $n = 2$ and $n \leq 0$.

Graphic representations

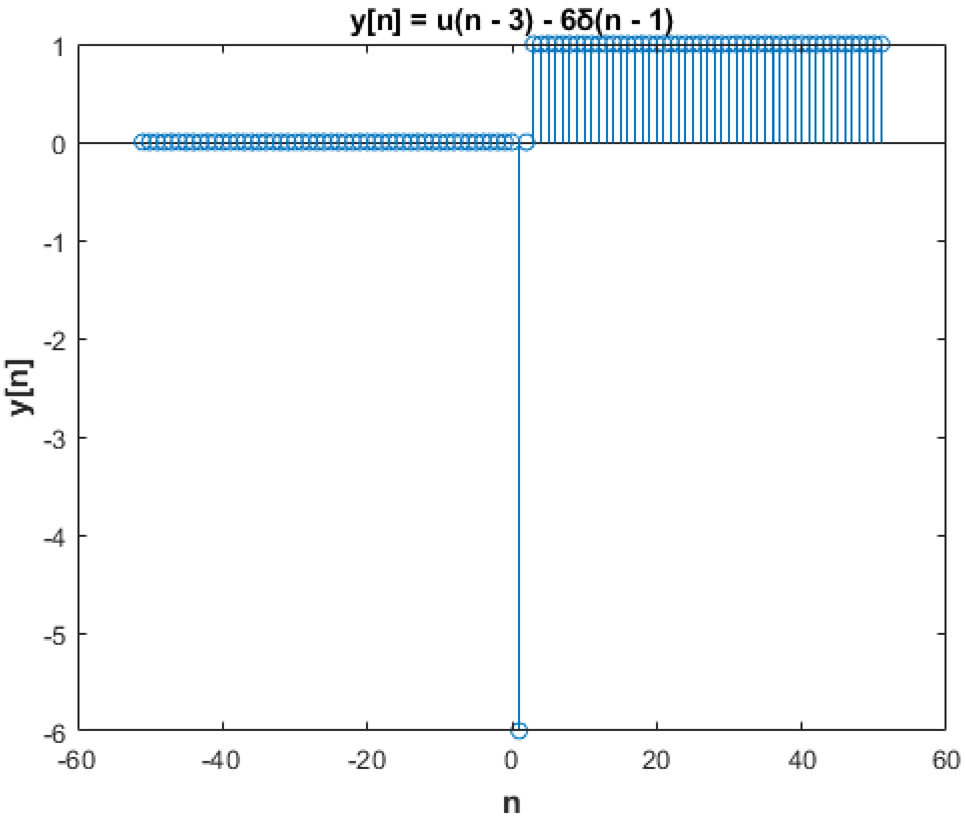

Figure 2.1 The discrete-time signal y [n]

b) z[n] = u (n + 2 + AM mod 5) − u (n − 2 − AM mod 4)

The discrete-time signal z [n] is:

AM = 19390005
z [n] = u (n + 2 + 19390005 mod 5)) – u (n – 2 – 19390005 mod 4) →
z [n] = u (n + 2 + 0) – u (n – 3) →
z [n] = u (n + 2) – u (n – 3)

Code at Matlab

Main program

```
AM = 19390005


n1 = -50 - mod(AM, 4);
n2 = 50 + mod(AM, 4);
n = n 1:n 2

u1 = stepseq (-(2 - mod (AM, 5)), n1, n2)
```

```
u2 = stepseq (2 + mod(AM, 4), n1, n2)
z = u1 - u2

stem (n, z)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'z[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'z[n] = u( n + 2) − u(n − 3)' )
```

Code her function stepseq

```
function [x, n] = stepseq (n0, n1, n2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end

n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> AM = 19390005	AM = 19390005	The student's registration number.
>> n1 = -50 - mod(AM, 4); >> n2 = 50 + mod(AM, 4); >> n = n1:n2	n = Columns 1 through 22 -51 -50 -49 -48 -47 -46 -45 -44 -43 -42 -41 -40 -39 -38 -37 -36 -35 -34 -33 -32 -31 -30 Columns 23 through 44 -29 -28 -27 -26 -25 -24 -23 -22 -21 -20 -19 -18 -17 -16 -15 -14 -13 -12 -11 -10 -9 -8 Columns 45 through 66 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 Columns 67 through 88 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 Columns 89 through 103 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51	Define the time interval of the signal y [n] with step 1.

>> u1 = stepseq (-(2 - mod (AM, 5)), n1, n2)	u1 = 1×103 logical array Columns 1 through 33 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 34 through 66 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 67 through 99 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 100 through 103 1 1 1 1	Define the time interval of the step sequence u (n + 2 + AM mod 5) by calling the function " stepseq ", where the signal is equal to 0 for each n belonging to the interval for -51 ≤ n ≤ -3
>> u2 = stepseq (2 + mod(AM, 4), n1, n2)	u2 = 1×103 logical array Columns 1 through 33 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 34 through 66 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 Columns 67 through 99 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 100 through 103 1 1 1 1	Define the time interval of the step sequence u (n – 2 – AM mod 4) by calling the function " stepseq ", where the signal is equal to 0 for every n belonging to the interval for -51 ≤ n ≤ 1
>> z = u1 - u2	z = Columns 1 through 22 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 23 through 44 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 45 through 66	Definition of the signal z[n] by the difference of the signals u 1 and u 2.

	0 0 0 0 0 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 Columns 67 through 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 89 through 103 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	
>> stem (n, z)	Figure 2 . 2 (Page 19)	Plotting the discrete-time signal z [n] with the stem command .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 2 . 2 (Page 19)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('z[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 2 . 2 (Page 19)	Addition her tag "z[n]" in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title (' z[n] = u(n + 2) – u(n – 3) ')	Figure 2 . 2 (Page 19)	Add a title to the plot named " z [n] = u (n + 2) – u (n – 3)".

Commenting on the resolution

The solution is based on the gradual creation of the discrete signal z [n], constructing one by one the two signals that in the end with their difference will represent z [n] . To begin with, the step sequence u (n + 2) is constructed which is shifted by 2 time instants to the right. Separately, the step sequence u (n – 3) shifted by 3 time instants to the right on the horizontal axis is also constructed. The intervals are defined, where the signal will be equal to 1 and equal to 0 respectively. Hence the use of the zero vector zeros for the interval, where the step sequence is equal to 0 and the use of the unit vector ones , where the step sequence is equal to 1. Finally, the difference of the signals of the two step sequences are stored in the signal z , to represent the discrete-time signal z [n] with the stem command .

Commenting on the results

For n ϵ [-51, 51], the vector z has dimensions 1 x 103. The vector n contains the values that will be represented on the horizontal axis, that is, the n ϵ [-51, 51], while the vector z contains the values to be represented on the vertical axis, that is, z [n] for n ϵ [-51, 51].
The vector u A contains the elements of the step sequence u (n + 2) in the domain -51 ≤ n ≤ 51, where it is 1 for n ≥ -2 and 0 for n < 2. The same is true for the step sequence u (n – 3) (u B) in the field -51 ≤ n ≤ 51,where it is 0 for n ≥ 3 and 1 for n < 3. Finally, the discrete-time signal z [n] defined by the difference of signals u A and u B, is 1 for -2 ≤ n ≤ 2 and 0 for -51 ≤ n ≤ -3 and 3 ≤ n ≤ 51.

Graphic representations

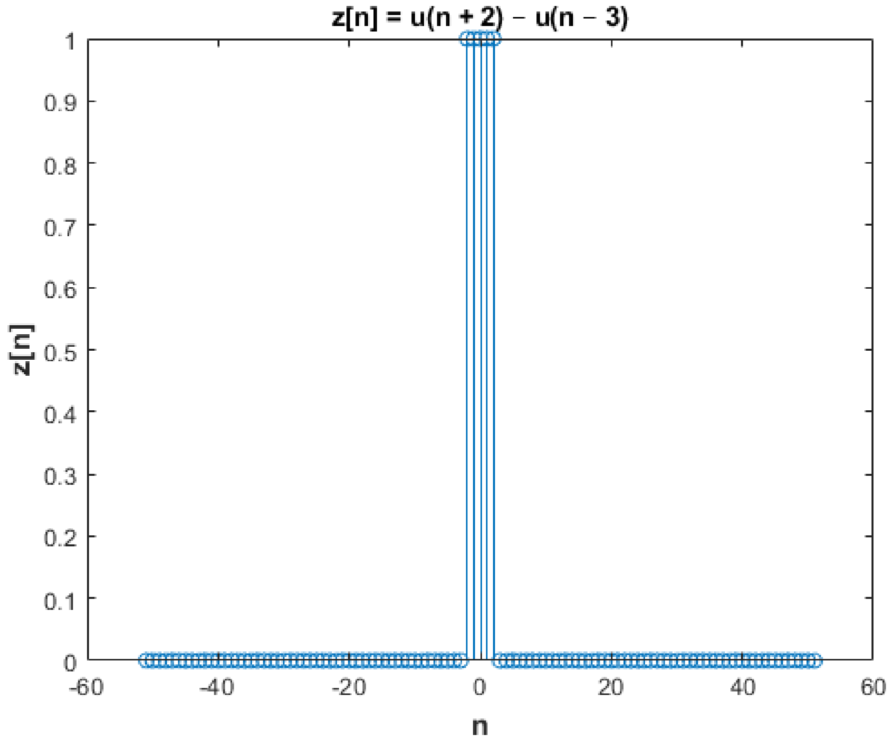

Figure 2.2 The discrete-time signal z [n]

EXERCISE 3

Plot the following signals and, for those that are periodic, determine their fundamental period and graphically verify the periodicity.

a) $y1[n] = e^{j\pi n/8} \cos\left(\frac{n\pi}{11}\right)$

b) $y\,2[\,n\,] = 2\cos(\pi/(3 + AM \bmod 4) + 0.4\,n)$

c) $y\,3[\,n\,] = (AM/100)\cos(0.25\pi\,n)$

d) $y\,4[\,n\,] = 3\cos^2\left(\frac{\pi}{6}n + \pi/10\right)$

e) $y\,5[\,n\,] = e^{j\pi n/2} + e^{-j\pi n/2}$

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$\boxed{\textbf{a) } y1[n] = e^{j\pi n/8}\cos(n\pi/11)}$

<u>Code in Matlab</u>

```
n = -10:10

yA = exp ((j * pi * n) / 8)
```

```
yB = cos ((n * pi) / 11)
y1 = yA . * yB

stem (n, real (y1))
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y1[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' ' )
title ( ' y1[n] = e^j^n ^ π ^/^8 cos(n π /11) ' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = -10:10	n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Define the time interval of the signal y1[n] with step 1.
>> y A = exp ((j * pi * n) / 8)	yA = Columns 1 through 7 -0.7071 + 0.7071i - 0.9239 + 0.3827i - 1.0000 - 0.0000i -0.9239 - 0.3827i - 0.7071 - 0.7071i -0.3827 - 0.9239i 0.0000 - 1.0000i Columns 8 through 14 0.3827 - 0.9239i 0.7071 - 0.7071i 0.9239 - 0.3827i 1.0000 + 0.0000i 0.9239 + 0.3827i 0.7071 + 0.7071i 0.3827 + 0.9239i Columns 15 through 21 0.0000 + 1.0000i - 0.3827 + 0.9239i - 0.7071 + 0.7071i -0.9239 + 0.3827i - 1.0000 + 0.0000i -0.9239 - 0.3827i - 0.7071 - 0.7071i	Definition of the first term of the signal y 1[n], that is, the term $e^{j\pi n/8}$, with the help of the exponential function exp and the complex number j .
>> y B = cos ((n * pi) / 11)	yB = Columns 1 through 13 -0.9595 -0.8413 -0.6549 -0.4154 - 0.1423 0.1423 0.4154 0.6549 0.8413 0.9595 1.0000 0.9595 0.8413 Columns 14 through 21 0.6549 0.4154 0.1423 -0.1423 - 0.4154 -0.6549 -0.8413 -0.9595	Definition of the second term of the signal y 1[n], that is, the term cos $(\frac{n\pi}{11})$ with the help of the cosine function cos .
>> y1 = yA .* yB	y1 = Columns 1 through 7	Element-by-element multiplication of the two terms yA and yB in order to define the signal y 1[n].

	0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i 0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 8 through 14 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 15 through 21 0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i	
>> stem (n, real (y1))	Figure 3.1 (Page 2 7)	Plotting the real part of the discrete-time signal y 1[n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.1 (Page 2 7)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y1[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.1 (Page 2 7)	Addition her label 'y1[n]' in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title (' y1[n] = e^j^n ^ π ^/^8 cos(n π /11)')	Figure 3.1 (Page 2 7)	Add a title to the graph named " y 1[n] = e ^ j ^ n ^π^/^8 cos (n π/11) ".

Commenting on the resolution

The solution is based on the gradual generation of the discrete signal y 1[n], constructing one by one the two signals that in the end with their product will represent y 1 [n]. To begin with, the exponential signal is constructed$e^{j\pi n/8}$ having in the exponent the complex number j = 1 i . Separately, the cosine signal cos $(\frac{n\pi}{11})$ is also constructed . Both signals are defined in the same time interval -10 ≤ n ≤ 10. Finally, to define the signal y 1[n] the element-by-element multiplication of the two defined signals is performed.

Commenting on the results

For n ϵ [-10, 10], the vector y 1 has dimensions 1 x 21. The vector n contains the values that will be represented on the horizontal axis, that is, the n ϵ [-10, 10], while the vector y 1 contains the values to be represented on the vertical axis, i.e., y 1[n] for n ϵ [-10, 10].

The vector yA contains the elements of the exponential sequence$e^{j\pi n/8}$ in the field -10 ≤ n ≤ 10, while the vector yB includes the elements of the cosine sequence cos $(\frac{n\pi}{11})$ in the field -10 ≤ n ≤ 10. Observed from the table of results in the chapter "Explanation of the used commands" on the command line " y 1 = yA .* yB ?" that the values y 1[n] are repeated after the value 1.0000 + 0.0000 i . This verifies the periodicity of the signal.

Theoretical Solution

The signal will be theoretically examined to determine whether it is periodic or aperiodic. The signal is the product of two subsequences. Both sequences are periodic because, there exists an integer k for which $\Omega = 2\pi\frac{k}{N}$. The sequencee $^{j\pi n}/_{8}$ has period $N_1 = 16$, while the sequence cos $(\frac{n\pi}{11})$ has period $N_2 = 22$. Therefore, the product y 1[n] will also be periodic with period :

$$N = \frac{N1 \times N2}{MK\Delta\ (N1, N2)} \rightarrow N = \frac{16 \times 22}{MK\Delta\ (16, 22)} \rightarrow N = \frac{352}{2} \rightarrow N = 176$$

Matlab code for graphical verification of periodicity

Main program

```
N = 5

[ x , n ] = Create _ y 1 (-10, 10)

y 1 = repmat (x, 1, N)
nP = 0:length (y1) - 1

stem ( nP , real (y1))
xlabel ( ' nP ' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y1[ nP ]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' ' )
title ( ' Graphics verification her periodicity of signal y1[n]' )
```

Code her function Create_y1

```
function [x, n] = Create_y1 (n0, n1)

if (n0 >= n1)
error ( 'Arguments must satisfy n0 < n1' );
end

n = n 0:n 1;
yA = exp ((j * pi * n) / 8);
yB = cos ((n * pi) / 11);
x = yA . * yB ;
```

Explanation of the commands used

Commands	Results	Comments
>> N = 5	N = 5	Define the frequency of the discrete-time signal y 1[n].

>> [x, n] = Create_y1 (-10, 10)	x = Columns 1 through 7 0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i 0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 8 through 14 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 15 through 21 0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Call the function Create _ y 1 with arguments the interval $-10 \leq n \leq 10$, where it is checked with an if if the arguments satisfy this inequality. The rest of the code of the function is the same as that of constructing the signal y 1[n] (see Explanation of the commands used, pages 17-19). The function obviously returns the signal y 1[n].
>> y 1 = repmat (x, 1, N)	y1 = Columns 1 through 7 0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i 0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 8 through 14 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 15 through 21 0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i Columns 22 through 28 0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i	repmat function to graphically verify the periodicity of the discrete-time signal y 1[n] with frequency N = 5.

	0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 29 through 35 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 36 through 42 0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i Columns 43 through 49 0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i 0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 50 through 56 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 57 through 63 0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i Columns 64 through 70 0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i 0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 71 through 77 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 78 through 84	

	0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i Columns 85 through 91 0.6785 - 0.6785i 0.7772 - 0.3219i 0.6549 + 0.0000i 0.3838 + 0.1590i 0.1006 + 0.1006i - 0.0545 - 0.1315i 0.0000 - 0.4154i Columns 92 through 98 0.2506 - 0.6050i 0.5949 - 0.5949i 0.8865 - 0.3672i 1.0000 + 0.0000i 0.8865 + 0.3672i 0.5949 + 0.5949i 0.2506 + 0.6050i Columns 99 through 105 0.0000 + 0.4154i - 0.0545 + 0.1315i 0.1006 - 0.1006i 0.3838 - 0.1590i 0.6549 - 0.0000i 0.7772 + 0.3219i 0.6785 + 0.6785i Columns 15 through 21 0.0000 + 1.0000i - 0.3827 + 0.9239i -0.7071 + 0.7071i -0.9239 + 0.3827i -1.0000 + 0.0000i - 0.9239 - 0.3827i -0.7071 - 0.7071i	
>> nP = 0:length(y1) -1	nP = Columns 1 through 22 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 Columns 23 through 44 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 Columns 45 through 66 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 Columns 67 through 88 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 Columns 89 through 105	Define the time interval to graphically verify the periodicity of the signal y 1[n].

	88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104	
>> stem (nP , real (y1))	Figure 3. 2 (Page 2 8)	Plotting the real part of the discrete-time signal y 1[n] with the command stem .
>> xlabel (' nP ', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3. 2 (Page 2 8)	Add the label " nP " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) bold .
>> ylabel ('y1[nP]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3. 2 (Page 2 8)	Addition her label "y1[nP]" in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('Graphical verification of the periodicity of the signal y 1[n]')	Figure 3. 2 (Page 2 8)	Added title to plot named " Graphical verification of periodicity of signal y 1[n] ".

Commenting on the resolution

The solution is based on plotting the periodicity of the signal y 1[n] constructed with the function Create _ y 1. The repmat command overtly plots the periodic signal for N times (N = 5) and easily plots the periodicity of the signal.

Commenting on the results

For n ϵ [-10, 10], it is observed from the table of results in the chapter "Explanation of the commands used" on the command line " y 1 = repmat (x , 1, N)" that the values y 1[n] are repeated 5 times. This verifies the periodicity of the signal and the usefulness of the repmat command .

Graphic representations

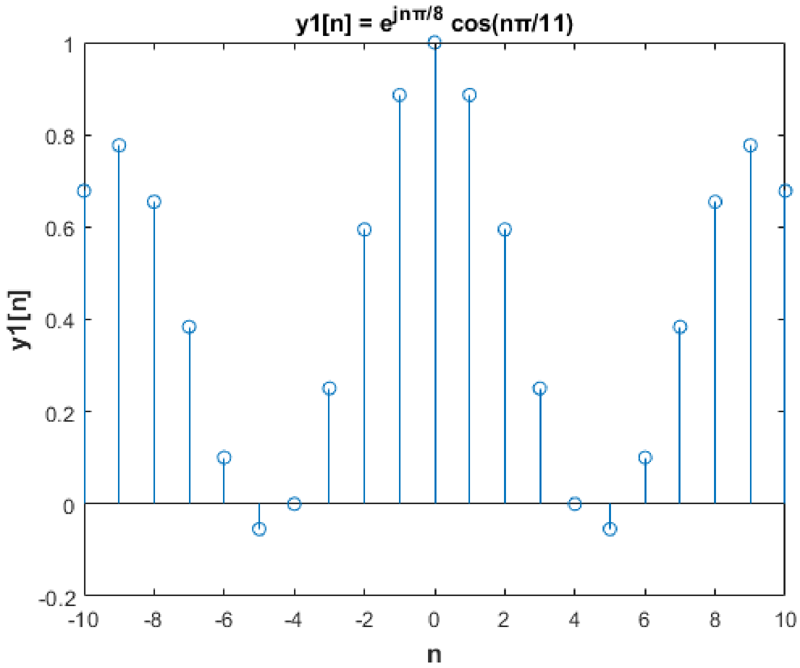

Figure 3.1 The discrete-time signal y 1[n]

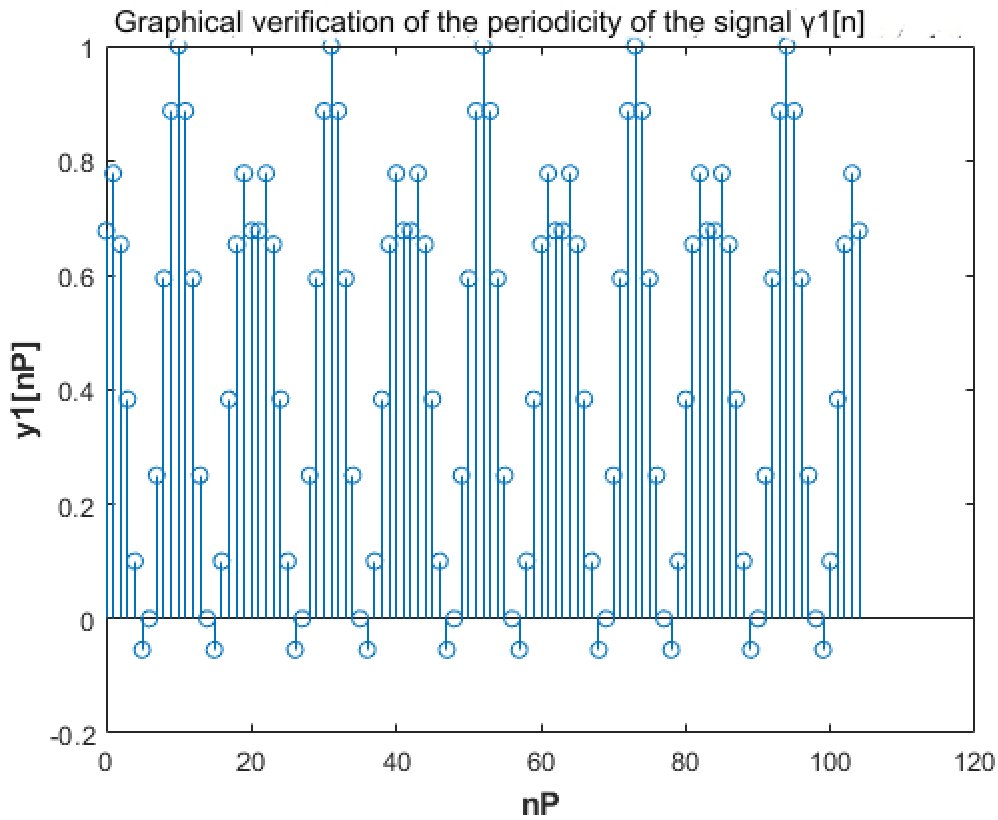

Figure 3.2 Graphical verification of the periodicity of the signal y 1[n] for N = 5

b) y2[n] = 2 cos(π /(3 + AM mod 4) + 0.4n)

The discrete-time signal y 2[n] is as follows:

AM = 19390005
y 2[n] = 2 cos (π / (3 + 19390005 mod 4) + 0.4 n) →
y 2[n] = 2 cos (π / (3 + 1) + 0.4 n) →
y 2[n] = 2 cos ((π / 4) + 0.4 n)

Code in Matlab

```
n = -10:10


y2 = 2 * cos ((pi / 4) + 0.4 * n)

stem (n, y2)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y2[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( ' y2[n] = 2 cos( π /4 + 0.4n) ' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = -10:10	n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Define the time interval of the signal y2[n] with step 1.
>> y2 = 2 * cos ((pi / 4) + 0.4 * n)	y2 = Columns 1 through 13 -1.9947 -1.8940 -1.4944 -0.8588 -0.0876 0.6974 1.3723 1.8306 1.9998 1.8533 1.4142 0.7519 -0.0292 Columns 14 through 21 -0.8057 -1.4549 -1.8745 -1.9981 -1.8062 -1.3292 -0.6424 0.1459	Definition of the discrete time signal y 2[n] using the cosine function for -10 ≤ n ≤ 10.
>> stem (n, y 2)	Figure 3.3 (Page 30)	Plotting the discrete-time signal y 2[n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.3 (Page 30)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y2[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.3 (Page 30)	Addition her label 'y2[n]' in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .

>> title ('y2[n] = 2 cos(π /4 + 0.4n)')	Figure 3.3 (Page 30)	Add a title to the graph named " y 2[n] = 2 cos (π/4 + 0.4 n) ".

Commenting on the resolution

The solution is based on generating the signal y 2[n] with the function cos . The time interval over which the signal is defined was chosen to be the interval $-10 \leq n \leq 10$.

Commenting on the results

For n ϵ [-10, 10], the vector y 2 has dimensions 1 x 21. The vector n contains the values that will be represented on the horizontal axis, that is, the n ϵ [-10, 10], while the vector y 2 includes the values to be represented on the vertical axis, i.e., y 2[n] for n ϵ [-10, 10].
The vector y 2 includes the elements of the sequence 2 cos ((π / 4) + 0.4 n) in the field $-10 \leq n \leq 10$. It is observed from the table of results in the chapter "Explanation of the used commands" in the command line " y 2 = 2 * cos ((pi / 4) + 0.4 * n);" that the values y 2[n] are not repeated. This verifies the aperiodicity of the signal.

Theoretical Solution

For the signal to be periodic there should be N for which :

$y2[n] = y2[n + N] \leftrightarrow$
$2 * \cos((\pi / 4) + 0.4 * n) = 2 * \cos((\pi / 4) + 0.4 * (n + N))$

The term 0.4 n should be an explicit multiple of 2π. The number π is undefined and therefore there is no integer value N that verifies the equality y 2[n] = y 2[n + N]. Therefore, the signal y2[n] is aperiodic.

Graphic representations

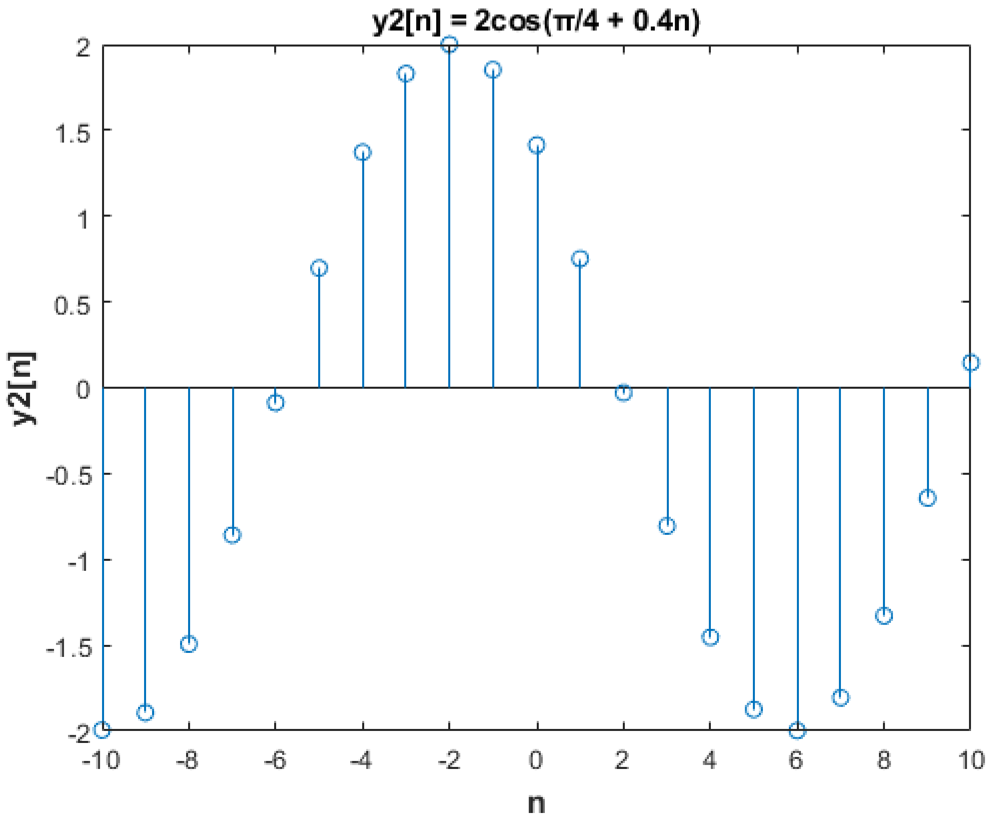

Figure 3.3 The discrete-time signal y 2[n]

c) y3[n] = (AM/100) cos(0.25 π n)

The discrete-time signal y 3[n] is as follows:

AM = 19390005
y3[n] = (19390005 / 100) cos (0.25 π n) →
y3[n] = 193900.05 cos (0.25 π n)

Code at Matlab

```
n = -10:10

y3 = ( 193900 0 05 / 100) * cos (0.25 * pi * n)

stem (n, y3)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y3[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y3[n] = 193900.05cos(0.25 π n)' )
```

Explanation of the commands used

Commands	Results	Comments

>> n = -10:10	n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Define the time interval of the signal y3[n] with step 1.
>> y3 = (19390005 / 100) * cos (0.25 * pi * n);	y3 = 1.0e+05 * Columns 1 through 13 0.0000 1.3711 1.9390 1.3711 -0.0000 -1.3711 -1.9390 -1.3711 0.0000 1.3711 1.9390 1.3711 0.0000 Columns 14 through 21 -1.3711 -1.9390 -1.3711 -0.0000 1.3711 1.9390 1.3711 0.0000	Definition of the discrete time signal y 3[n] using the cosine function for -10 ≤ n ≤ 10.
>> stem (n, y3)	Figure 3. 4 (Page 36)	Plotting the discrete-time signal y 3[n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3. 4 (Page 36)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y3[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3. 4 (Page 36)	Addition her label 'y3[n]' in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('y3[n] = 193900.05cos(0.25 π n)')	Figure 3. 4 (Page 36)	Adding a title to the graph named " y 3[n] = 193900.05 cos (0.25π n) ".

Commenting on the resolution

The solution is based on generating the signal y 3[n] with the function cos . The time interval over which the signal is defined was chosen to be the interval -10 ≤ n ≤ 10.

Commenting on the results

For n ϵ [-10, 10], the vector y 3 has dimensions 1 x 21. The vector n contains the values to be represented on the horizontal axis, that is, the n ϵ [-10, 10], while the vector y 3 includes the values to be represented on the vertical axis, i.e., y 2[n] for n ϵ [-10, 10].
The vector y 3 includes the elements of the sequence 193900.05 cos (0.25π n) in the field -10 ≤ n ≤ 10. It is observed from the table of results in the chapter "Explanation of the used commands" in the command line " y 3 = 193900.05 * cos (0.25 * pi * n);" that the values y 3[n] are repeated. This verifies the periodicity of the signal.

Theoretical Solution

In order to verify the periodicity of the signal y 3[n] it is sufficient that the equality holds:

$y\,3[\,n\,] = y\,3[\,n + N\,] \leftrightarrow$
$193900.05 \cos\left({}^{\pi}/_{4}\, n\right) = 193900.05 \cos\left({}^{\pi}/_{4}(n + N)\right) \leftrightarrow$
$193900.05 \cos\left({}^{\pi}/_{4}\, n\right) = 193900.05 \cos\left({}^{\pi}/_{4}\, n + {}^{\pi}/_{4}\, N\right) \leftrightarrow$
$\omega = {}^{2\pi k}/_{N}$

$\cos\left({}^{\pi}/_{4} n\right) = \cos\left({}^{\pi}/_{4}(n + 8)\right) \leftrightarrow \cos\left({}^{\pi}/_{4}\, n\right) = \cos(2\,\pi\, n)$

Therefore, the signal y 3[n] is periodic with period N = 8.

Matlab code for graphical verification of periodicity

Main program

```
N = 5

[ x , n ] = Create _ y 3 (-10, 10)

y3 = repmat (x, 1, N)
nP = 0:length (y3) - 1

stem ( nP , y3 )
xlabel ( ' nP ' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y3[ nP ]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( ' Graphics verification her periodicity of signal y3[n]' )
```

Code her function Create_y3

```
function [x, n] = Create_y3 (n0, n1)

if (n0 >= n1)
error ( 'Arguments must satisfy n0 < n1' );
end

n = n 0:n 1;
x = ( 193900 0 05 / 100) * cos (0.25 * pi * n);
```

Explanation of the commands used

Commands	Results	Comments
>> N = 5	N = 5	Define the frequency of the discrete-time signal y 3[n].
>> [x, n] = Create_y3 (-10, 10)	x =	Call the function Create _ y 3 with arguments the interval -10 ≤ n ≤ 10,

	1.0e+05 * Columns 1 through 13 0.0000 1.3711 1.9390 1.3711 -0.0000 -1.3711 -1.9390 -1.3711 0.0000 1.3711 1.9390 1.3711 0.0000 Columns 14 through 21 -1.3711 -1.9390 -1.3711 -0.0000 1.3711 1.9390 1.3711 0.0000 n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	where it is checked with an if if the arguments satisfy this inequality. The rest of the code of the function is the same as that of constructing the signal y 3[n] (see Explanation of the commands used, page 29). The function obviously returns the signal y 3[n].
>> y 3 = repmat (x, 1, N)	y3 = 1.0e+05 * Columns 1 through 13 0.0000 1.3711 1.9390 1.3711 -0.0000 -1.3711 -1.9390 -1.3711 0.0000 1.3711 1.9390 1.3711 0.0000 Columns 14 through 26 -1.3711 -1.9390 -1.3711 -0.0000 1.3711 1.9390 1.3711 0.0000 0.0000 1.3711 1.9390 1.3711 -0.0000 Columns 27 through 39 -1.3711 -1.9390 -1.3711 0.0000 1.3711 1.9390 1.3711 0.0000 -1.3711 -1.9390 -1.3711 -0.0000 1.3711 Columns 40 through 52 1.9390 1.3711 0.0000 0.0000 1.3711 1.9390 1.3711 -0.0000 -1.3711 -1.9390 -1.3711 0.0000 1.3711 Columns 53 through 65 1.9390 1.3711 0.0000 -1.3711 -1.9390 -1.3711 -0.0000 1.3711	repmat function to graphically verify the periodicity of the discrete-time signal y 3[n] with frequency N = 5.

	1.9390 1.3711 0.0000 0.0000 1.3711 Columns 66 through 78 1.9390 1.3711 -0.0000 -1.3711 -1.9390 -1.3711 0.0000 1.3711 1.9390 1.3711 0.0000 -1.3711 -1.9390 Columns 79 through 91 -1.3711 -0.0000 1.3711 1.9390 1.3711 0.0000 0.0000 1.3711 1.9390 1.3711 -0.0000 -1.3711 -1.9390 Columns 92 through 104 -1.3711 0.0000 1.3711 1.9390 1.3711 0.0000 -1.3711 -1.9390 -1.3711 -0.0000 1.3711 1.9390 1.3711 Column 105 0.0000	
>> nP = 0:length(y3) -1	nP = Columns 1 through 22 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 Columns 23 through 44 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 Columns 45 through 66 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 Columns 67 through 88 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 Columns 89 through 105 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104	Define the time interval to graphically verify the periodicity of the signal y 3[n].
>> stem (nP , y3)	Figure 3.5 (Page 36)	Plotting the discrete-time signal y 3[n] with the command stem .

>> xlabel (' nP ', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.5 (Page 36)	Add the label " nP " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) bold .
>> ylabel ('y3[nP]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.5 (Page 36)	Addition her tag "y3[nP]" on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('Graphical verification of the periodicity of the signal y 3[n]')	Figure 3.5 (Page 36)	Added title to plot named " Graphical verification of periodicity of signal y 3[n] ".

Commenting on the resolution

The solution is based on plotting the periodicity of the signal y 3[n] constructed with the function Create _ y 3. The repmat command overtly plots the periodic signal for N times (N = 5) and easily plots the periodicity of the signal.

Commenting on the results

For n ϵ [-10, 10], it is observed from the table of results in the chapter "Explanation of the used commands" on the command line " y 3 = repmat (x , 1, N)" that the values y 3[n] are repeated 5 times. This verifies the periodicity of the signal and the usefulness of the repmat command .

Graphic representations

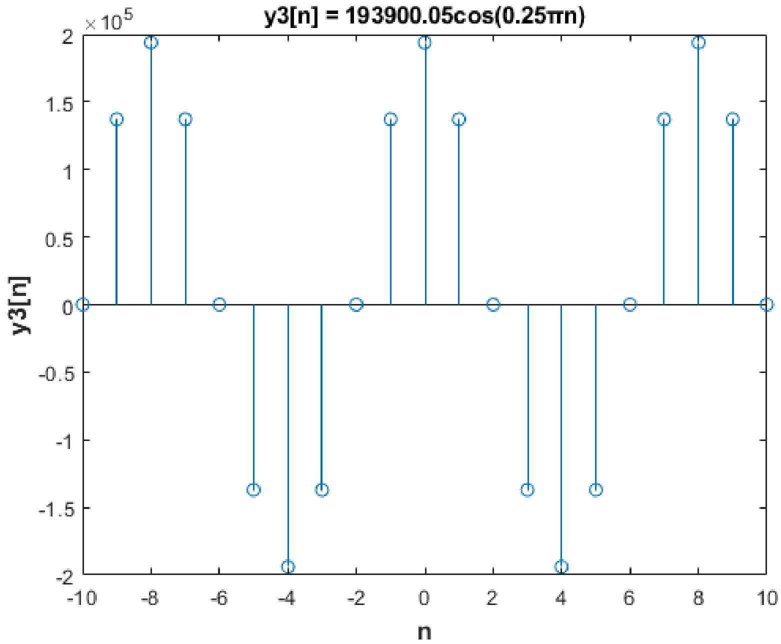

Figure 3.4 The discrete-time signal y 3[n]

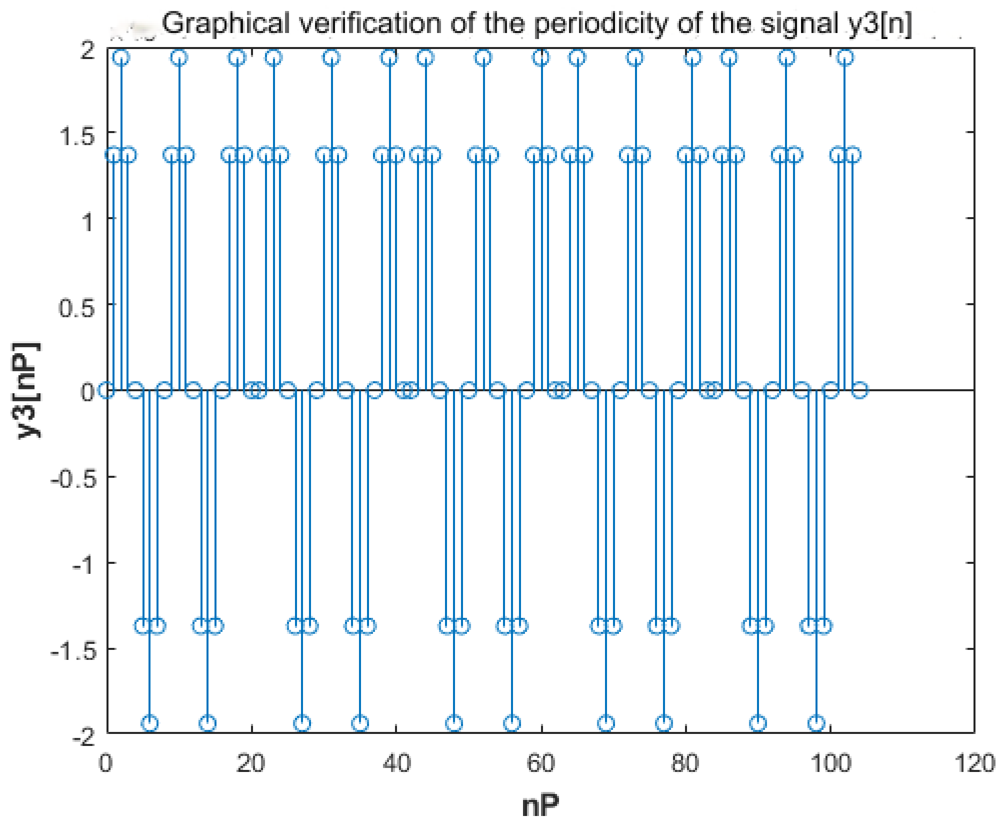

Figure 3.5 The signal y 3[n] is repeated for N = 5 times verifying its periodicity

d) y **4[n] = 3 cos²(** $\frac{\pi}{6}n + \pi/10$**)**

Code in Matlab

```
n = -10:10

y4 = 3 * cos ((pi / 6) * n + (pi / 10) ).^ 2

stem (n, y4)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y4[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y4[n] = 3cos^2( π /6 n + π /10)' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = -10:10	n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Define the time interval of the signal y4[n] with step 1.
>> y4 = 3 * cos ((pi / 6) * n + (pi / 10)).^ 2	y4 =	Definition of the discrete time signal y 4[n] using the

	Columns 1 through 13 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 Columns 14 through 21 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568	cosine function for -10 ≤ n ≤ 10.
>> stem (n, y4)	Figure 3.6 (Page 42)	Plotting the discrete-time signal y 4[n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.6 (Page 42)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y4[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.6 (Page 42)	Addition her label 'y4[n]' on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('y4[n] = 3cos^2(π /6 n + π /10)')	Figure 3.6 (Page 42)	Add a title to the graph named " y 4[n] = 3 cos ^2(π/6 n + π/10) ".

Commenting on the resolution

The solution is based on generating the signal y 4[n] with the function cos . The time interval over which the signal is defined was chosen to be the interval -10 ≤ n ≤ 10.

Commenting on the results

For n ϵ [-10, 10], the vector y 4 has dimensions 1 x 21. The vector n contains the values that will be represented on the horizontal axis, that is, the n ϵ [-10, 10], while the vector y 2 contains the values to be represented on the vertical axis, that is, y 4[n] for n ϵ [-10, 10].
The vector y 4 includes the elements of the sequence 3 * cos ((pi / 6) * n + (pi / 10)).^2 in the field -10 ≤ n ≤ 10. It is observed from the table of results in the chapter "Explanation of the used commands" in the command line " y 4 = 3 * cos ((pi / 6) * n + (pi / 10)).^2 ?" that the values y 4[n] are repeated. This verifies the periodicity of the signal.

Theoretical Solution

From the trigonometric identity $\cos^2\varphi = \frac{1+\cos 2\varphi}{2}$ we have:

$$3 \times \frac{1+\cos 2(\frac{\pi}{6}n + \frac{\pi}{10})}{2} \rightarrow \frac{3}{2} + \frac{3}{2}\cos 2\left(\frac{\pi}{6}n + \frac{\pi}{10}\right) \rightarrow \frac{3}{2} + \frac{3}{2}\cos\left(\frac{\pi}{3}n + \frac{\pi}{5}\right)$$

That is, the signal can be expressed as the sum of two signals $x1[n] = \frac{3}{2}$ and $x2[n] = \frac{3}{2}\cos\left(\frac{\pi}{3}n + \frac{\pi}{5}\right)$

x 1[n] can be expressed as $x1[n] = \frac{3}{2} \times 1^n$, so it is also periodic with period N = 1. x 2[n] is periodic because there exists k such that $\omega = {}^{2\pi k}/_{N} \rightarrow N = {}^{2\pi k}/_{\omega} \rightarrow N = {}^{2\pi}/_{\pi\,/\,3} \rightarrow N = 6$. Therefore, the period of signal y 4[n] is 6.

Matlab code for graphical verification of periodicity

Main program

```
N = 5


[ x , n ] = Create _ y 4 (-10, 10)

y4 = repmat ( x , 1 , N
nP = 0:length (y4) - 1

stem ( nP , y4 )
xlabel ( ' nP ' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y4[ nP ]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'Graphical verification of the periodicity of the signal y4[n]' )
```

Code her function Create_y4

```
function [x, n] = Create_y4 (n0, n1)


if (n0 >= n1)
error ( 'Arguments must satisfy n0 < n1' );
end

n = n 0:n 1;
x = 3 * cos ((pi / 6) * n + (pi / 10) ).^ 2;
```

Explanation of the commands used

Commands	Results	Comments
>> N = 5	N = 5	Define the frequency of the discrete-time signal y 4[n].
>> [x, n] = Create_y4(-10, 10)	x = Columns 1 through 13 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 Columns 14 through 21	Call the function Create _ y 4 with arguments the interval -10 ≤ n ≤ 10, where it is checked with an if if the arguments satisfy this inequality. The rest of the code of the function is the same as that of constructing the signal y 4[n] (see Explanation of the commands used, page 35). The function obviously returns the signal y 4[n].

	0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	
>> y4 = repmat (x, 1, N)	y4 = Columns 1 through 13 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 Columns 14 through 26 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 0.1297 0.2865 1.6568 2.8703 2.7135 Columns 27 through 39 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 Columns 40 through 52 0.1297 0.2865 1.6568 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 Columns 53 through 65 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 0.1297 0.2865 Columns 66 through 78 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 Columns 79 through 91	repmat function to graphically verify the periodicity of the discrete-time signal y 4[n] with frequency N = 5.

	2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 Columns 92 through 104 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 1.6568 2.8703 2.7135 1.3432 0.1297 0.2865 Column 105 1.6568	
>> nP = 0:length(y4) -1	nP = Columns 1 through 22 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 Columns 23 through 44 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 Columns 45 through 66 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 Columns 67 through 88 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 Columns 89 through 105 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104	Define the time interval to graphically verify the periodicity of the signal y 4[n].
>> stem (nP , y4)	Figure 3.7 (Page 43)	Plotting the discrete-time signal y 4[nP] with the command stem .
>> xlabel (' nP ', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.7 (Page 43)	Add the label " nP " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) bold .
>> ylabel ('y4[nP]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.7 (Page 43)	Addition her tag "y4[nP]" on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('Graphical verification of the periodicity of the signal y 4[n]')	Figure 3.7 (Page 43)	Add title to plot named " Graphical verification of periodicity of signal y 4[n] ".

Commenting on the resolution

The solution is based on plotting the periodicity of the signal y 4[n] constructed with the function Create _ y 4. The repmat command overtly plots the periodic signal for N times (N = 5) and easily plots the periodicity of the signal.

Commenting on the results

For n ϵ [-10, 10], it is observed from the table of results in the chapter "Explanation of the used commands" on the command line " y 4 = repmat (x , 1, N)" that the values y 4[n] are repeated 5 times. This verifies the periodicity of the signal and the usefulness of the repmat command .

Graphic representations

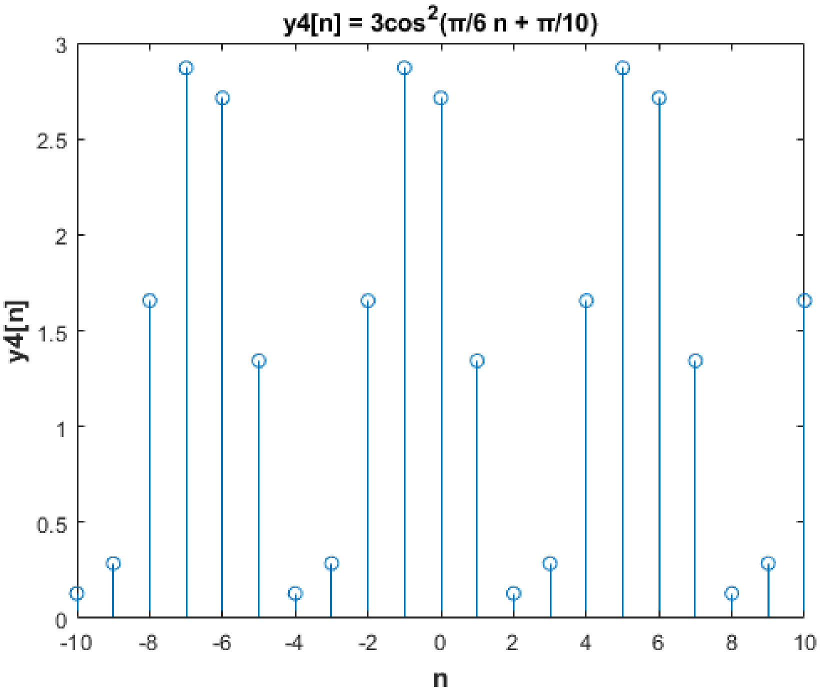

Figure 3.6 The discrete-time signal y 4[n]

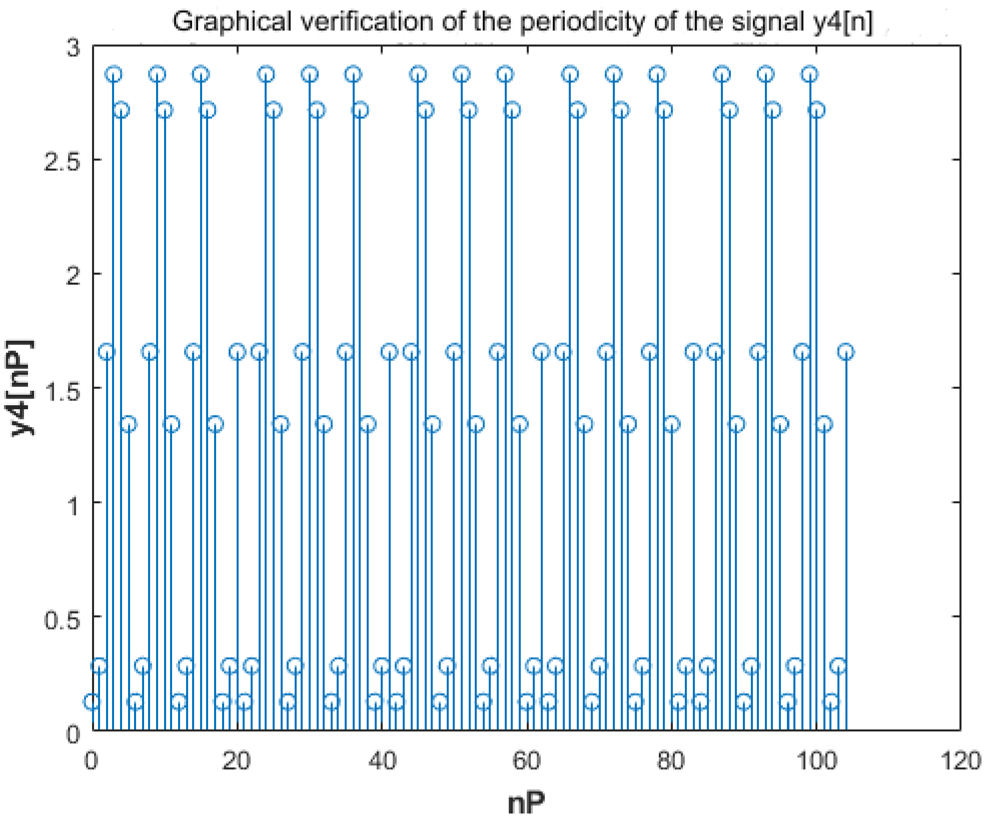

Figure 3.7 The signal y 4[n] is repeated for N = 5 times verifying its periodicity

e) $y\,5[\,n\,] = e^{j\pi n/2} + e^{-j\pi n/2}$

Code in Matlab

```
n = -10:10

yA = exp ((j * pi * n) / 2)
yB = exp ((-j * pi * n) / 2)
y5 = yA + yB

stem (n, real (y5))
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y5[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y5[n] = e^j^n ^ π ^/^2 + e^j^n ^ π ^/^-^2' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = -10:10	n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Define the time interval of the signal y5[n] with step 1.

>> yA = exp ((j * pi * n) / 2)	yA = Columns 1 through 7 -1.0000 - 0.0000i 0.0000 - 1.0000i 1.0000 + 0.0000i - 0.0000 + 1.0000i -1.0000 - 0.0000i 0.0000 - 1.0000i 1.0000 + 0.0000i Columns 8 through 14 -0.0000 + 1.0000i - 1.0000 - 0.0000i 0.0000 - 1.0000i 1.0000 + 0.0000i 0.0000 + 1.0000i -1.0000 + 0.0000i -0.0000 - 1.0000i Columns 15 through 21 1.0000 - 0.0000i 0.0000 + 1.0000i -1.0000 + 0.0000i -0.0000 - 1.0000i 1.0000 - 0.0000i 0.0000 + 1.0000i -1.0000 + 0.0000i	Definition of the 1st term of $e^{j\pi n / 2}$ the signal y 5[n] with the help of the exponential function exp .
>> yB = exp ((-j * pi * n) / 2)	yB = Columns 1 through 7 -1.0000 + 0.0000i 0.0000 + 1.0000i 1.0000 - 0.0000i - 0.0000 - 1.0000i -1.0000 + 0.0000i 0.0000 + 1.0000i 1.0000 - 0.0000i Columns 8 through 14 -0.0000 - 1.0000i - 1.0000 + 0.0000i 0.0000 + 1.0000i 1.0000 + 0.0000i 0.0000 - 1.0000i -1.0000 - 0.0000i -0.0000 + 1.0000i Columns 15 through 21 1.0000 + 0.0000i 0.0000 - 1.0000i -1.0000 - 0.0000i -0.0000 + 1.0000i 1.0000 + 0.0000i 0.0000 - 1.0000i -1.0000 - 0.0000i	Definition of the 2nd term of $e^{-j\pi n / 2}$ the signal y 5[n] with the help of the exponential function exp .
>> y5 = yA + yB	y5 = Columns 1 through 13 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 0.0000 -2.0000 Columns 14 through 21	Definition of the signal y 5[n] as the sum of the two individual signals $e^{j\pi n / 2}$ and $e^{-j\pi n / 2}$.

	-0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000	
>> stem (n, real (y5))	Figure 3.8 (Page 49)	Plotting the real part of the discrete-time signal y 5[n] with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.8 (Page 49)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y5[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.8 (Page 49)	Addition her label 'y5[n]' on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('y5[n] = e^j^n ^ π ^/^2 + e^j^n ^ π ^/^-2')	Figure 3.8 (Page 49)	Add title to graph named " y5[n] = e^j^n^π^/^2 + e^j^n^π^/^-2 ".

Commenting on the resolution

The solution is based on the gradual creation of the discrete signal y 5[n], constructing one by one the two signals that in the end with their sum will depict y 5 [n]. To begin with, the exponential signal is constructed$e^{j\pi n/2}$ having in the exponent the complex number j = 1 i . Separately, the exponent sign is also made $e^{-j\pi n/2}$. Both signals are defined in the same time interval $-10 \leq n \leq 10$. Finally, to define the signal y 5[n] a sum of the two defined signals is performed.

Commenting on the results

For n ϵ [-10, 10], the vector y 5 has dimensions 1 x 21. The vector n contains the values that will be represented on the horizontal axis, that is, the n ϵ [-10, 10], while the vector y 5 contains the values to be represented on the vertical axis, i.e., y 5[n] for n ϵ [-10, 10].

The vector yA contains the elements of the exponential sequence$e^{j\pi n/2}$ in the field $-10 \leq n \leq 10$, while the vector yB includes the elements of the exponential sequence$e^{-j\pi n/2}$ in the field $-10 \leq n \leq 10$. It is observed from the table of results in the chapter "Explanation of the used commands" on the command line " y 5 = yA + yB ?" that the values y 5[n] are repeated. This verifies the periodicity of the signal.

Theoretical Solution

The signal y 5[n] is the partial sum of the two periodic signals $e^{j\pi n/2}$and $e^{-j\pi n/2}$, since there is an integer k for which $\omega = 2\pi k/N \rightarrow N = 2\pi k/\omega \rightarrow N = 2\pi/\pi/2 \rightarrow N = 4$. Each signal has a period of N = 4, therefore the signal y 5[n] will also have a period of N = 5.

Matlab code for graphical verification of periodicity

Main program

```
N = 5


[ x , n ] = Create _ y 5 (-10, 10)

y5 = repmat (x, 1, N)
nP = 0:length (y5) - 1

stem ( nP , real (y5))
xlabel ( ' nP ' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y5[ nP ]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'Graphical verification of the periodicity of the signal y5[n]' )
```

Code her function Create_y5

```
function [x, n] = Create_y5 (n0, n1)

if (n0 >= n1)
error ( 'Arguments must satisfy n0 < n1' );
end

n = n 0:n 1;
j = 1i;
yA = exp ((j * pi * n) / 2);
yB = exp ((-j * pi * n) / 2);
x = yA + yB ;
```

<u>Explanation of the commands used</u>

Commands	Results	Comments
>> N = 5	N = 5	Define the frequency of the discrete-time signal y 5[n].
>> [x, n] = Create_y5 (-10, 10)	x = Columns 1 through 13 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 0.0000 -2.0000 Columns 14 through 21 -0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000 n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Call the function Create _ y 5 with arguments the interval -10 ≤ n ≤ 10, where it is checked with an if if the arguments satisfy this inequality. The rest of the code of the function is the same as that of constructing the signal y 5[n] (see Explanation of the commands used, pages 42 – 43). The function obviously returns the signal y 5[n].

>> y 5 = repmat (x, 1, N)	y5 = Columns 1 through 13 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 0.0000 -2.0000 Columns 14 through 26 -0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 Columns 27 through 39 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000 -0.0000 Columns 40 through 52 2.0000 0.0000 -2.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 Columns 53 through 65 2.0000 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000 -2.0000 0.0000 Columns 66 through 78 2.0000 -0.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000 Columns 79 through 91 0.0000 -2.0000 -0.0000 2.0000 0.0000 -2.0000 -2.0000 0.0000 2.0000 -0.0000 -2.0000 0.0000 2.0000 Columns 92 through 104 -0.0000 -2.0000 0.0000 2.0000 0.0000 -2.0000 -0.0000 2.0000	repmat function to graphically verify the periodicity of the discrete-time signal y 5[n] with frequency N = 5.

	0.0000 -2.0000 -0.0000 2.0000 0.0000 Column 105 -2.0000	
>> nP = 0:length(y5) -1	nP = Columns 1 through 22 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 Columns 23 through 44 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 Columns 45 through 66 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 Columns 67 through 88 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 Columns 89 through 105 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104	Define the time interval to graphically verify the periodicity of the signal y 5[n].
>> stem (nP , real (y 5))	Figure 3.9 (Page 50)	Plotting the real part of the discrete-time signal y 5[nP] with the command stem .
>> xlabel (' nP ', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.9 (Page 50)	Add the label " nP " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) bold .
>> ylabel ('y5[nP]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 3.9 (Page 5 0)	Addition her tag "y5[nP]" on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('Graphical verification of the periodicity of the signal y 5[n]')	Figure 3.9 (Page 50)	Add title to plot named " Graphical verification of periodicity of signal y 5[n] ".

Commenting on the resolution

The solution is based on plotting the periodicity of the signal y 5[n] constructed with the Create _ y 5 function. The repmat command overtly plots the periodic signal for N times (N = 5) and easily plots the periodicity of the signal.

Commenting on the results

For n ϵ [-10, 10], it is observed from the table of results in the chapter "Explanation of the used commands" on the command line " y 5 = repmat (x , 1, N)" that the values y 5[n] are repeated 5 times. This verifies the periodicity of the signal and the usefulness of the repmat command .

Graphic representations

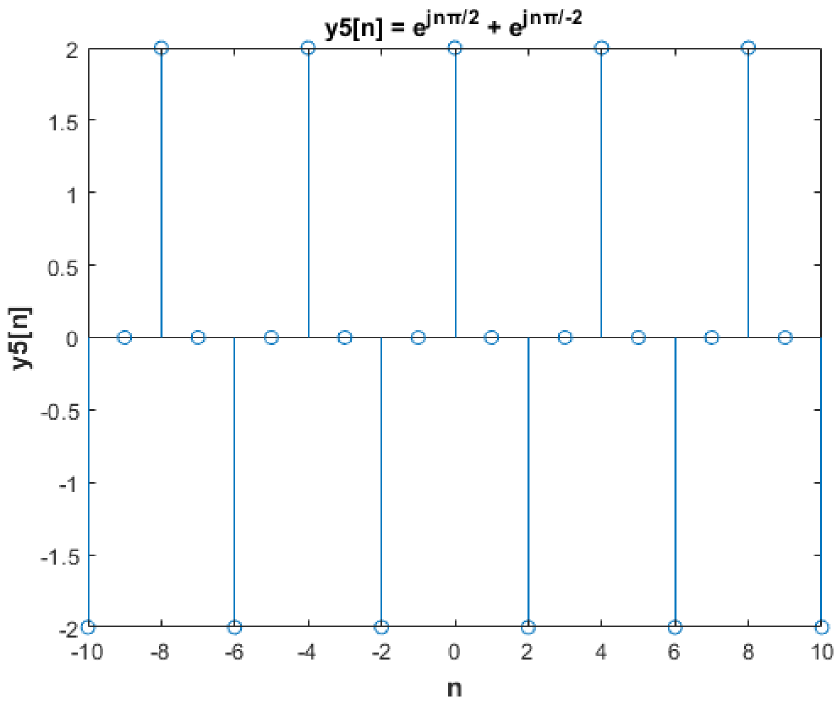

Figure 3.8 The discrete-time signal y 5[n]

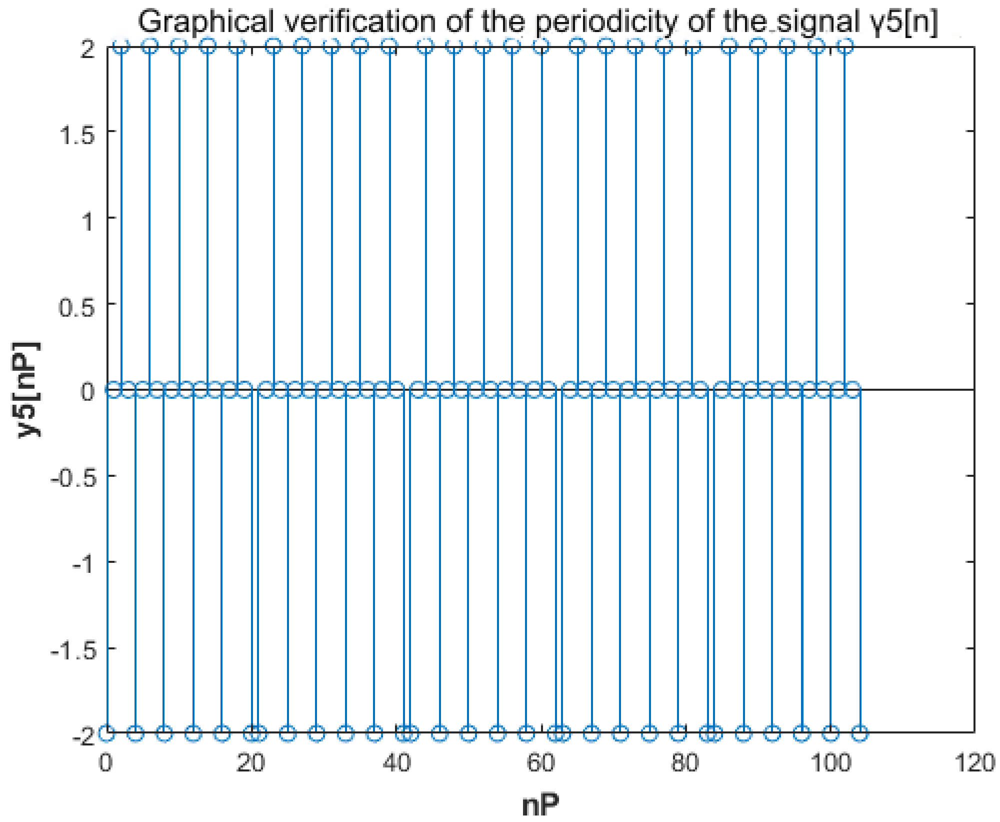

Figure 3.9 The signal y 5[n] is repeated for N = 5 times verifying its periodicity

EXERCISE 4

Write a function that accepts as an argument a discrete time signal and the time interval in which it is defined and returns the even and odd part of the signal, as well as the time axis in which they are defined. Finally, the function should make the graph of the two output signals, on the correct time axis, e.g. [xe,xo,m] = ev_od(x,n). Then using the function, calculate the even and odd part of the sequence x[n] = [the digits of AM, 1, the digits of AM], where the unit in the center corresponds to the time instant n=0. Make the graphic representations. (for example, for AM = 17098, x[n] = [1 7 0 9 8 1 1 7 0 9 8])

--- --- --------------------------------
-

The sequence by which the even and odd part will be calculated and plotted is as follows:

AM = 19390005
x [n] = [digits of AM, 1, digits of AM] →
x [n] = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5]

Code in Matlab

Main program

```
n = -8:8


x = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5]
[ xe , xo, m] = ev_od (x, n)
```

Code her function ev_od

```
function [ xe , xo, m] = ev_od (x, n)

len = length (n)
xr = x( len :- 1:1)
xe = 1/2 * (x + xr )
xo = 1/2 * (x - xr )
x_sum = xe + xo
m = n

subplot (4, 1, 1)
stem (m, x)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'x[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'x[n] = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5]' )
subplot (4, 1, 2)
stem (m, xe )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( ' xe [n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'The even part of the signal x[n]' )
subplot (4, 1, 3)
stem (m, xo)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'xo[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'The odd part of the signal x[n]' )
subplot (4, 1, 4)
stem (m, xe + xo)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( ' xe [n] + xo[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'The sum of the even and the odd part of the signal x[n]' )
```

Explanation of the commands used

Explanation of the main program commands used

Commands	Results	Comments
>> n = -8:8	n = -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8	Define the time interval of the signal x [n] with step 1.
>> x = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5]	x = 1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5	Definition of the discrete time signal y 4[n] using the cosine function for $-10 \leq n \leq 10$.
>> [xe , xo, m] = ev_od (x, n)	xe =	Call the function ev _ od with arguments the time interval $-8 \leq$

	Columns 1 through 13 3.0000 4.5000 1.5000 4.5000 4.5000 1.5000 4.5000 3.0000 1.0000 3.0000 4.5000 1.5000 4.5000 Columns 14 through 17 4.5000 1.5000 4.5000 3.0000 xo = Columns 1 through 13 -2.0000 4.5000 1.5000 4.5000 -4.5000 -1.5000 -4.5000 2.0000 0 -2.0000 4.5000 1.5000 4.5000 Columns 14 through 17 -4.5000 -1.5000 -4.5000 2.0000 m = -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8	$n \leq 8$. The function returns the even and odd part of the signal x[n], the time interval defined and plots the graphs. In detail, the explanation of the function code in the chapter "Explanation of the used commands of the ev _ od function " on pages .

Explanation of the used commands of the ev _ od function

Commands	Results	Comments
>> function [xe , xo, m] = ev_od (x, n)		ev _ od function that takes as an argument a discrete-time signal x [n] and returns the even part of the signal (xe), the odd part (xo) of the signal, and the time interval specified. To solve the exercise, the signal x [n] = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5] in the interval $-8 \leq n \leq 8$ was given as an argument.
>> len = length(n)	len = 17	The size of the time interval $-8 \leq n \leq 8$ given as an argument, that is, the number of values in the time domain defined by the signal x [n].
>> xr = x(len :- 1:1)	xr = 5 0 0 0 9 3 9 1 1 5 0 0 0 9 3 9 1	Define the signal x [- n] to plot the sequences xe [n] = $\frac{x[n]+x[-n]}{2}$ and xo [n] = $\frac{x[n]-x[-n]}{2}$, where xe [n] the even part of the signal x [n] and xo [n] the odd part of the signal.
>> xe = 1/2 * (x + xr)	xe =	Define the signal xe [n], that is, the even part of the signal x [n] by

	Columns 1 through 13 3.0000 4.5000 1.5000 4.5000 4.5000 1.5000 4.5000 3.0000 1.0000 3.0000 4.5000 1.5000 4.5000 Columns 14 through 17 4.5000 1.5000 4.5000 3.0000	plotting the sequence xe [n] = $\frac{x[n]+x[-n]}{2}$.
>> xo = 1/2 * (x - xr)	xo = Columns 1 through 13 -2.0000 4.5000 1.5000 4.5000 -4.5000 -1.5000 -4.5000 2.0000 0 -2.0000 4.5000 1.5000 4.5000 Columns 14 through 17 -4.5000 -1.5000 -4.5000 2.000	Define the signal xo [n], that is, the odd part of the signal x [n] by plotting the sequence xo [n] = $\frac{x[n]-x[-n]}{2}$.
>> x_sum = xe + xo	x_sum = 1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5	Define the signal x _ sum as the sum of the even and the odd part of the signal x [n].
>> m = n	m = -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8	Assign the time interval defined by the even and odd part of the signal x [n], to the variable returned by the function.
>> subplot(4, 1, 1)	Figure 4.1 / 1 º diagram	Drawing of the 1st of 4 diagrams that will be represented one below the other.
>> stem (m, x)	Figure 4.1 / 1 º diagram	Plot the signal x [n] in the interval -8 ≤ n ≤ 8.
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 1 º diagram	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('x[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 1 º diagram	Add the label " x [n]" to the vertical y- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> title ('x[n] = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5]')	Figure 4.1 / 1 º diagram	Add a title to the plot named " x[n] = [1 9 3 9 0 0 0 5 1 1 9 3 9 0 0 0 5] ".
>> subplot(4, 1, 2)	Figure 4. 1 / 2 the diagram	Drawing of the 2nd of 4 diagrams that will be represented one below the other.
>> stem (m, xe)	Figure 4.1 / 2 the diagram	Plot the even part xe [n] of the signal x [n] in the interval -8 ≤ n ≤ 8.
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 2 the diagram	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .

>> ylabel (' xe [n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 2 the diagram	Addition her tag " xe [n]" on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('The even part of the signal x[n]')	Figure 4.1 / 2 the diagram	Add title to plot named " The even part of signal x[n]' ".
>> subplot(4, 1, 3)	Figure 4.1 / 3 ° diagram	Drawing the 3rd of 4 diagrams that will be represented one below the other.
>> stem (m, xo)	Figure 4.1 / 3 ° diagram	Plot the odd part xo [n] of the signal x [n] in the interval -8 ≤ n ≤ 8.
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 3 ° diagram	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('xo[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 3 ° diagram	Addition her tag "xo[n]" on vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('The odd part of the signal x[n]')	Figure 4.1 / 3 ° diagram	Add title to plot named " The redundant part of the signal x[n] ".
>> subplot(4, 1, 4)	Figure 4.1 / 4 ° diagram	Drawing the 4th of the 4 diagrams that will be represented one below the other.
>> stem (m, x_sum)	Figure 4.1 / 4 ° diagram	Plot the sum of the even and the odd part (xe [n] + xo [n]) of the signal x [n] in the interval -8 ≤ n ≤ 8.
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 4.1 / 4 ° diagram	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel (' xe [n] + xo [n]', ' FontSize ', 12, ' FontWeight ', ' bold ')	Figure 4.1 / 4 ° diagram	Addition her tag " xe [n] + xo[n]" in vertical y axis , with size label (FontSize) 12 and weight (FontWeight) bold .
>> title ('The sum of the even and the odd part of the signal x[n]')	Figure 4.1 / 4 ° diagram	Add a title to the plot named "The sum of the even and the odd part of the signal x[n]"

Commenting on the resolution

The solution is based on plotting the sequences xe [n] = $\frac{x[n]+x[-n]}{2}$ and xo [n] = $\frac{x[n]-x[-n]}{2}$ to plot the even part and the odd part of the signal x [n] respectively. Drawing the diagram of the sum of the even and the odd part of the signal x [n] aims to prove the equality x [n] = xe [n] + xo [n], that is, that the signal x [n] can be expressed and as the sum of its even and odd parts.

Commenting on the results

For n ϵ [-8, 8] it is observed that the values xe [n] for the even part of the signal x [n] show a symmetry with respect to the vertical axis. For example, xe [-8] = xe [8] = 3, xe [-7] = xe [7] = 4.5 and so on.
For n ϵ [-8, 8] it is observed that the values xo [n] for the odd part of the signal x [n] exhibit a symmetry with

respect to the origin of the axes 0. For example, xo [-8] = -2 | xo [8] = 2, xo [-7] = 4.5 | xo [7] = -4.5 and so on For n ϵ [-8, 8] it is observed that the values x _ sum [n] for the sum of the even part and the odd part of the signal x [n] are the same as those of the signal x [n]. For example, x [-8] = x _ sum [8] = 1, x [-7] = x _ sum [7] = 9 and so on.

Graphic representations

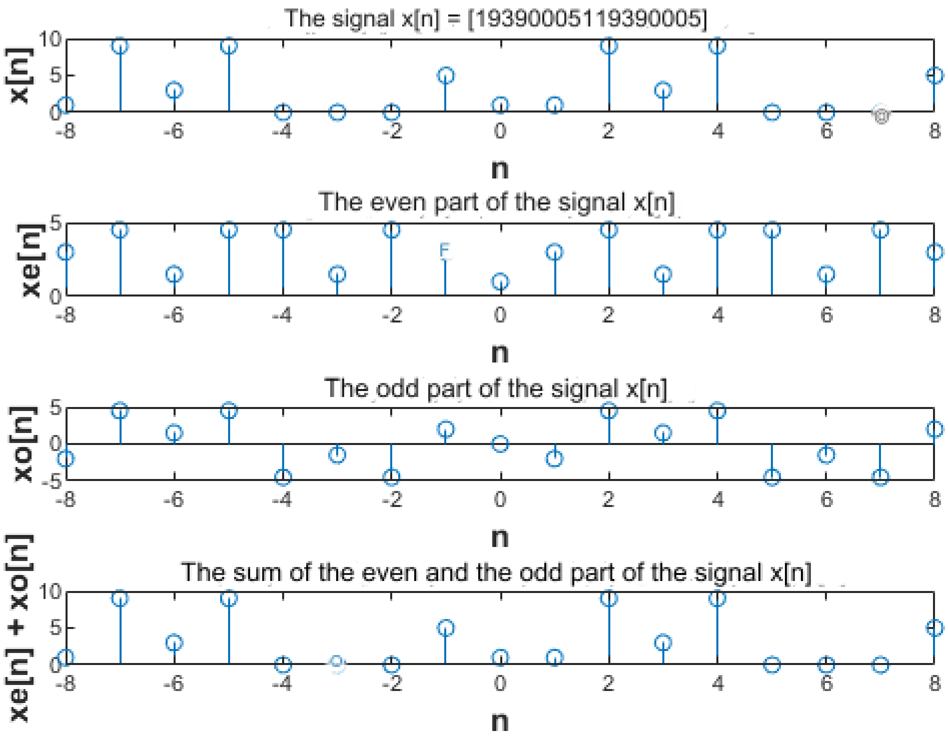

Figure 4.1 The discrete-time signal x [n], the even part xe [n], the odd part xo [n] and the sum xe [n] + xo [n]

EXERCISE 5

Calculate the LTI system output:

a) with shock response $h[n] = |n-3|(u[n] - u[n-6])$ and input

$$x[n] = \begin{cases} -1, & n = -1 \\ 1, & n = 0 \\ 2, & n = 1 \\ -1, & n = 2 \end{cases}$$

Setting $-10 \leq n \leq 10$.

b) with impulse response $h[n] = 0.6\, nu[n]$ and input $x[n] = u[n] - u[n-10]$

Setting $-10 \leq n \leq 30$.

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-

a)

Code in Matlab

Main program

```
n = -10:10
n _ conv = -20:20

m = length(n)
x( 10:13) = [-1 1 2 -1]
x(14:m) = 0
h1 = abs(n - 3)
h2 = stepseq (0, -10, 10)
h3 = stepseq (6, -10, 10)
h = h 1 . * (h2 - h3)
y1 = conv(h, x)

stem ( n_conv , y1 )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y1[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'The system output y1[n]' )
```

Code of the stepseq function

```
function [ x , n ] = stepseq ( n 0, n 1, n 2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end
n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> n = -10:10;	n = -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Defining the time interval of the input signal x [n] and the impulse response h [n] = $\lvert n-3 \rvert (u[n] - u[n-6])$, with step 1.
>> n_conv = -20:20	n_conv = Columns 1 through 22 -20 -19 -18 -17 -16 -15 -14 -13 -12 -11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1	Define the time interval of the system output y 1[n] with input signal x [n] and shock deviation h [n] = $\lvert n-3 \rvert (u[n] - u[n-6])$, with step 1.

	Columns 23 through 41 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20	
>> m = length(n)	m = 21	The size of the time interval -10 ≤ n ≤ 10, that is, the number of values in the time domain defined by the input signal x [n] and the impulse response h [n] = $\| n - 3 \|(u [n] - u [n - 6])$.
>> x(10:13) = [-1 1 2 -1]	x = Columns 1 through 22 0 0 0 0 0 0 0 0 0 -1 1 2 -1 1 1 1 1 1 1 1 0 0	Define the values of the input signal x [n] in the interval -10 ≤ n ≤ 10, where from the interval -1 ≤ n ≤ 2 with step 1, the input signal takes the values -1, 1, 2, -1 respectively.
>> x(14:m) = 0	x = Columns 1 through 22 0 0 0 0 0 0 0 0 0 -1 1 2 -1 0 0 0 0 0 0 0 0 0	Zero values from the interval 3 ≤ n ≤ 10 that were initialized with aces in the command x (10:13) = [-1 1 2 -1].
>> h1 = abs(n - 3)	h1 = 13 12 11 10 9 8 7 6 5 4 3 2 1 0 1 2 3 4 5 6 7	Definition of the signal h1 = | n – 3| in the interval -10 ≤ n ≤ 10.
>> h2 = stepseq(0, -10, 10)	h2 = 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1	Define the signal h2 = u [n] on the interval -10 ≤ n ≤ 10 by calling the stepseq function whose code is discussed in Lab 1 “Introduction to Discrete Signals” on pages 7 – 8.
>> h3 = stepseq(6, -10, 10)	h3 = 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1	Define the signal h3 = u [n – 6] on the interval -10 ≤ n ≤ 10 by calling the function stepseq whose code is discussed in Lab 1 “Introduction to Discrete Signals” on pages 7 – 8.
>> h = h 1 . * (h2 - h3)	h = 0 0 0 0 0 0 0 0 0 0 3 2 1 0 1 2 0 0 0 0 0	Definition of the shock response h [n] = $\| n - 3 \|(u [n] - u [n - 6])$ in the interval -10 ≤ n ≤ 10.
>> y1 = conv(h, x)	y1 = Columns 1 through 22 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 -3 1 7 Columns 23 through 44 2 -1 -2 4 3 -2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 45 through 61 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	Computing the convolution of the input x [n] with the impulse response h [n] = $\| n - 3 \|(u [n] - u [n - 6])$ to calculate the output y 1[n] of the system. The convolution is calculated by calling the conv function .

>> stem (n_conv, y1)	Figure 5.5 (page 62)	Plot the output y 1[n] of the system in the interval -20 ≤ n ≤ 20 with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 5.5 (page 62)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y1[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 5.5 (page 62)	Add the label " y 1[n]" to the vertical y- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> title ('The system output y1[n]')	Figure 5.5 (page 62)	Add a title to the plot named "The output of the system y1[n]"

Commenting on the resolution

The solution is based on the stepwise design of the input signal x [n] and the impulse response h [n] = $|n-3|(u[n]-u[n-6])$. The step sequences u [n] and u [n – 6] are defined using the stepseq function from Lab 1 “Introduction to Discrete Signals” on pages 7-8. To calculate the output of the system we convolve the input with the shock response in twice the time.

Commenting on the results

For n ϵ [-20, 20] it is observed that the number of non-zero values of the system output is the sum of the non-zero values of the input signal x [n] and those of the impulse response h [n]. The results alone do not lead to any comment without the intervention of a theoretical solution.

Theoretical Solution

The output of the system y 1[n] with input signal the discrete time signal x [n] and with impulse response the signal h [n] = $|n-3|(u[n]-u[n-6])$ is calculated from the convolution of the two signals and specifically from the formula:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{k=+\infty} x(k)h(n-k)$$

Recalling the shock response h [n] and neglecting the zero values (Figure 5.1), we have :

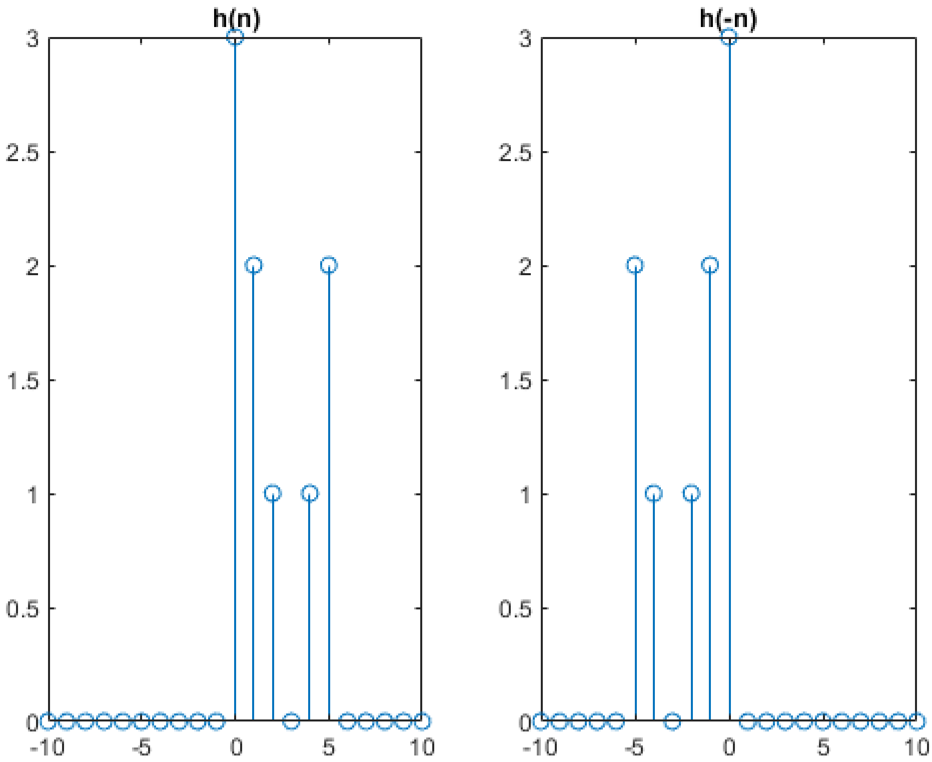

Figure 5.1 The shock response h [n] and its reflection h [- n]

For -20 ≤ n ≤ -2 and 8 ≤ n ≤ 20 we have no overlap at any point.
For -1 ≤ n ≤ 7 we have an overlap at one or more points. In more detail:
For n = -1 we have overlap at one point (Figure 5.2).

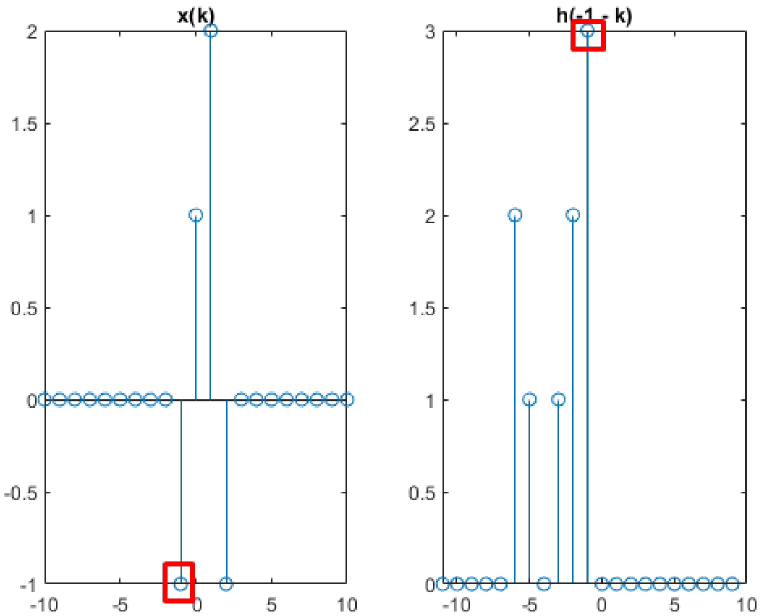

Figure 5.2 Overlap of a point for n = -1

Therefore, the output of the system from the convolution formula is y [-1] = x [-1] h [-1] → y [-1] = -1 × 3 → y [-1] = -3

For n = 0 we have an overlap at two points (Figure 5.3).

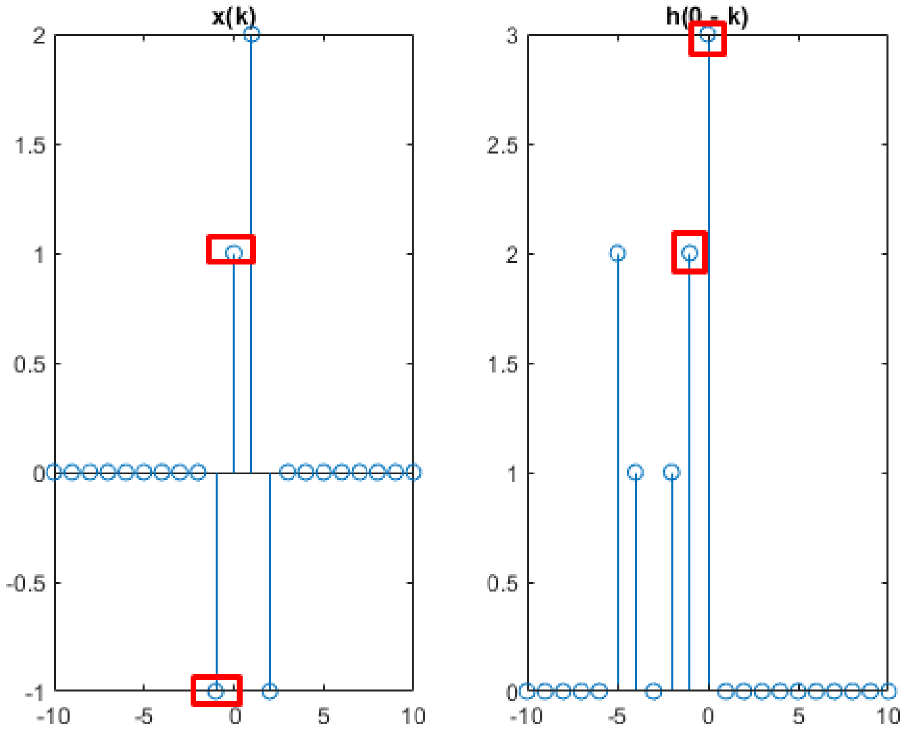

Figure 5.3 Overlap of two points for n = 0

Therefore, the output of the system from the convolution formula is y [0] = x [0] h [0] + x [-1] h [-1] → y [0] = 3 × 1 + 2 × (- 1) → y [0] = 3 – 2 → y [0] = 1

For n = 1 we have an overlap in three points (Figure 5.4).

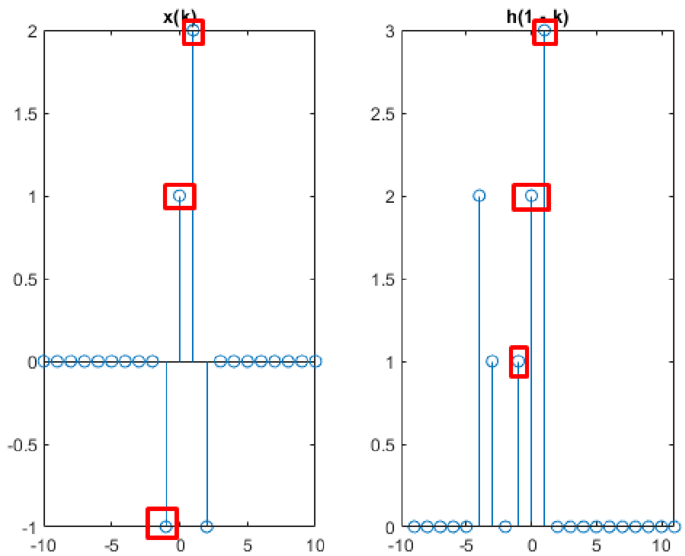

Figure 5.4 Overlap of three points for n = 1

Therefore, the output of the system from the convolution formula is y [1] = x [1] h [1] + x [0] h [0] + x [-1] h [-1] → y [1] = 2 × 3 + 3 × 1 + 2 × (-1) → y [1] = 6 + 1 → y [0] = 7 and so on.

The results of the command y1 = conv (h, x) come from the above calculations.

Graphic representations

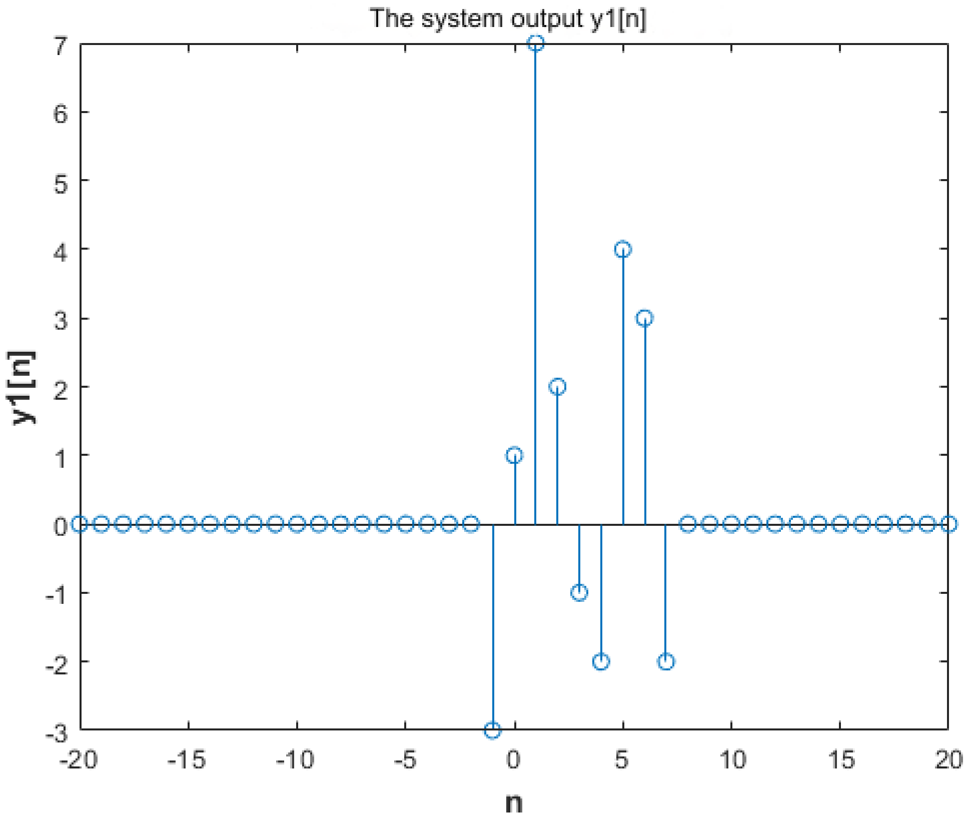

Figure 5.5 The output of the system y 1[n]

b)

Code in Matlab

Main program

```
n = -10:30
n _ conv = -30:50

x1 = stepseq (0, -10, 30)
x2 = stepseq (10, -10, 30)
x = x1 - x2
h1 = 0. 6.^ n
h2 = stepseq (0, -10, 30)
h = h 1 . *h2
```

```
y2 = conv(x,h)

stem ( n_conv , y2 )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y2[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'The system output y2[n]' )
```

Code of the stepseq function

```
function [ x , n ] = stepseq ( n 0, n 1, n 2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end
n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> n = -10:30	n = Columns 1 through 22 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 Columns 23 through 41 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30	Define the time interval of the input signal x [n] = u [n] – u [n – 10] and the impulse response h [n] = 0.6^n u [n] with step 1.
>> n_conv = - 3 0 : 5 0	n_conv = Columns 1 through 22 -30 -29 -28 -27 -26 -25 -24 -23 -22 -21 -20 -19 -18 -17 -16 -15 -14 -13 -12 -11 -10 -9 Columns 23 through 44 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 Columns 45 through 66 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 Columns 67 through 81 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50	Defining the time interval of the system output y 1[n] with input signal x [n] = u [n] – u [n – 10] and the impulse response h [n] = 0.6^n u [n] with step 1 .

>> x1 = stepseq (0, -10, 30)	x1 = Columns 1 through 33 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 34 through 41 1 1 1 1 1 1 1 1	Define the signal $x1 = u[n]$ on the interval $-10 \leq n \leq 30$ by calling the function stepseq whose code is discussed in Lab 1 "Introduction to Discrete Signals" on pages 7 – 8.
>> x2 = stepseq (10, -10, 30)	x2 = Columns 1 through 33 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 34 through 41 1 1 1 1 1 1 1 1	Define the signal $x2 = u[n-10]$ on the interval $-10 \leq n \leq 30$ by calling the stepseq function whose code is discussed in Lab 1 "Introduction to Discrete Signals" on pages 7 – 8.
>> x = x1 - x2	x = Columns 1 through 22 0 0 0 0 0 0 0 0 0 -1 1 2 -1 0 0 0 0 0 0 0 0 0	Define the input signal $x[n] = u[n] - u[n-10]$ in the interval $-10 \leq n \leq 30$.
>> h1 = 0. 6.^ n	h1 = Columns 1 through 13 165.3817 99.2290 59.5374 35.7225 21.4335 12.8601 7.7160 4.6296 2.7778 1.6667 1.0000 0.6000 0.3600 Columns 14 through 26 0.2160 0.1296 0.0778 0.0467 0.0280 0.0168 0.0101 0.0060 0.0036 0.0022 0.0013 0.0008 0.0005 Columns 27 through 39 0.0003 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 Columns 40 through 41 0.0000 0.0000	Define the signal $h1 = 0.6^{n}$ in the interval $-10 \leq n \leq 30$.
>> h2 = stepseq (0, -10, 30)	h2 = Columns 1 through 33	Define the signal $h2 = u[n]$ on the interval $-10 \leq n \leq 30$ by calling the function stepseq whose code is discussed in Lab 1 "Introduction to Discrete Signals" on pages 7 – 8.

	0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Columns 34 through 41 1 1 1 1 1 1 1 1	
>> h = h 1 . *h2	h = Columns 1 through 13 0 0 0 0 0 0 0 0 0 0 1.0000 0.6000 0.3600 Columns 14 through 26 0.2160 0.1296 0.0778 0.0467 0.0280 0.0168 0.0101 0.0060 0.0036 0.0022 0.0013 0.0008 0.0005 Columns 27 through 39 0.0003 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 Columns 40 through 41 0.0000 0.0000	Define the impulse response h [n] = 0.6^{n} u [n] in the interval $-10 \leq n \leq 30$.
>> y2 = conv(x, h)	y = Columns 1 through 13 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 14 through 26 0 0 0 0 0 0 0 1.0000 1.6000 1.9600 2.1760 2.3056 2.3834 Columns 27 through 39 2.4300 2.4580 2.4748 2.4849 1.4909 0.8946 0.5367 0.3220 0.1932 0.1159 0.0696 0.0417 0.0250 Columns 40 through 52 0.0150 0.0090 0.0054 0.0032 0.0019 0.0012 0.0007 0.0004 0.0003 0.0002 0.0001 0.0001 0.0000 Columns 53 through 65 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0 0 0 0 0	Calculation of the convolution of the input x [n] with the impulse response h [n] = 0.6^{n} u [n] to calculate the output y 2 [n] of the system. The convolution is calculated by calling the conv function.

	Columns 66 through 78 0 0 0 0 0 0 0 0 0 0 0 0 0 Columns 79 through 81 0 0 0	
>> stem (n_conv , y2)	Figure 5.9 (page 69)	Plot the output y 2[n] of the system in the interval -30 ≤ n ≤ 50 with the command stem .
>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 5.9 (page 69)	Add the label " n " to the horizontal x- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> ylabel ('y2[n]', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 5.9 (page 69)	Add the label " y 2[n]" to the vertical y- axis , with a label size (FontSize) of 12 and a weight (FontWeight) of bold .
>> title ('The system output y2[n]')	Figure 5.9 (page 69)	Add a title to the plot named "The output of the system y2[n]"

Commenting on the resolution

The solution is based on the stepwise design of the input signal x [n] = u [n] – u [n – 10] and the impulse response h [n] = 0.6 ^{n} u [n]. The step sequences u [n] and u [n – 10] are defined using the stepseq function from Lab 1 “Introduction to Discrete Signals” on pages 7-8. To calculate the output of the system we convolve the input with the shock response in twice the time.

Commenting on the results

For n ϵ [-30, 50] it is observed that the number of non-zero values of the system output is the sum of the non-zero values of the input signal x [n] and those of the impulse response h [n]. The results alone do not lead to any comment without the intervention of a theoretical solution.

Theoretical Solution

The output of the system y 2[n] with input signal the discrete time signal x [n] = u [n] – u [n – 10] and with impulse response the signal h [n] = 0.6 ^{n} u [n] is calculated from the convolution of the two signals and specifically from the formula:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{k=+\infty} x(k)h(n-k)$$

Recalling the shock response h [n] and neglecting the zero values (Figure 5.6), we have :

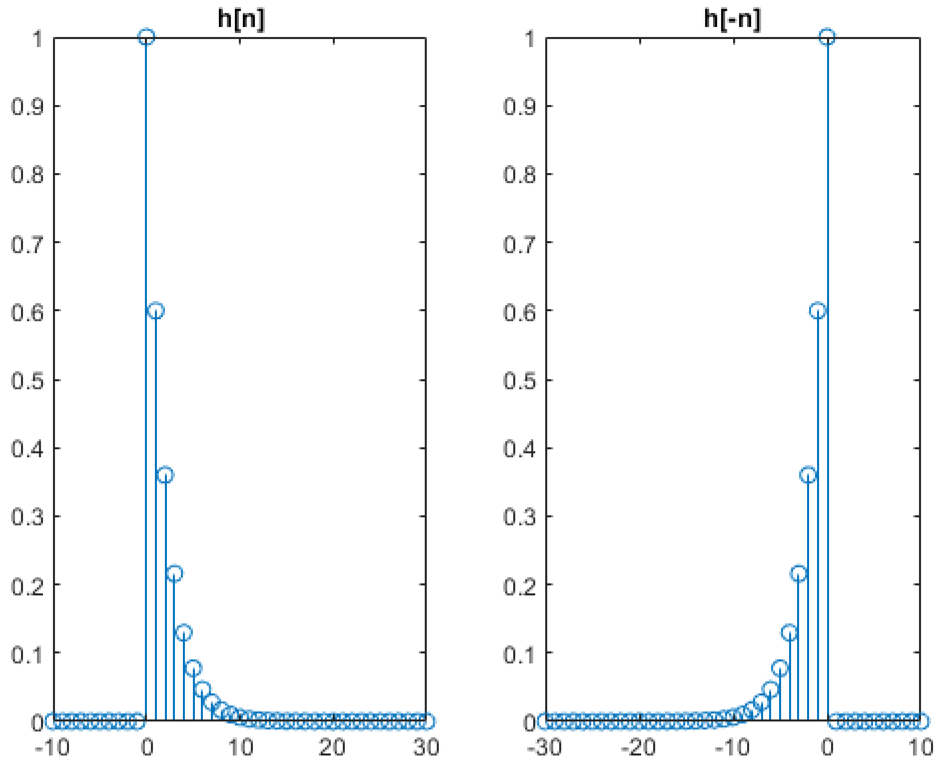

Figure 5.6 The shock response h [n] and its reflection h [- n]

For -30 ≤ n ≤ -11 and 61 ≤ n ≤ 80 we have no overlap at any point.
For -10 ≤ n ≤ 60 we have an overlap at one or more points. In more detail:
For n = 0 we have overlap at one point (Figure 5.7).

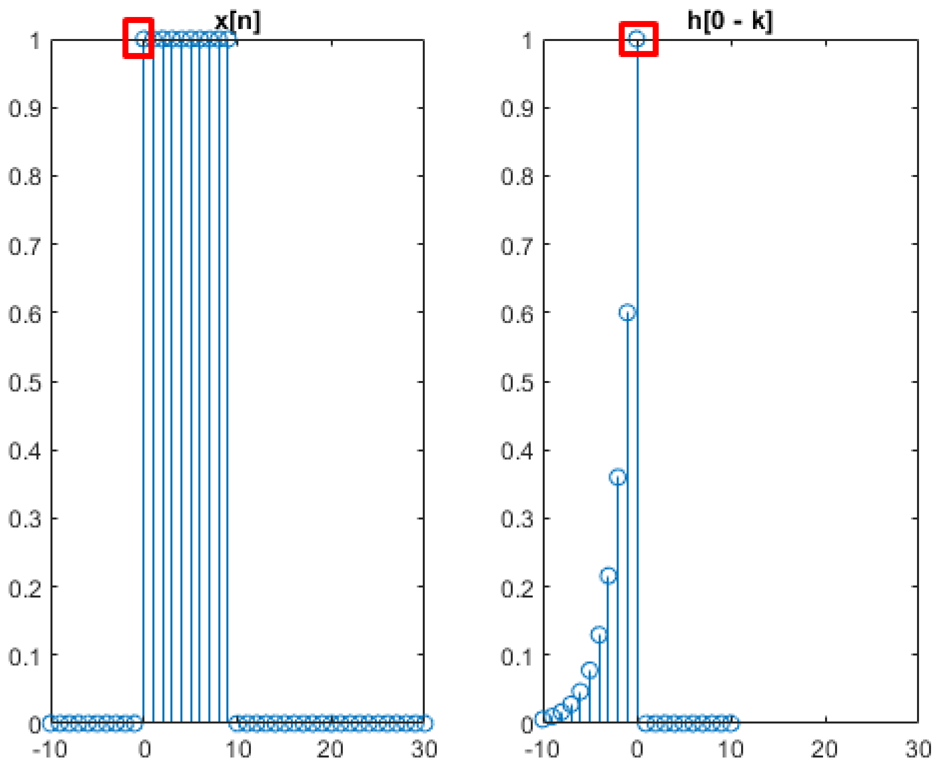

Figure 5.7 Overlap of a point for n = 0

Therefore, the system output from the convolution formula is y [-10] = x [0] h [0] → y [-10] = 1 × 1 → y [-10] = 1
For n = 1 we have an overlap at two points (Figure 5.8).

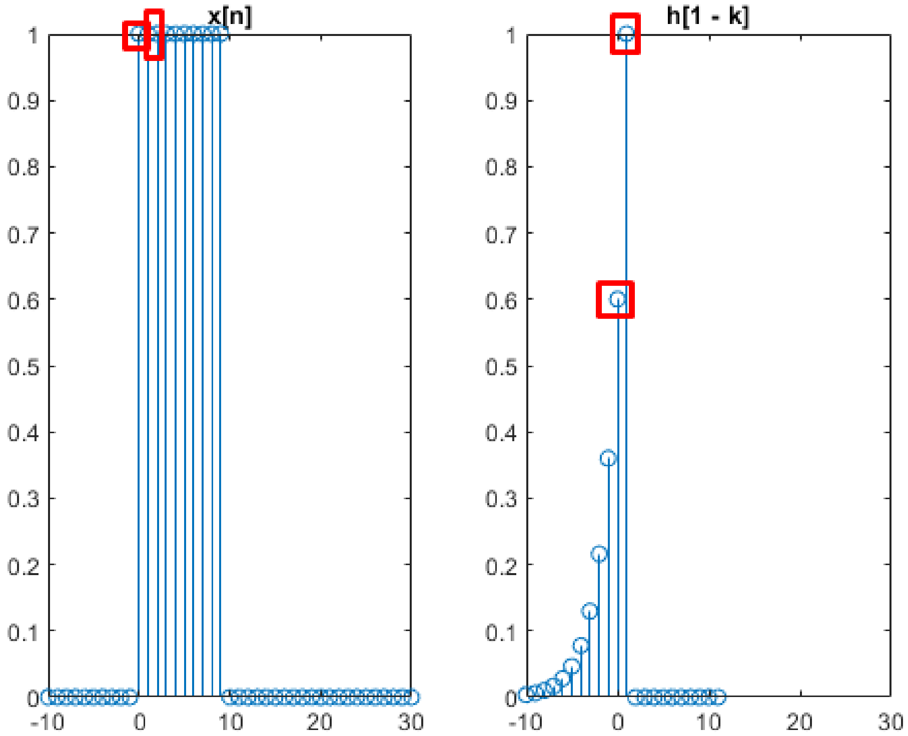

Figure 5.8 Overlap of two points for n = 1

Therefore, the output of the system from the convolution formula is y [-9] = x [0] h [0] + x [1] h [1] → y [-9] = 1 × 1 + 1 × 0.6 → y [-9] = 1.6 and so on

The results of the command y2 = conv (h, x) come from the above calculations.

Graphic representations

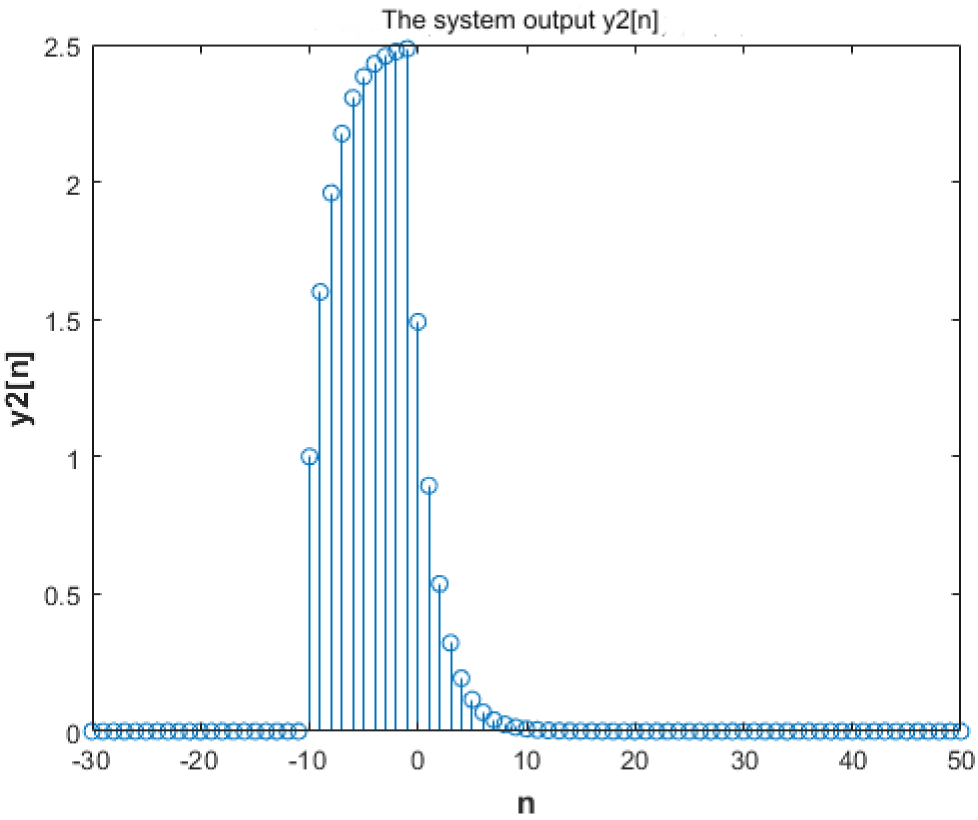

Figure 5.9 The output of the system y2[n]

EXERCISE 6

Plot the convolution of the sequences x [n] * y [n] where

$$x[n] = -\left(\frac{n}{4}\right)(u(n) - u(n-4))$$

$$y[n] = \left(1 - \frac{n}{5}\right)(u(n) - u(n-5))$$

The convolution of the two sequences should be written in the form of a Toeplitz matrix product of a vector. Write a function that implements the convolution of two sequences by calculating the convolution as the product of a Toeplitz vector matrix by a vector.

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Code in Matlab

Main program

```
n = 0:5
```

```
x = -(n / 4) .* ( stepseq (0, 0, 5) - stepseq (4, 0, 5))
y = (1 - n / 5 ) . * ( stepseq (0, 0, 5) - stepseq (5, 0, 5))
[toe, nt ] = my_toeplitz (x, n, y)

stem ( nt , toe)
xlabel ( ' nt ' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'x[n] * y[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'Convolution Using Toeplitz Matrix' )
```

Code of the function my _ toeplitz

```
function [ t , n _ conv ] = my _ toeplitz ( x , n , y )


v = [ y( 2) y(3) y(4) y(5) y(6)];
r = [ x( 2) 0 0 0 0];
c = [ x( 2) x(3) x(4) 0 0 0 0];
T = toeplitz (c, r)
t = T * v'
n _ conv = 1:7;
```

Code her function stepseq

```
function [x, n] = stepseq (n0, n1, n2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end
n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Explanation of the main program commands used

Commands	Results	Comments
>> n = 0:5;	n = 0 1 2 3 4 5	Define the time interval of the sequences x [n] and y [n] with step 1.
>> x = -(n / 4) .* (stepseq (0, 0, 5) - stepseq (4, 0, 5));	x = 0 -0.2500 -0.5000 -0.7500 0 0	Define the signal x [n] = - (n / 4) (u [n] – u [n – 4]) for $0 \leq n \leq 5$.
>> y = (1 - n / 5) .* (stepseq (0, 0, 5) - stepseq (5, 0, 5));	y = 1.0000 0.8000 0.6000 0.4000 0.2000 0	Definition of the signal y [n] = (1 - n / 5) (u [n] – u [n – 5]) for $0 \leq n \leq 5$.
>> [toe, nt] = my_toeplitz (x, n, y);	Toeplitz =	Call the function m y _ toeplitz with arguments the time interval $0 \leq n \leq 5$ and the signals x [n] and y [n]. The function calculates the convolution of the two sequences by calculating

	-0.2500 0 0 0 0 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 0 0 0 0 -0.7500 Convolution = -0.2000 -0.5500 -1.0000 -0.7000 -0.4000 -0.1500 0 toe = -0.2000 -0.5500 -1.0000 -0.7000 -0.4000 -0.1500 0 nt = 1 2 3 4 5 6 7	the convolution as a Toeplitz vector matrix product . In detail, the explanation of the function code in the chapter "Explanation of the used commands of the function my _ toeplitz ".
>> stem (nt , toe) >> xlabel (' nt ', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('x[n] * y[n]', ' FontSize ', 12, ' FontWeight ', 'bold') title ('Convolution Using Toeplitz Matrix')	Figure 6.1 (page)	Plot the convolution of the two sequences x [n] and y [n] by computing the convolution as a Toeplitz vector matrix product by vector on the interval 1 ≤ n ≤ 7.

Explanation of the used commands of the function my _ toeplitz

Commands	Results	Comments
>> function [t, n_conv] = my_toeplitz (x, n, y)		The prototype of the function my _ toeplitz which takes as arguments the two sequences x [n] = -(n / 4) (u [n] – u [n – 4]) and y [n] = (1 - n / 5) (

		u [n] – u [n – 5]) and the interval 0 ≤ n ≤ 5, where they are defined. The function calculates the convolution of the two sequences by calculating the convolution as a Toeplitz vector matrix product .
>> v = [y(2) y(3) y(4) y(5) y(6)];	v = 0.8000 0.6000 0.4000 0.2000 0	The vector to be multiplied by the vector Toeplitz matrix .
>> r = [x(2) 0 0 0 0];	r = -0.2500 0 0 0 0	The first row of the vector Toeplitz matrix .
>> c = [x(2) x(3) x(4) 0 0 0 0];	c = -0.2500 -0.5000 -0.7500 0 0 0 0	The first column of the vector Toeplitz matrix .
>> T = toeplitz (c, r)	Toeplitz = -0.2500 0 0 0 0 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 -0.2500 0 0 0 -0.7500 -0.5000 0 0 0 0 -0.7500	Defining the vector Toeplitz matrix by calling the function t oeplitz .
>> t = T * v'	Convolution = -0.2000 -0.5500 -1.0000 -0.7000 -0.4000 -0.1500 0	Computing the convolution of the two sequences x [n] and y [n] as a Toeplitz vector matrix product by vector. Convolution is returned to the main program.
>> n_conv = 1:7;	n_conv = 1 2 3 4 5 6 7	Set the time interval over which the convolution is defined and return to the main program to plot the convolution over that time.

Commenting on the resolution

The solution is based on the stepwise design of the sequences x [n] = -(n / 4) (u [n] – u [n – 4]) and y [n] = (1 - n / 5) (u [n] – u [n – 5]). The elements of the sequence x [n] participate in the creation of the Toeplitz vector matrix , while the elements of the sequence y [n] in unit vector form are multiplied by the elements of the matrix. The convolution is written as a Toeplitz vector-by-vector matrix product form .

Commenting on the results

The result of the product returned by the function my _ toelitz gives us equivalently the convolution x [n] * y [n].

Graphic Representations

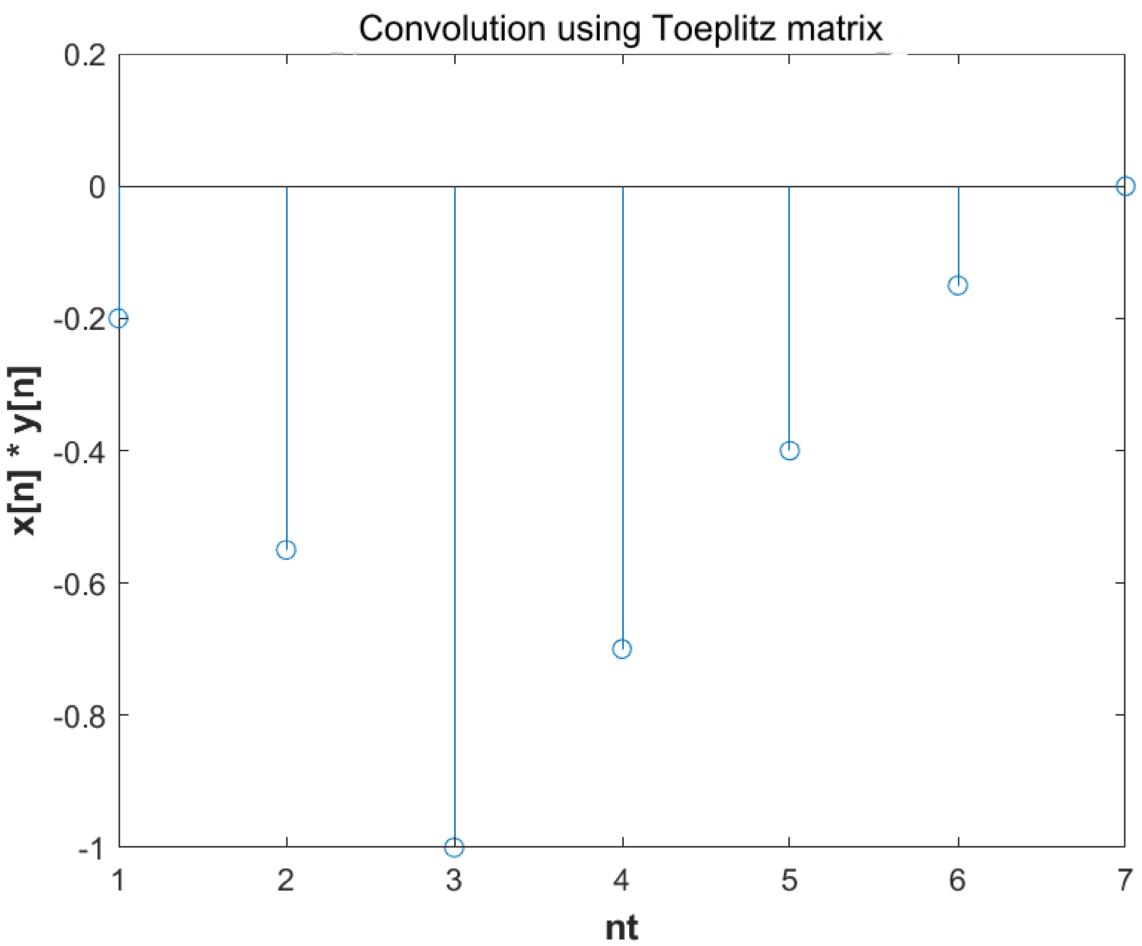

Figure 6.1 The convolution of the sequences x [n] and y [n] using the Toeplitz matrix

EXERCISE 7

Investigate graphically the linearity of the system described by the relationship:

$$y[n] = 3x[n] + 4.$$

Consider the following signals as inputs:

$$x1[n] = [1\ 2\ 3\ 4\ 5], \quad 0 \le n \le 4, \qquad x2[n] = u[n] - u[n-6].$$

Let α = 2 and β = 3 be considered as constants and the investigation should be done in the interval n = [-5 :10]. Draw the inputs x1, x2 and the two outputs in subwindows of the same graphics window.

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Code in Matlab

Main program

```
n = -5:10

x 1 = [0 0 0 0 0 1 2 3 4 5 0 0 0 0 0 0]
x2 = stepseq (0, -5, 10) - stepseq (6, -5, 10)
x = 2 . * x1 + 3 .* x2
y1 = 3 . * x1 + 4
y2 = 3 . * x2 + 4
y = 2 . * y1 + 3 .* y2

subplot (3, 2, 1)
stem (n, x1)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'x1[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'x1[n] = [1 2 3 4 5]' )
subplot (3, 2, 2)
stem (n, y1)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y1[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y1[n] = 3 * x1[n] + 4' )
subplot (3, 2, 3)
stem (n, x2)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'x2[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'x2[n] = u[n] - u[ n - 6]' )
subplot (3, 2, 4)
stem (n, y2)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y2[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y2[n] = 3 * x2[n] + 4' )
subplot (3, 2, 5)
stem (n, x)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'x[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'x[n] = 2 * x1[n] + 3 * x2[n]' )
subplot (3, 2, 6)
stem (n, y)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'y[n] = 2 * y1[n] + 3 * y2[n]' )
```

Code of the stepseq function

```
function [x, n] = stepseq (n0, n1, n2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
```

```
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end
n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> n = - 5 : 10	n = -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10	Definition of the time interval of the input signals x 1[n] = [1 2 3 4 5], x 2[n] = u [n] – u [n – 6] and the outputs of the systems y 1[n] = 3 * x 1[n] + 4 and y 2[n] = 3 * x 2[n] + 4.
>> x1 = [0 0 0 0 0 1 2 3 4 5 0 0 0 0 0 0]	x1 = 0 0 0 0 0 1 2 3 4 5 0 0 0 0 0 0	Define the input signal x 1[n] = [1 2 3 4 5] in the interval -5 ≤ n ≤ 10.
>> x2 = stepseq (0, -5, 10) - stepseq (6, -5, 10)	x2 = 0 0 0 0 0 1 1 1 1 1 1 0 0 0 0 0	Define the input signal x 2[n] = u [n] – u [n – 6] in the interval -5 ≤ n ≤ 10.
>> x = 2 . * x1 + 3 .* x2	x = 0 0 0 0 0 5 7 9 11 13 3 0 0 0 0 0	Definition of the linear combination of the input signals x 1[n] and x 2[n], x [n] = α * x 1[n] + β * x 2[n], where α = 2 and β = 3.
>> y1 = 3 . * x1 + 4	y1 = 4 4 4 4 4 7 10 13 16 19 4 4 4 4 4 4	The system output y 1[n] = 3 * x 1[n] + 4 in the interval -5 ≤ n ≤ 10.
>> y2 = 3 . * x2 + 4	y2 = 4 4 4 4 4 7 7 7 7 7 7 4 4 4 4 4	The system output y 2[n] = 3 * x 2[n] + 4 in the interval -5 ≤ n ≤ 10.
>> y = 2 . * y1 + 3 .* y2	y = 20 20 20 20 20 35 41 47 53 59 29 20 20 20 20 20	Definition of the linear combination of the outputs of the input signals y 1[n] and y2[n], y[n] = α * y1[n] + β * y2[n], where α = 2 and β = 3.
>> subplot (3, 2, 1) >> stem (n, x1) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('x1[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('x1[n] = [1 2 3 4 5]')	Figure 7.1 (page 72)	Plot the input signal x 1[n] = [1 2 3 4 5] on the interval -5 ≤ n ≤ 10.
>> subplot(3, 2, 2) >> stem (n, y1) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('y1[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('y1[n] = 3 * x1[n] + 4')	Figure 7.1 (page 72)	Plot the output of the system y 1[n] = 3 * x 1[n] + 4 on the interval -5 ≤ n ≤ 10.
>> subplot(3, 2, 3) >> stem (n, x2)	Figure 7.1 (page 72)	Plot the input signal x 2[n] = u [n] – u [n – 6] on the interval -5 ≤ n ≤ 10.

>> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('x2[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('x2[n] = u[n] - u[n - 6]')		
>> subplot (3, 2, 4) >> stem (n, y2) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('y2[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('y2[n] = 3 * x2[n] + 4')	Figure 7.1 (page 72)	Plot the output of the system y 2[n] = 3 * x 2[n] + 4 on the interval -5 ≤ n ≤ 10.
>> subplot (3, 2, 5) >> stem (n, x) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('x[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('x[n] = 2 * x1[n] + 3 * x2[n]')	Figure 7.1 (page 72)	Plot the linear combination of the input signals x 1[n] and x 2[n], x [n] = α * x 1[n] + β * x 2[n], where α = 2 and β = 3.
>> subplot (3, 2, 6) >> stem (n, y) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('y[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('y[n] = 2 * y1[n] + 3 * y2[n]')	Figure 7.1 (page 72)	Plot the linear combination of the outputs of the input signals y 1[n] and y2[n], y[n] = α * y1[n] + β * y2[n], where α = 2 and β = 3.

Commenting on the resolution

The solution is based on plotting the sequences x [n] = α * x 1[n] + β * x 2[n] to prove whether the linear combination of the input signals gives the output system y [n] = α * y 1 [n] + β * y 2[n], where α = 2 and β = 3.

Commenting on the results

For n ϵ [-5, 10] it is observed that the input x [n] = α * x 1[n] + β * x 2[n] does not give the output y [n] = α * y 1[n] + β * y 2[n], where α = 2 and β = 3. For example,

For n = 1:

x [1] = 2 * x 1[1] + 3 * x 2[1] → x [1] = 2 * 2 + 3 * 1 → x [1] = 7

y [1] = 2 * y 1[1] + 3 * y 2[1] → y [1] = 2 * 10 + 3 * 7 → y [1] = 41

Theoretical Solution

For input x 1[n] → y 1[n] and x 2[n] → y 2[n]

The system y [n] = 3 x [n] + 4 will be a linear function α x 1[n] + β x 2[n] → α y 1[n] + β y 2[n].

For input α x 1[n] + β x 2[n] → 3(α x 1[n] + β x 2[n]) + 4 → 3α x 1[n] + 3β x 2[n] + 4 → α(3 x 1[n] + 4) + β(3 x 2[n]) ≠ α y 1[n] + β y 2[n].

Therefore, the system is not linear.

Graphic representations

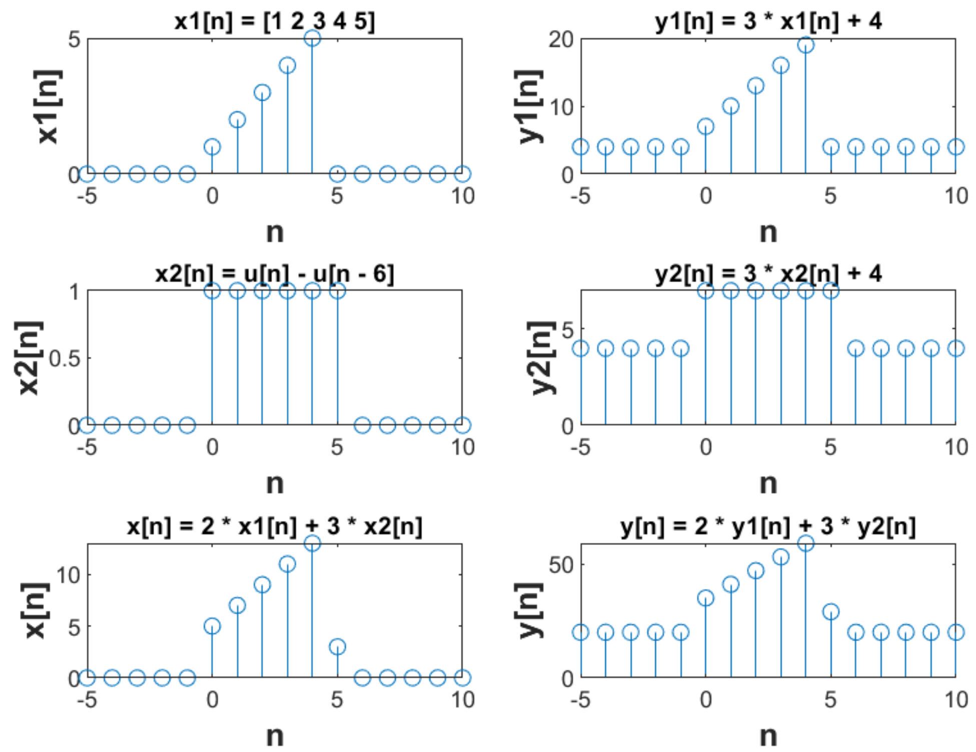

Figure 7.1 The input signals x 1[n], x 2[n], the outputs of the input signals y 1[n], y 2[n], the input signal 2 x 1[n] + 3 x 2[n] and the signal output y [n]

EXERCISE 8

The difference equation of a system is given:

$$y[n] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$$

a) Calculate and graph the impact response.

b) From this graph explain whether the system is BIBO stable or not.

c) Compute the output for input the sequence x[n]=[2 1 1 1 0 0 1 1 0 0 2] using the filter function and the conv function and explain what you see in relation to what we commented on in the handout .

d) Repeat question (c) for input x[n] = 2cos(3π n).

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a)

Code in Matlab

```
n = 0:10

b = [0.5 0 1]
a = [1 3 5]
d = zeros (size (n))
d( 1) = 1
h = filter (b, a, d)

stem (n, h)
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'h[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'Shock Response of Difference Equation System' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = 0 : 10	n = 0 1 2 3 4 5 6 7 8 9 10	Definition of the time interval of the impulse response of the system given by the difference equation $y[n] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$
>> b = [0.5 0 -1]	b = 0.5000 0 - 1.0000	Define the sample coefficients of the input x [n], x [n – 1] and x [n – 2].
>> a = [1 3 - 5]	a = 1 3 - 5	Define the sample coefficients of the output y [n], y[n – 1] and y[n – 2].
>> d = zeros (size (n)) >> d(1) = 1	d = 1 0 0 0 0 0 0 0 0 0 0	Define the input vectors in the interval $0 \le n \le 10$.
>> h = filter (b, a, d)	h = 1.0e+05 * 0.0000 -0.0000 0.0001 -0.0003 0.0011 -0.0045 0.0187 -0.0786 0.3293 -1.3808 5.7891	Calculation of the shock response of the system using the filter function .
>> stem (n, h) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 8.1 (page)	Plot the impulse response of the system $y[n] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$ in the interval $0 \le n \le 10$.

>> ylabel ('h[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('Shock Response of Difference Equation System')		

Commenting on the resolution

The solution is based on plotting the input vectors (d), input coefficients (b) and output coefficients (a), where are the necessary parameters to calculate the shock response of the system $y[n] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$.

Commenting on the results

For n ϵ [0, 10] it is observed that the shock response of the system $y[n] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$ is finite duration.

Graphic representations

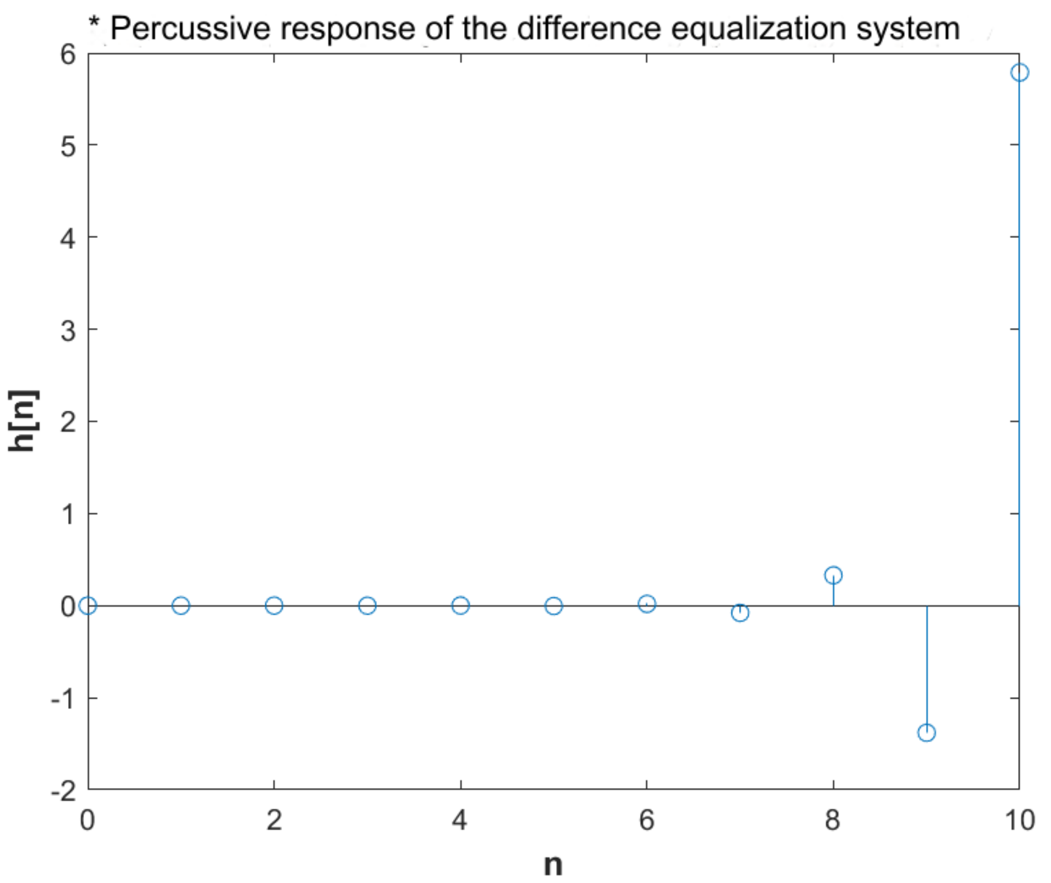

Figure 8.1 The shock response of the system $y[\;] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$

b)

The system is BIBO stable, as for every instant the outputs of the system remain blocked and do not approach infinity.

c)

Code in Matlab

```
n = 0:10

b = [0.5 0 -1]
a = [1 3 -5]
d = n == 0
h = filter (b, a, d)
x = [2 1 1 1 0 0 1 1 0 0 2]
y_conv = conv (h, x)
y_filter = filter (b, a, x)

stem ( y_conv )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'Output of signal x = [2 1 1 1 0 0 1 1 0 0 2]' )
hold on
stem ( y_filter , 'r*' )
legend ( 'conv' , 'filter' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = 0 : 10	n = 0 1 2 3 4 5 6 7 8 9 10	Defining the time interval of the input signal x [n] = [2 1 1 1 0 0 1 1 0 0 2], where its output is given by the difference equation $y[n] + 3\,y[n-1] - 5\,y[n-2] = 0.5\,x[n] - x[n-2]$
>> b = [0.5 0 -1]	b = 0.5000 0 - 1.0000	Define the sample coefficients of the input x [n], x [n – 1] and x [n – 2].
>> a = [1 3 -5]	a = 1 3 - 5	Define the sample coefficients of the output y [n], y[n – 1] and y[n – 2].
>> d = n == 0	d = 1 0 0 0 0 0 0 0 0 0 0	Define the input vectors in the interval 0 ≤ n ≤ 10.
>> h = filter (b, a, d)	h = 1.0e+05 * 0.0000 -0.0000 0.0001 -0.0003 0.0011 -0.0045 0.0187 -0.0786 0.3293 -1.3808 5.7891	Calculation of the shock response of the system using the filter function .

>> x = [2 1 1 1 0 0 1 1 0 0 2]	x = 2 1 1 1 0 0 1 1 0 0 2	Define the input signal x = [2 1 1 1 0 0 1 1 0 0 2] in the interval 0 ≤ n ≤ 10.
>> y_conv = conv(h, x)	y_conv = 1.0e+06 * Columns 1 through 13 0.0000 -0.0000 0.0000 -0.0000 0.0002 -0.0008 0.0034 -0.0142 0.0594 -0.2492 1.0449 0.4734 0.4423 Columns 14 through 21 0.5729 0.0253 -0.1060 0.4446 0.5632 0.0659 -0.2762 1.1578	Compute the output of the input signal x [n] with the impulse response h [n] of the system, by calling conv .
>> y_filter = filter(b, a, x)	y_filter = 1.0e+06 * 0.0000 -0.0000 0.0000 -0.0000 0.0002 -0.0008 0.0034 -0.0142 0.0594 -0.2492 1.0449	Calculation of the output of the input signal x [n] with the shock response h [n] of the system, by calling filter .
>> stem (y_conv) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('y[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('Output of signal x = [2 1 1 1 0 0 1 1 0 0 2]') >> hold on >> stem (y_filter , 'r*') >> legend('conv', 'filter')	Figure 8.2 (page)	Plotting the two outputs of the input signal x [n] = [2 1 1 1 0 0 1 1 0 0 2] using the conv and filter functions and displaying in the same graphical environment as the hold command on .

Commenting on the resolution

The solution is based on plotting the input vectors (d), input coefficients (b) and output coefficients (a), where are the necessary parameters to calculate the shock response of the system $y[n] + 3y[n-1] - 5y[n-2] = 0.5x[n] - x[n-2]$. With conv the output of the input signal x [n] = [2 1 1 1 0 0 1 1 0 0 2] is calculated by the convolution of the signals x [n] and the impulse response of the system h [n], while with filter calculates the output of the signal x [n] with the filter response.

Commenting on the results

For n ϵ [0, 10] it is observed in the results of conv and filter , as well as from the graphical representation (Figure 8.2), that there is an identity in the first 11 terms . Given that the system is BIBO stable by subquestion (b), that is, there are no infinite elements, both outputs are correct.

Graphic representations

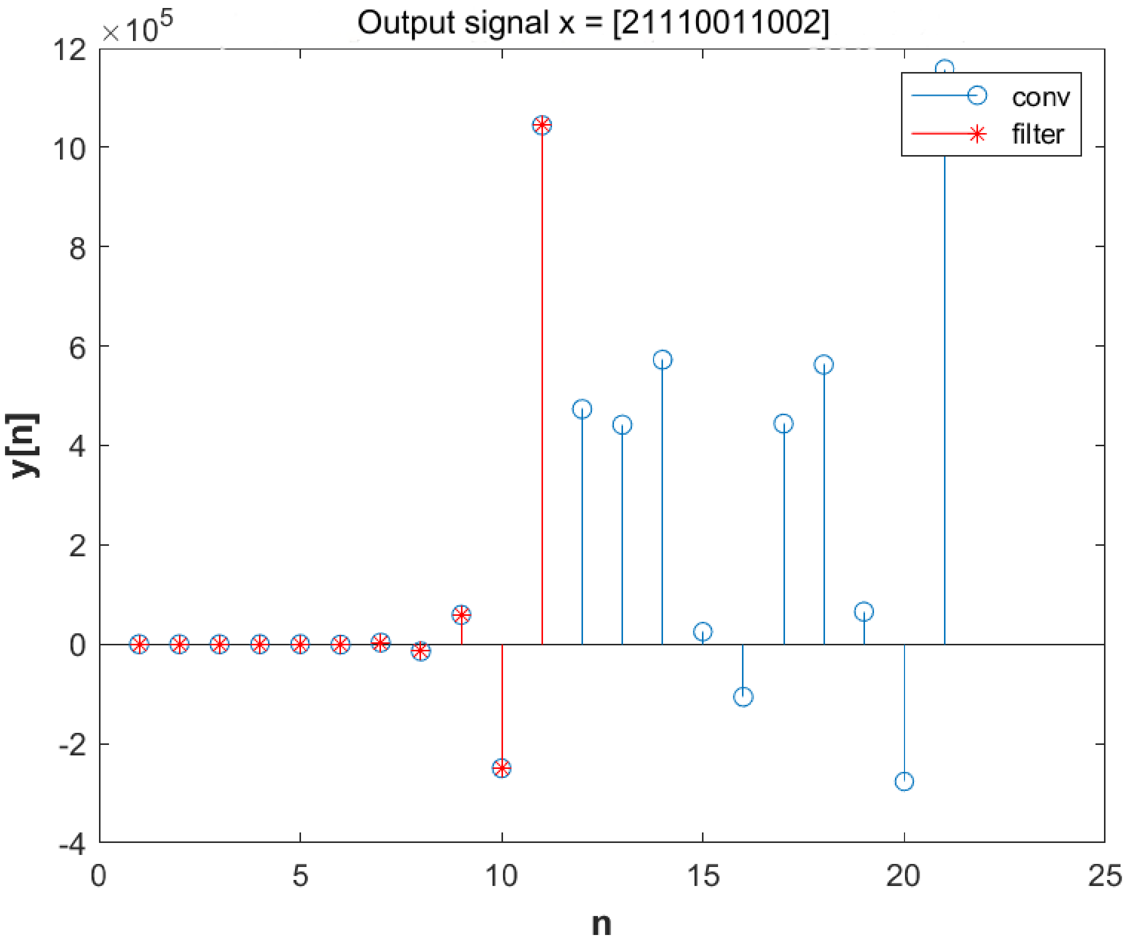

Figure 8.2 The outputs of the input signal x [n] = [2 1 1 1 0 0 1 1 0 0 2] using the filter and conv functions

d)

Code in Matlab

```
n = 0:10

b = [0.5 0 -1]
a = [1 3 -5]
d = n == 0
h = filter (b, a, d)
x = 2 * cos (3 * pi * n)
y_conv = conv (h, x)
y_filter = filter (b, a, x)

stem ( y_conv )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'Output of signal x = 2cos(3πn)' )
hold on
stem ( y_filter , 'r*' )
legend ( 'conv' , 'filter' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = 0 : 10 ;	n = 0 1 2 3 4 5 6 7 8 9 10	Defining the time interval of the input signal x [n] = 2 cos (3π n), where its output is given by the difference equation $y[n] + 3\,y[n-1] - 5\,y[n-2] = 0.5\,x[n] - x[n-2]$
>> b = [0.5 0 -1];	b = 0.5000 0 - 1.0000	Define the sample coefficients of the input x [n], x [n – 1] and x [n – 2].
>> a = [1 3 -5];	a = 1 3 - 5	Define the sample coefficients of the output y [n], y[n – 1] and y[n – 2].
>> d = n == 0;	d = 1 0 0 0 0 0 0 0 0 0 0	Define the input vectors in the interval 0 ≤ n ≤ 10.
>> h = filter(b, a, d);	h = 1.0e+05 * 0.0000 -0.0000 0.0001 -0.0003 0.0011 -0.0045 0.0187 -0.0786 0.3293 -1.3808 5.7891	Calculation of the shock response of the system using the filter function .
>> x = 2 * cos (3 * pi * n);	x = 2 -2 2 -2 2 -2 2 -2 2 -2 2	Define the input signal x [n] = 2 cos (3π n) in the interval 0 ≤ n ≤ 10.
>> y_conv = conv(h, x)	y_conv = Columns 1 through 11 1 -4 16 -67 280 -1174 4921 -20632 86500 -362659 1520476 Columns 12 through 21 -1520475 1520472 -1520460 1520409 -1520196 1519302 -1515555 1499844 -1433976 1157817	Compute the output of the input signal x [n] with the impulse response h [n] of the system, by calling conv .
>> y_filter = filter(b, a, x)	y_filter = 1 -4 16 -67 280 -1174 4921 -20632 86500 -362659 1520476	Calculation of the output of the input signal x [n] with the shock response h [n] of the system, by calling filter .
>> stem (y_conv) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('y[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('Output of signal x = [2 1 1 1 0 0 1 1 0 0 2]')	Figure 8. 3 (page)	Plot the two outputs of the input signal x [n] = 2 cos (3π n) using the conv and filter functions and display in the same graphical environment as the hold command on .

>> hold on >> stem (y_filter , 'r*') >> legend('conv', 'filter')		

Commenting on the resolution

The solution is based on plotting the input vectors (d), input coefficients (b) and output coefficients (a), where are the necessary parameters to calculate the shock response of the system $y\,[\,n\,] + 3\,y\,[\,n-1] - 5\,y\,[\,n-2] = 0.5\,x\,[\,n\,] - x\,[\,n-2]$. With conv the output of the input signal x [n] = 2 cos (3π n) is calculated by the convolution of the signals x [n] and the impulse response of the system h [n], while with the filter the output of the signal is calculated x [n] with the filter response.

Commenting on the results

For n ϵ [0, 10] it is observed in the results of conv and filter , as well as from the graphical representation (Figure 8.3), that there is an identity only in the first 11 terms . Given that the system is BIBO stable by subquestion (b), that is, there are no infinite elements, both outputs are correct.

Graphic representations

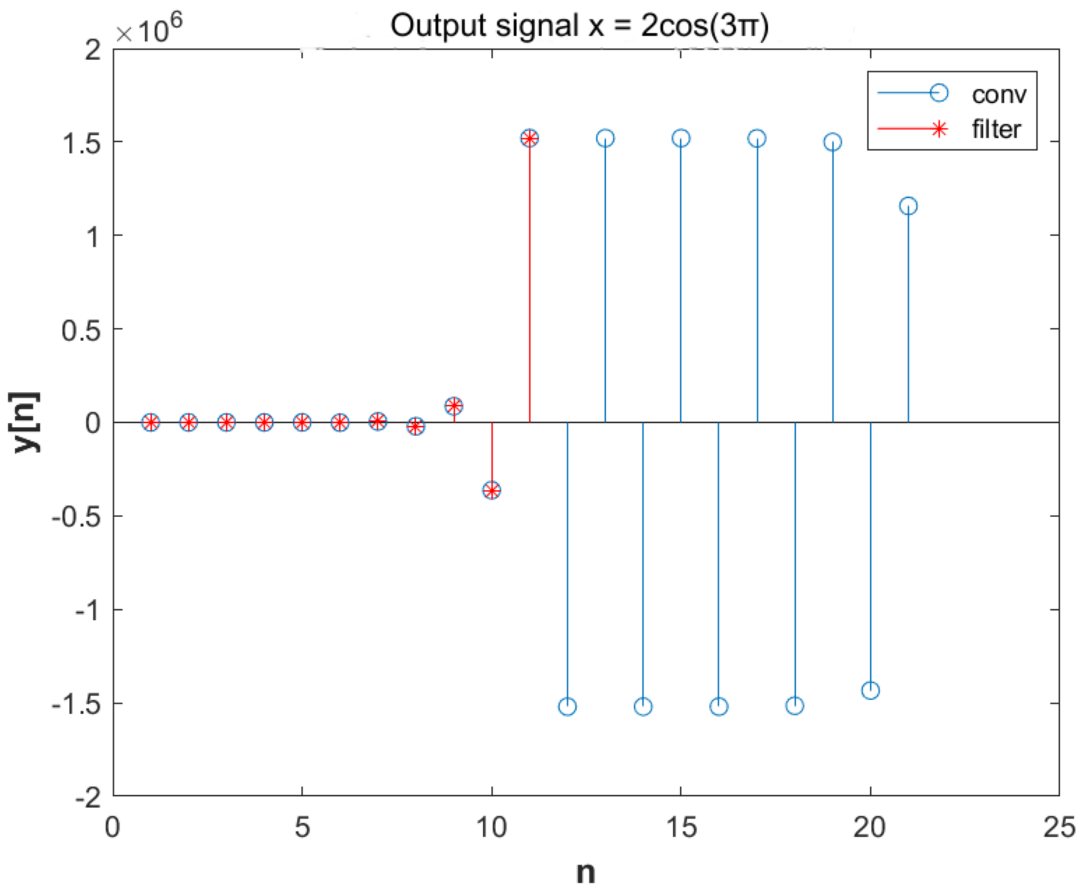

Figure 8.3 The outputs of the input signal x [n] = 2 cos (3π n) using the filter and conv functions

EXERCISE 9

A system described by ED is given

$$y[n] = 0.4\,y[n-1] + x[n] - 0.7\,x[n-2]$$

a) Find and plot the output for time n=[-20 : 20] when the input is

$$x[n] = 2\,\delta[n+1] + 4\,\delta[n] + 8\,\delta[n-1] + 9\,\delta[n-2]$$

b) Find and plot for n=[-20 : 20] the shock response of the system using the impz function (help impz for syntax).

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EXERCISE 10

The signal is given

$$s[n] = 2 * \cos(\pi n / 8), \text{ n=0:60.}$$

Gaussian noise with a standard deviation of 0.5 is added to this signal and the noisy signal is input to the system described by the difference equation:

$$y[n] = \frac{1}{M}\sum_{m=0}^{M-1} x[n-m]$$

This system is an average value filter of order M. Write a program that calculates the output of the system for M=4. Make the graphs of the quiet signal, the noisy signal and the output of the system. What do you think is the system's response to the noisy signal?

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EXERCISE 11

Plot the step responses of the systems described by the following input-output relationships:

a) $y(n) = x(n + 2) + x(n - 2)$

b) $y(n) = x^2(n - 2)$

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EXERCISE 12

Find sequences that have the following Z-transforms, then compute their first 6 terms

$$X1(z) = \frac{z}{z-2}$$

$$X2(z) = \frac{3z^2}{(z^2 - 1.5z + 0.5)(z - 0.5)}$$

$$X3(z) = \frac{2z^2 + 7z}{z^2 + z - 2}$$

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a) X 1[z] = z / z - 2

Code in Matlab

```
syms  n z

X1 = z / (z - 2)
seq1 = iztrans (X1, n)
n = 0:5
seq1 = iztrans (X1, n)
```

Explanation of the commands used

Commands	Results	Comments
>> syms n z		Declaration of symbolic variables ' n ' and ' z '.
>> X1 = z / (z - 2)	X1 = z/(z - 2)	The Z-transform of the requested sequence
>> seq1 = iztrans (X1, n)	seq1 = 2^n	The sequence with a call to the " iztrans " function that calculates the inverse Z-transform.
>> n = 0:5 >> seq1 = iztrans (X1, n)	n = 0 1 2 3 4 5 seq1 = [1, 2, 4, 8, 16, 32]	Calculation of the first 6 terms of the sequence by calling " iztrans " and defining a time interval from 0 ≤ n ≤ 5.

Theoretical Solution

$$\mathrm{seq1[n] = IZT[X1(z)]} = 2^{n} \times \mathrm{u[n]}$$

b) X2[z] = 3 z^2/ (z^2- 1.5z + 0.5) (z – 0.5)

Code at Matlab

```
syms nz

X2 = (3 * z^2) / ((z^2 - 1.5 * z + 0.5) * (z - 0.5));
disp ( "X2 = " )
pretty (X2)
seq2 = iztrans (X2, n);
disp ( "seq2 = " )
pretty (seq2)
n = 0:5
seq2 = iztrans(X2, n)
```

Explanation of the commands used

Commands	Results	Comments
>> syms n z		Declaration of symbolic variables ' n ' and ' z '.
>> X2 = (3 * z^2) / ((z^2 - 1.5 * z + 0.5) * (z - 0.5)); >> disp (“X2 = “) >> pretty (X2)	X2 = 2 3 z -------------------------- / 1 \ / 2 3 z 1 \ | z - - | | z - --- + - | \ 2 / \ 2 2 /	The Z-transform of the requested sequence
>> seq2 = iztrans (X2, n); >> disp ("seq2 = ") >> pretty (seq2)	seq2 = / 1 \n / 1 \n 12 - 6 | - | (n - 1) - 18 | - | \ 2 / \ 2 /	The sequence with a call to the " iztrans " function that calculates the inverse Z-transform.
>> n = 0:5 >> seq2 = iztrans (X2, n)	n = 0 1 2 3 4 5 seq2 = [0, 3, 6, 33/4, 39/4, 171/16]	Calculation of the first 6 terms of the sequence by calling " iztrans " and defining a time interval from 0 ≤ n ≤ 5.

c) X3[z] = 2 z^2 + 7z / z^2 + z - 2

Code in Matlab

```
syms nz

X3 = (2 * z^2 + 7 * z) / (z^2 + z - 2);
disp ( "X3 = " )
pretty (X3)
seq3 = iztrans (X3,n)
n = 0:5
seq3 = iztrans (X3,n)
```

Explanation of the commands used

Commands	Results	Comments
>> syms n z		Declaration of symbolic variables ' n ' and ' z '.
>> X3 = (2 * z^2 + 7 * z) / (z^2 + z - 2) >> disp("X3 = ") >> pretty (X3)	X3 = 2 2 z + 7 z ---------- 2 z + z - 2	The Z-transform of the requested sequence
>> seq3 = iztrans (X3, n);	seq3 = 3 - (-2)^n	The sequence with a call to the " iztrans " function that calculates the inverse Z-transform.
>> n = 0:5 >> seq3 = iztrans (X3, n)	n = 0 1 2 3 4 5 seq3 = [2, 5, -1, 11, -13, 35]	Calculation of the first 6 terms of the sequence by calling " iztrans " and defining a time interval from 0 ≤ n ≤ 5.

EXERCISE 13

Analyze the explicit function into a sum of simple fractions:

$$X(z) = \frac{2z^2 + 3z - 1}{z^3 - 5z^2 + 8z - 4}$$

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Code at Matlab

```
num = [2 3 -1]
den = [1 -5 8 -4]
[r, p, k] = residue (num, den)
```

Explanation of the commands used

Commands	Results	Comments
>> num = [2 3 -1]	num = 2 3 -1	The coefficients of the numerator.
>> den = [1 -5 8 -4]	den = 1 -5 8 -4	The coefficients of the denominator.
>> [r, p, k] = residue (num, den)	r = -2.0000 13,0000 4,0000 p = 2,0000 2,0000 1,0000 k = []	Analysis of the function X(z) into a sum of simple fractions by calling the " residue " function, where " r " are the coefficients of the numerator, " p " are the coefficients of the denominator and " k " are the constants.

EXERCISE 14

Given an LTI system with impulse response h(n)=0.9nu(n). To find its transfer function, to investigate if it is stable, to make the pole zero diagram.

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Code at Matlab

Main program

```
syms nz

h = 0.9^n * stepseq(0, 0, 0)
H = ztrans (h, z);
disp ( "H = " )
pretty (H)
num = [1.0]
den = [1.0 -0.9]
zplane (num, den)
```

Code her function stepseq

```
function [x, n] = stepseq (n0, n1, n2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end
n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> sims nz		Declaration of symbolic variables ' n ' and ' z '.
>> h = 0.9^n * stepseq (0, 0, 0)	h = (9/10)^n	The shock response of the system
>> H = ztrans (h, z); >> disp("H = ") >> pretty (H)	H = z ------ 9 z - -- 10	The system transfer function that is equivalent to the Z-transform of the impulse response by calling ' ztrans '.
>> num = [1.0]	num = 1	The numerator coefficients of the transfer function.
>> den = [1.0 -0.9]	den =	The coefficients of the denominator of the transfer function.

	1.0000 -0.9000	
>> zplane (num, den)	Figure 14.1	Plotting the zero-pole diagram by calling " zplane "

Graphic representations

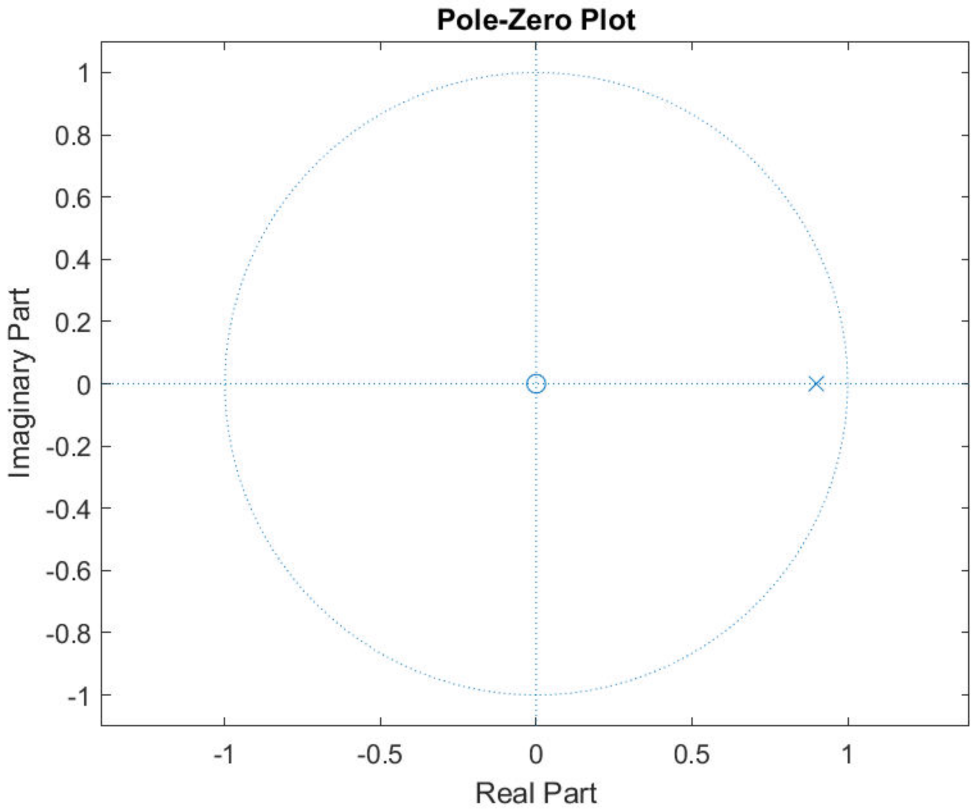

Figure 14.1 The zero pole diagram

Theoretical Solution

The system is stable, as the poles lie within the unit circle (Figure 14.1).

EXERCISE 15

Analyze the explicit function into a sum of simple fractions:

$$X(z) = \frac{3z^4 - 1.1z^3 + 0.88z^2 - 2.396z + 1.348}{z^3 - 0.7z^2 - 0.14z + 0.048}$$

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Code in Matlab

```
num = [3.0 -1.1 0.88 -2.396 1.348]
```

```
den = [1.0 -0.7 -0.14 0.048]
[r, p, k] = residue (num, den)

R = [1.0 4.0 -3.0]
P = [0.8 -0.3 0.2]
K = [3.0 1.0]
[NUM, DEN] = residue (R, P, K)
```

Explanation of the commands used

Commands	Results	Comments
>> num = [3.0 -1.1 0.88 -2.396 1.348]	num = 3.0000 -1.1000 0.8800 -2.3960 1.3480	The coefficients of the numerator.
>> den = [1.0 -0.7 -0.14 0.048]	den = 1.0000 -0.7000 -0.1400 0.0480	The coefficients of the denominator.
>> [r, p, k] = residue (num, den)	r = 1,0000 4,0000 -3.0000 p = 0.8000 -0.3000 0.2000 k = 3.0000 1.0000	Analysis of the function X(z) into a sum of simple fractions by calling the " residue " function, where " r " are the coefficients of the numerators, " p " are the coefficients of π and " k " are the constants.
>> R = [1.0 4.0 -3.0]	R = 1 4 -3	The coefficients of the numerators.
>> P = [0.8 -0.3 0.2]	P = 0.8000 -0.3000 0.2000	The coefficients of misnomers.
>> K = [3.0 1.0]	K = 3 1	The constants.
>> [NUM, DEN] = residue (R, P, K)	NUM = 3.0000 -1.1000 0.8800 -2.3960 1.3480 DEN = 1.0000 -0.7000 -0.1400 0.0480	Verification of the analysis of the function X(z) into a sum of simple fractions.

EXERCISE 16

Find the Transfer Function of the system described by the Difference Equation

$$y(n) + 1.5\,y(n-1) + 0.5\,y(n-2) = x(n) + x(n-1)$$

in two ways a) using tf() b) using solve(). To check the stability, make a pole zero diagram

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a)

Code at Matlab

```
sunt_x = [1.0 1.0 0.0]
sunt_y = [1.0 1.5 0.5]
Ts = 0.5
H = tf ( sunt_x , sunt_y , Ts);
H = simplify (H)
pzmap (H)
```

Explanation of the commands used

Commands	Results	Comments
>> sunt_x = [1.0 1.0 0.0];	sunt_x = 1 1 0	The coefficients of the inputs x .
>> sunt_y = [1.0 1.5 0.5];	sunt_y = 1.0000 1.5000 0.5000	The coefficients of the costs y .
>> Ts = 0.5;	Ts = 0.5000	Define the sampling period.
>> H = tf (sunt_x , sunt_y , Ts); >> H = simplify (H)	H = z^2 + z ----------------- z^2 + 1.5 z + 0.5	Calculation of the transfer function of the system described by the difference equation $y(n) + 1.5\,y(n-1) + 0.5\,y(n-2) = x(n) + x(n-1)$, by calling the function “ tf »
>> pzmap (H)	Figure 16.1	Plotting the pole-zero diagram by calling the ' pzmap ' function.

Graphic representations

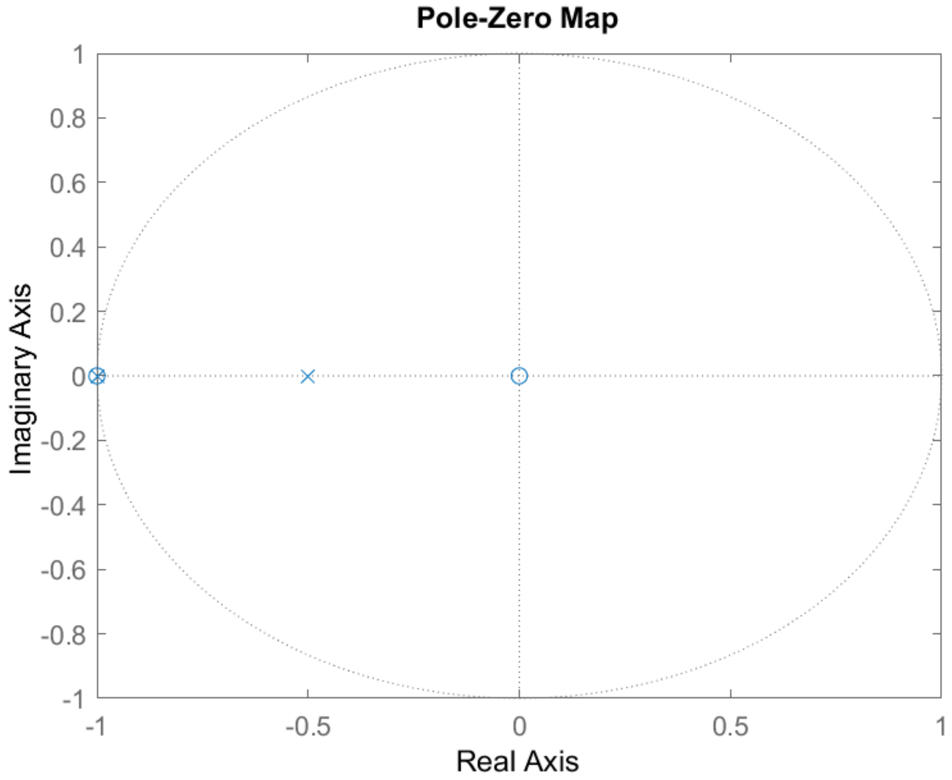

Figure 16.1 The zero-pole plot by invoking " pzmap "

b)

Code in Matlab

```
syms nz XY

Y1 = (z^-1) * Y;
Y2 = (z^-2) * Y;
X1 = (z^-1) * X;
G = Y + 1.5 * Y1 + 0.5 * Y2 - X - X1;
disp ( "G = " )
pretty (G)
Sol = solve (G, Y);
disp ( "Sol = " )
pretty (Sol)
H = Sol / X;
disp ( "H = " )
pretty (H)
num = [2 0]
den = [2 1]
zplane (num, den)
```

Explanation of the commands used

Commands	Results	Comments
>> syms nz XY	sunt_x = 1 1 0	Declaration of symbolic variables ' n ', ' z ', ' Y ' and ' X '.
>> Y1 = (z^-1) * Y; >> Y2 = (z^-2) * Y; >> X1 = (z^-1) * X; >> G = Y + 1.5 * Y1 + 0.5 * Y2 - X - X1; >> disp ("G = ") >> pretty (G)	G = X 3 YY Y - X - - + --- + ---- z 2 z 2 2 z	The difference equation $y(n) + 1.5\,y(n-1) + 0.5\,y(n-2) = x(n) + x(n-1)$.
>> Sol = solve(G, Y); >> disp ("Sol = ") >> pretty (Sol)	Sol = 2 X z ------- 2 z + 1	Solving the equation in terms of the " Y " variable, by calling the " solve " function.
>> H = Sol / X; >> disp ("H = ") >> pretty (H)	H = 2 z ------- 2 z + 1	Calculation of the transfer function of the system described by the difference equation $y(n) + 1.5\,y(n-1) + 0.5\,y(n-2) = x(n) + x(n-1)$, by dividing by “X ” of the solution computed by the “ solve ” function.
>> num = [2 0]	num = 20	The numerator coefficients of the transfer function.
>> den = [2 1]	den = 2 1	The coefficients of the denominator of the transfer function.
>> zplane (num, den)	Figure 16.2	Plotting the zero-pole diagram by calling the ' zplane ' function.

Graphic representations

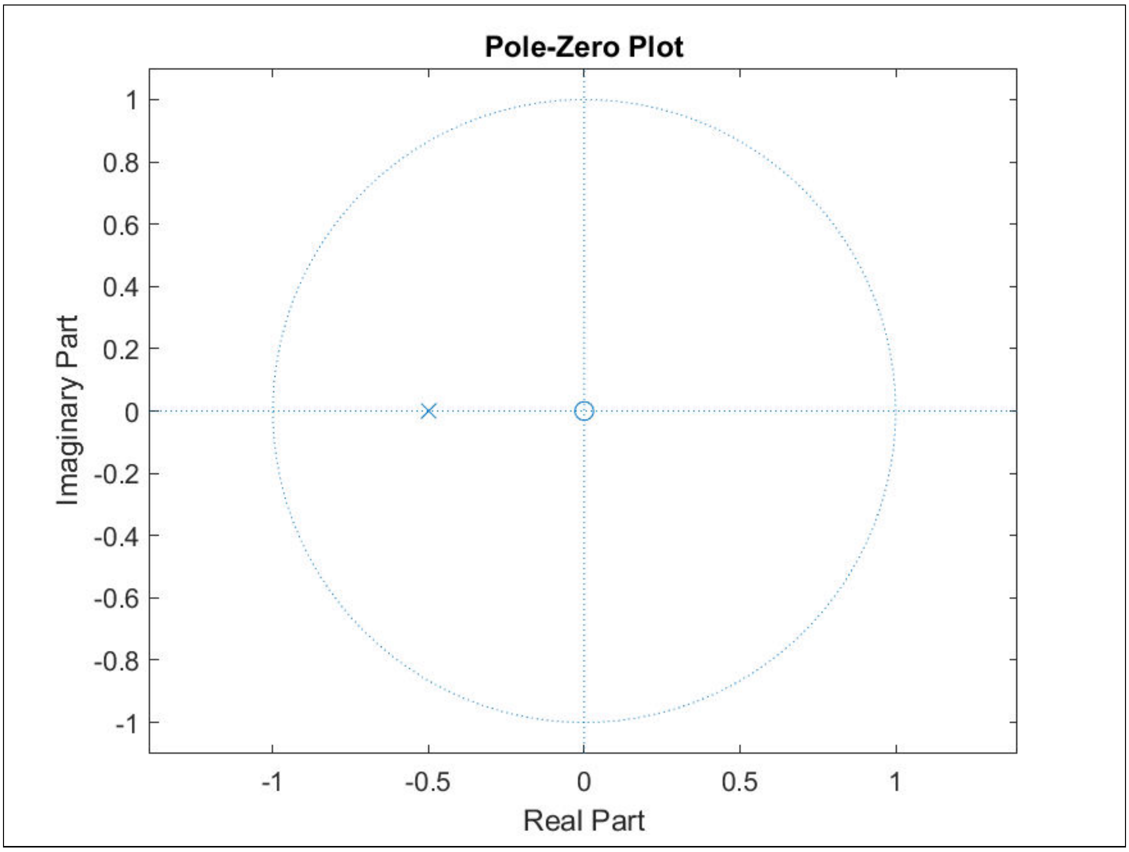

Figure 16.2 The zero-pole diagram by calling " zplane "

Theoretical Solution

From Figures 16.1 and 16.2, it is found that the system described by the difference equation $y(n) + 1.5\,y(n-1) + 0.5\,y(n-2) = x(n) + x(n-1)$, is stable because the poles are inside the unit circle.

EXERCISE 17

Calculate the transfer function and the shock response of the first-order discrete-time causal GNSS system, which is described by the difference equation:

$$y(n) - 0.9\,y(n-1) = x(n)$$

Make the zero pole diagram and determine the PS.

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Code in Matlab

```
sunt_x = [1.0 0.0]
sunt_y = [1.0 -0.9]
Ts = 0.5
H = tf ( sunt_x , sunt_y , Ts);
H = simplify (H)
pzmap (H)
metro = abs (pole (H))
```

Explanation of the commands used

Commands	Results	Comments
>> sunt_x = [1.0 0.0]	sunt_x = 1 0	The coefficients of the inputs x .
>> sunt_y = [1.0 -0.9]	sunt_y = 1.0000 -0.9000	The coefficients of the costs y .
>> Ts = 0.5	Ts = 0.5000	Define the sampling period.
>> H = tf (sunt_x , sunt_y , Ts); >> H = simplify (H)	H = z ------- z - 0.9	Calculation of the transfer function of the system described by the difference equation $y(n) - 0.9\, y(n-1) = x(n)$ by calling the " tf " function .
>> pzmap (H)	Figure 1 7 . 1	Plotting the pole-zero diagram by calling the ' pzmap ' function.
>> metro = abs (pole (H))	subway = 0.9000	Determining the region of convergence by calling the " pole " function.

Graphic representations

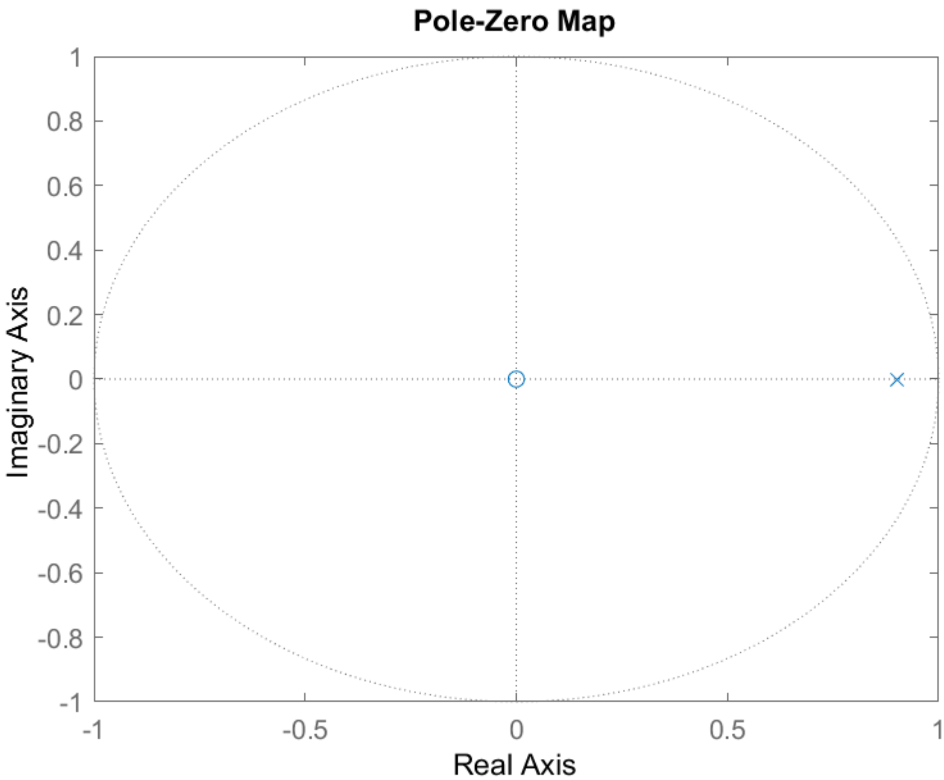

Figure 17.1 The zero pole diagram

EXERCISE 18

Given the discrete-time causal GNSS system whose input and output are related by the difference equation

$$y(n) - 0.5\, y(n-1) = x(n)$$

Calculate the output of the system if the input signal is x(n) = u(n) and the system is at rest.

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Code in Matlab

Main program

```
syms n z Y

X = ztrans ( stepseq ( 0 , 0 , 0 ), z );
disp ( "X = " )
pretty (G)
```

```
Y1 = z ^( -1) * Y;
G = Y - 0.5 * Y1 - X;
disp ( "G = " )
pretty (G)
Sol = solve( G,Y );
disp ( "Sol = " )
pretty (Sol)
y = iztrans (Sol, n);
disp ( "y = " )
pretty

n = 0:10
yn = subs( y, n)
stem ( n,yn )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( 'y[n] - 0.5y[n - 1] = x[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'System Output' )
```

Code of the stepseq function

```
function [ x , n ] = stepseq ( n 0, n 1, n 2)

if ((n0 < n1) | (n0 > n2) | (n1 > n2))
error ( 'arguments must satisfy n1 <= n0 <= n2' )
end
n = [n1:n2];
x = [(n-n0) >= 0];
```

Explanation of the commands used

Commands	Results	Comments
>> syms n z Y		Declaration of symbolic variables ' n ', ' z ' and ' Y '.
>> X = ztrans (stepseq (0 , 0 , 0), z); >> disp ("X = ") >> pretty (X)	X = z ----- z - 1	The Z-transform of the step function u [n] (stepseq) by calling the " ztrans " function.
>> Y1 = z ^(-1) * Y; >> G = Y - 0.5 * Y1 - X; >> disp ("G = ") >> pretty (G)	G = z Y Y - ----- - --- z - 1 2 z	The difference equation $y(n) - 0.5\,y(n-1) = x(n)$ describing the system.
>> Sol = solve(G,Y); >> disp ("Sol = ") >> pretty (Sol)	Sol = 2 2 z -------------- 2 2z - 3z + 1	Solving the equation in terms of the " Y " variable, by calling the " solve " function.
>> y = iztrans (Sol, n); >> disp ("y = ") >> pretty (y)	y = / 1 \n 2 - | - | \ 2 /	The inverse Z-transform of the solution of the equation in terms of the " Y " variable, by calling the " solve " function.

>> n = 0:10	n = 0 1 2 3 4 5 6 7 8 9 10	Define the time interval for which the system output will be defined.
>> yn = subs(y, n)	yn = [1, 3/2, 7/4, 15/8, 31/16, 63/32, 127/64, 255/128, 511/256, 1023/512, 2047/1024]	Calculation of system output.
>> stem (n,yn) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('y[n] - 0.5y[n - 1] = x[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('System Output')	Figure 18.1	Designing the system output by calling the `` stem '' function.

Graphic representations

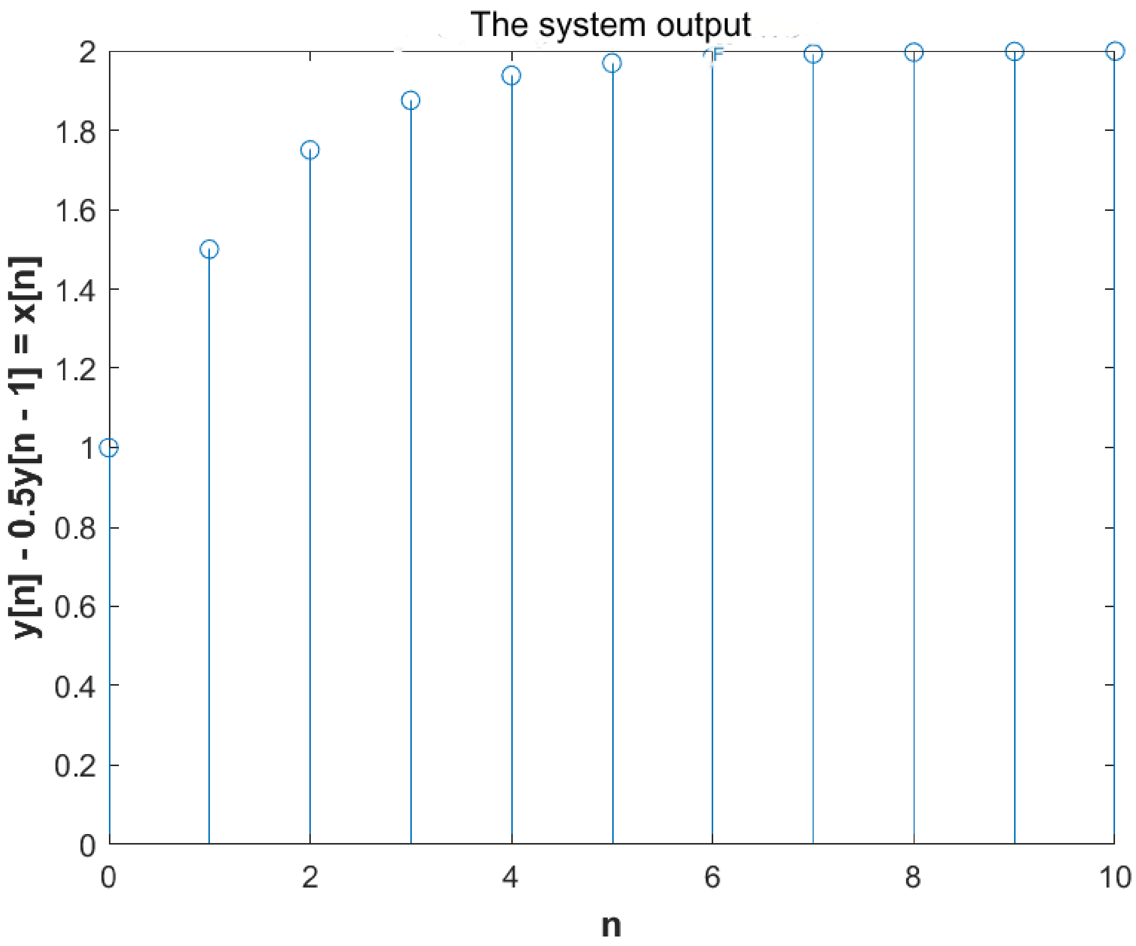

Figure 18.1 The system output

EXERCISE 19

A discrete GFA system is described by the difference equation:

$$y[n] = x[n] - x[n-1] + x[n-2]$$

Plot: a) the impulse response and b) the gain and phase diagrams as a function of frequency. Is shock response calculation required?

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Code in Matlab

```
syms nz XY

X1 = (z^-1) * X;
X2 = (z^-2) * X;
G = Y - X + X1 - X2;
disp ( "G = " )
pretty (G)
Sol = solve (G, Y);
disp ( "Sol = " )
pretty (Sol)
H = Sol / X;
disp ( "H = " )
pretty (H)
num = [1 -1 1]
den = [1 0 0]
h = iztrans (H, n);
disp ( "h = " )
pretty
n = 0:5
hn = subs( h, n)
figure
    dimpulse (num, den)
figure
subplots (121)
stem (n, abs ( hn ))
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( 'h[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( ' Chart profit ' )
subplots (122)
stem (n, angle ( hn ))
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( 'h[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    title ( 'Phase Diagram' )
```

Explanation of the commands used

Commands	Results	Comments
>> syms nz XY		Declaration of symbolic variables ' n ', ' z ' ' X ' and ' Y '.
>> X1 = (z^-1) * X; >> X2 = (z^-2) * X; >> G = Y - X + X1 - X2; >> disp ("G = ") >> pretty (G)	G = XX Y - X + - - -- z 2 z	The difference equation $y[n] = x[n] - x[n-1] + x[n-2]$ that describes the system.
>> Sol = solve(G, Y); >> disp ("Sol = ") >> pretty (Sol)	Sol = XX X - - + -- z 2 z	Solving the equation in terms of the " Y " variable, by calling the " solve " function.
>> H = Sol / X; >> disp ("H = ") >> pretty (H)	H = XX X - - + -- z 2 z ---------- X	Calculation of the transfer function of the system described by the difference equation $y[n] = x[n] - x[n-1] + x[n-2]$, by dividing by “X” the solution calculated by the function “ solve ».
>> num = [1 -1 1]	num = 1 -1 1	The numerator coefficients of the transfer function.
>> den = [1 0 0]	den = 1 0 0	The coefficients of the denominator of the transfer function.
>> h = iztrans (H, n); >> disp ("h = ") >> pretty (h)	h = kroneckerDelta (n - 2, 0) - kroneckerDelta (n - 1, 0) + kroneckerDelta (n, 0)	The inverse Z-transform of the transfer function that gives the shock response of the system.
>> n = 0:5	n = 0 1 2 3 4 5	Define the time interval for which the shock response will be defined.
>> hn = subs(h, n)	hn = [1, -1, 1, 0, 0, 0]	The impulse response values in the time interval $0 \leq n \leq 5$.
>> figure >> dimpulse (num, den)	Figure 19.1	Designing the impulse response of the system by calling the `` dimpulse " function
>> figure >> subplot (121) >> stem (n, abs (hn)) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('h[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title (' Chart profit ')	Figure 19.2	Plotting the gain plot of the system by calling the `` stem " and `` abs " functions.

>> stem (n, angle (hn)) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('h[n]', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('Phase Diagram')	Figure 19.2	Plotting the phase diagram of the system by calling the function " stem " and " angle ".

Graphic representations

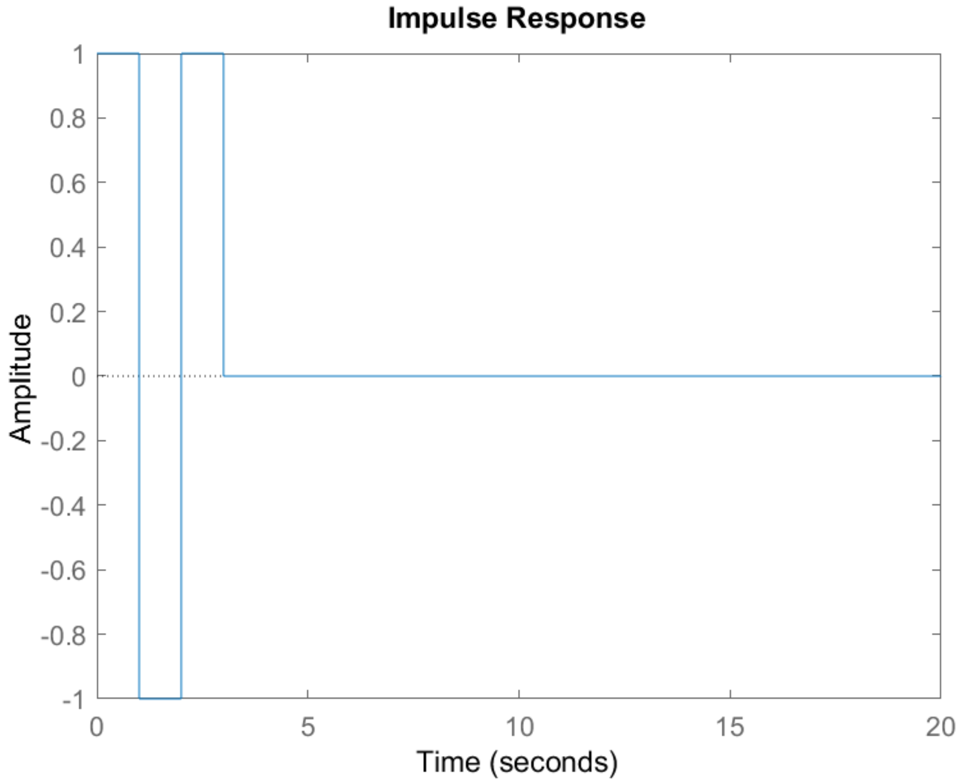


Figure 19.1 The shock response of the system

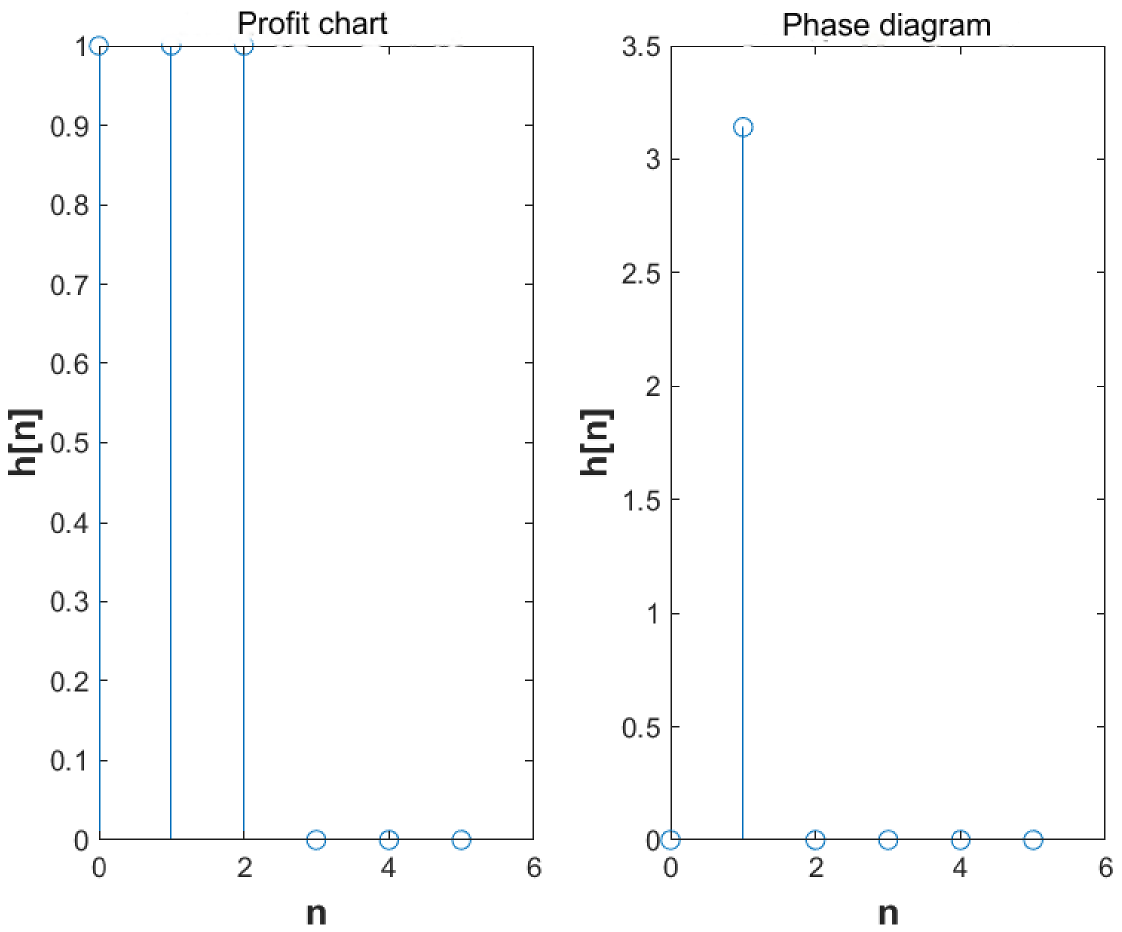
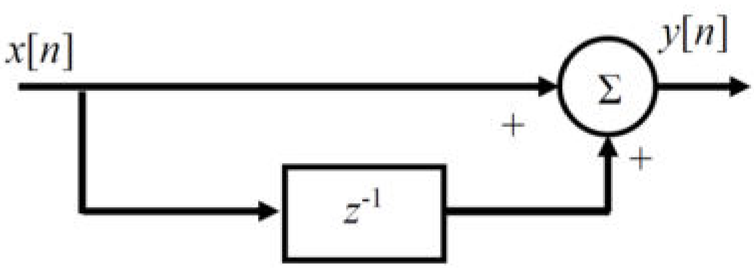


Figure 19.2 The gain and phase diagrams

EXERCISE 20

Plot the magnitude and phase as a function of frequency of the discrete-time GNSS system described in the figure



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EXERCISE 21

In the above system input x[n] = u[n] is applied. Plot the output y[n] for 0≤n≤200. How is the system characterized in terms of frequency passing?

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EXERCISE 22

The sequence is given

$$x(n) = 0.7^{n}, \quad 0 \le n \le 19$$

Plot the measure of the DTFT for 0≤ω≤2π and the DFT over the frequencies in the same figure:

$$\omega = 2\pi k/N, \qquad k = 0,\dots N\text{-}1.$$

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Code in Matlab

Main program

```
syms zw

n = 0:19
x = (0.7 ).^ n
Hz = ztrans (x, z)
H = subs (Hz, z, exp(j*w))
num = [0 7 0]
den = [10 -20 10]
w = 0:.1:2 * pi
DTFT = freqz (num, den, w)
dft = dft (x)
plot( w, abs(DTFT))
hold on
stem ( 0:length (n) -1, abs (DFT), 'r' )
legend( 'DTFT' , 'DFT' )
xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
ylabel ( ' The measure of DTFT and of the DFT of x[n] = 0.7^n' , ' FontSize ' , 12, '
FontWeight ' , 'bold' )
title ( 'Exercise 22' )
```

Code of the dft function

```
function Xk = dft(x);
N=length(x);
for k= 0:N -1
 for n= 0:N -1
X(n+ 1)= x(n+1)*exp(-j*2*pi*k*n/N);
 end
 Xk (k+ 1)= sum(X);
end
end
```

Explanation of the commands used

Commands	Results	Comments
>> syms zw		Declaration of symbolic variables 'w' and ' z '.
>> n = 0:19	n = 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19	The time interval defined by the sequence $x(n) = 0.7^n$.
>> x = (0.7).^ n	x = Columns 1 through 13 1.0000 0.7000 0.4900 0.3430 0.2401 0.1681 0.1176 0.0824 0.0576 0.0404 0.0282 0.0198 0.0138 Columns 14 through 20 0.0097 0.0068 0.0047 0.0033 0.0023 0.0016 0.0011	The sequence $x(n) = 0.7^n$.
>> Hz = ztrans (x, z)	Hz = [z /(z - 1), (7*z)/(10*(z - 1)) , ...]	The terms of the Z-transform of the sequence $x(n) = 0.7^n$ in the interval $0 \le n \le 19$, by calling the function " ztrans ".
>> H = subs (Hz, z, exp(j*w))	H = [exp(w*1i)/(exp(w*1i) - 1), (7*exp(w*1i))/(10*(exp(w*1i) - 1)), ...]	Replace z with $e^{j\omega}$.
>> num = [0 7 0]	num = 0 7 0	Coefficients of the numerator of the 1st term of the Z-transform of the sequence $x(n) = 0.7^n$
>> den = [10 -20 10]	den = 10 -20 10	Coefficients of the denominator of the 1st term of the Z-transform of the sequence $x(n) = 0.7^n$
>> w = 0:.1:2 * pi	w = Columns 1 through 13 0 0.1000 0.2000 0.3000 0.4000 0.5000 0.6000 0.7000 0.8000 0.9000 1.0000 1.1000 1.2000	Definition of the interval of angular frequencies $0 \le \omega \le 2\pi$.

	Columns 14 through 26 1.3000 1.4000 1.5000 1.6000 1.7000 1.8000 1.9000 2.0000 2.1000 2.2000 2.3000 2.4000 2.5000 Columns 27 through 39 2.6000 2.7000 2.8000 2.9000 3.0000 3.1000 3.2000 3.3000 3.4000 3.5000 3.6000 3.7000 3.8000 Columns 40 through 52 3.9000 4.0000 4.1000 4.2000 4.3000 4.4000 4.5000 4.6000 4.7000 4.8000 4.9000 5.0000 5.1000 Columns 53 through 63 5.2000 5.3000 5.4000 5.5000 5.6000 5.7000 5.8000 5.9000 6.0000 6.1000 6.2000	
>> DTFT = freqz (num, den, w)	DTFT = 1.0e+02 * Columns 1 through 6 Inf + 0.0000i - 0.7006 + 0.0000i -0.1756 + 0.0000i -0.0784 + 0.0000i -0.0443 - 0.0000i -0.0286 + 0.0000i Columns 7 through 12 -0.0200 + 0.0000i - 0.0149 + 0.0000i -0.0115 + 0.0000i - 0.0092 + 0.0000i -0.0076 - 0.0000i -0.0064 - 0.0000i Columns 13 through 18 -0.0055 + 0.0000i - 0.0048 + 0.0000i -0.0042 - 0.0000i - 0.0038 - 0.0000i -0.0034 + 0.0000i -0.0031 - 0.0000i Columns 19 through 24 -0.0029 + 0.0000i - 0.0026 + 0.0000i -0.0025 + 0.0000i - 0.0023 + 0.0000i -0.0022 + 0.0000i -0.0021 + 0.0000i Columns 25 through 30 -0.0020 + 0.0000i - 0.0019 + 0.0000i -0.0019 + 0.0000i - 0.0018 + 0.0000i -0.0018 + 0.0000i -0.0018 + 0.0000i Columns 31 through 36 -0.0018 - 0.0000i - 0.0018 + 0.0000i -0.0018 + 0.0000i - 0.0018 + 0.0000i -0.0018 + 0.0000i -0.0018 - 0.0000i Columns 37 through 42	Computing the DTFT of the sequence $x(n) = 0.7^{n}$ by calling the " freqz " function.

	-0.0018 + 0.0000i - 0.0019 - 0.0000i -0.0020 + 0.0000i -0.0020 + 0.0000i -0.0021 + 0.0000i -0.0022 + 0.0000i Columns 43 through 48 -0.0023 + 0.0000i - 0.0025 + 0.0000i -0.0027 + 0.0000i -0.0029 - 0.0000i -0.0031 + 0.0000i -0.0035 - 0.0000i Columns 49 through 54 -0.0038 + 0.0000i - 0.0043 - 0.0000i -0.0049 + 0.0000i -0.0056 + 0.0000i -0.0066 - 0.0000i -0.0079 + 0.0000i Columns 55 through 60 -0.0096 + 0.0000i - 0.0120 + 0.0000i -0.0156 + 0.0000i -0.0212 - 0.0000i -0.0306 - 0.0000i -0.0483 + 0.0000i Columns 61 through 63 -0.0879 + 0.0000i - 0.2092 - 0.0000i -1.0122 - 0.0000i	
>> dft = dft (x)	DFT = Columns 1 through 6 3.3307 + 0.0000i 2.1069 - 1.3635i 1.2126 - 1.1504i 0.8815 - 0.8482i 0.7406 - 0.6291i 0.6706 - 0.4694i Columns 7 through 12 0.6321 - 0.3460i 0.6098 - 0.2447i 0.5968 - 0.1568i 0.5899 - 0.0766i 0.5878 + 0.0000i 0.5899 + 0.0766i Columns 13 through 18 0.5968 + 0.1568i 0.6098 + 0.2447i 0.6321 + 0.3460i 0.6706 + 0.4694i 0.7406 + 0.6291i 0.8815 + 0.8482i Columns 19 through 20 1.2126 + 1.1504i 2.1069 + 1.3635i	Calculating the DFT of the sequence $x(n) = 0.7^{n}$ by calling the “ dft ” function.
>> plot(w, abs(DTFT))	Figure 22.1	Plotting the measure of the DTFT of the sequence $x(n) = 0.7^{n}$ at the frequencies $0 \leq \omega \leq 2\pi$, by calling the " plot " function
>> hold on >> stem (0:length (n) - 1, abs (DFT), 'r') legend('DTFT','DFT') >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold')	Figure 22. 1	Plotting the measure of the DFT of the sequence $x(n) = 0.7^{n}$ at the frequencies $0 \leq \omega \leq 2\pi$, by calling the “ stem ” function

>> ylabel (' The measure of DTFT and of DFT of x[n] = 0.7^n', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('Exercise 22')		

Graphic representations

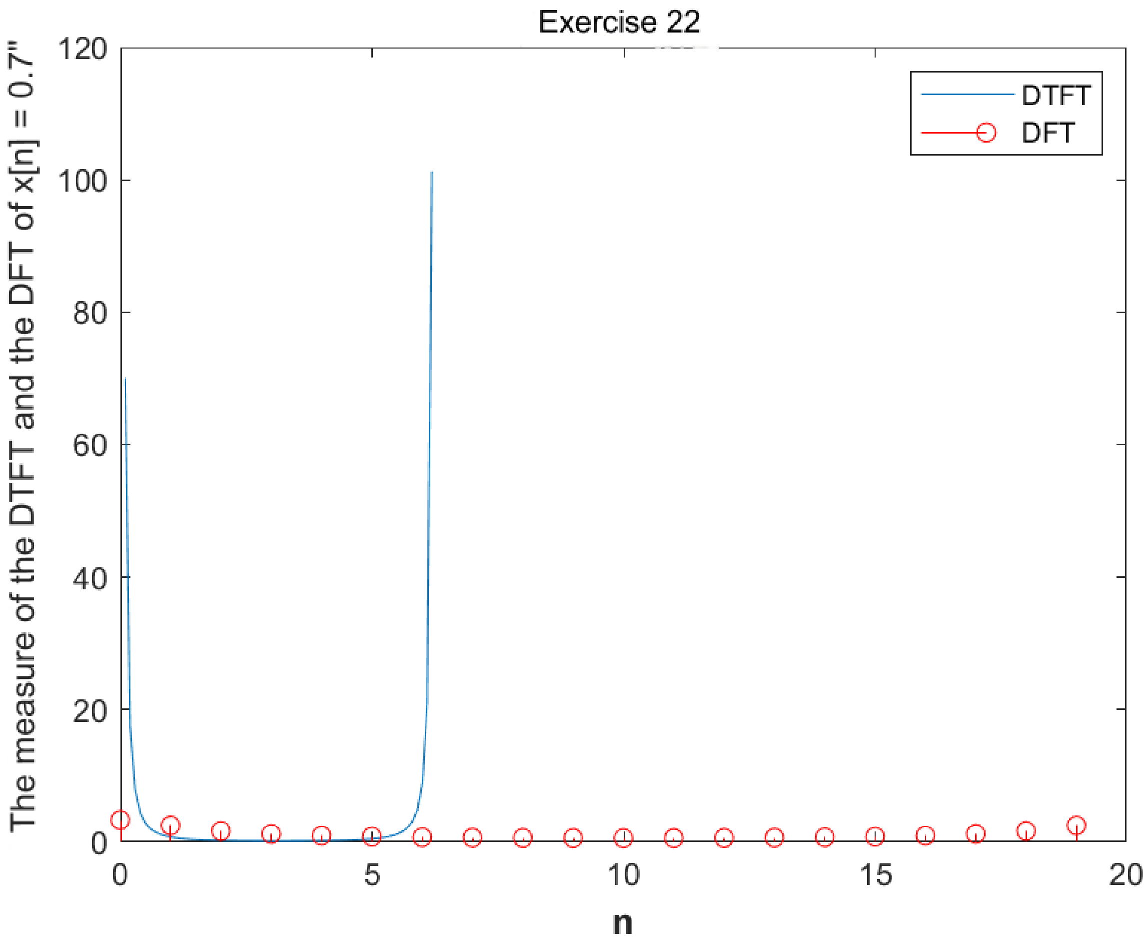

Figure 22.1 The measure of the DFT and the DFT of the sequence $x(n) = 0.7^n$

EXERCISE 23

Write a function that implements the linear convolution of two sequences using the functions fft(), ifft().

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EXERCISE 24

Calculate through the above function the linear convolution of the sequences:

$$x[n] = [1\ 2\ 3\ 4],\ 0 \le n \le 3,\ \kappa\alpha\iota\ y[n] = [7\ 6\ 5\ 4\ 3\ 2\ 1],\ 0 \le n \le 6.$$

Confirm the result using conv().

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EXERCISE 25

The sequences are given:

$$x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]\ \kappa\alpha\iota\ w[n] = [1,\ 1{,}2,\ -1\ ],\ 0 \le n \le 3$$

Calculate their circular convolution a) in the time domain b) using fft(), ifft().

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Code in Matlab

```
n = 0:3

x = gauspuls (n) - 2 * gauspuls (n - 2) + 4 * gauspuls (n - 5)
w = [1 1 2 -1]
C1 = cconv ( x , w , 4 )
X = fft ( x, 4 )
W = fft (w)
C2 = ifft (X.*W)
```

Explanation of the commands used

Commands	Results	Comments
>> n = 0:3	n = 0 1 2 3	Defining the time interval defined by the signals $x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]\ \kappa\alpha\iota\ w[n] = [1,\ 1{,}2,\ -1\ ]$.
>> x = gauspuls (n) - 2 * gauspuls (n - 2) + 4 * gauspuls (n - 5)	x = 1 0 -2 0	Computation of the sequence $x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]$ in the field $0 \le n \le 3$.
>> w = [1 1 2 -1]	w = 1 1 2 -1	Computation of the sequence $w[n] = [1,\ 1{,}2,\ -1\ ]$ in the field $0 \le n \le 3$.
>> C1 = cconv (x , w , 4)	C1 =	Calculation of the circular convolution of the two sequences $x[n] = \delta[n] - 2\,\delta[n$

	-3 3 0 -3	$-2]+4\,\delta[\,n-5]\;\kappa\alpha\iota\;w[\,n\,]=[1,\,1{,}2,\,-1\,]$ in the time domain by calling the " cconv " function.
>> X = fft (x, 4)	X = -1 3 -1 3	Fourier transform of the sequence $x[\,n\,]=\delta[\,n\,]-2\,\delta[\,n-2]+4\,\delta[\,n-5]$ by calling the " fft " function .
>> W = fft (w)	W = 3.0000 + 0.0000i -1.0000 - 2.0000i 3.0000 + 0.0000i - 1.0000 + 2.0000i	The fast Fourier transform of the sequence $w[\,n\,]=[1,\,1{,}2,\,-1\,]$by calling the " fft " function.
>> C2 = ifft (X.*W)	C2 = -3 3 0 -3	Calculation of the circular convolution of the two sequences $x[\,n\,]=\delta[\,n\,]-2\,\delta[\,n-2]+4\,\delta[\,n-5]\;\kappa\alpha\iota\;w[\,n\,]=[1,\,1{,}2,\,-1\,]$ by calling the " ifft " function.

EXERCISE 26

For the two sequences of the previous exercise:

a) Calculate the circular convolution of 8 points

b) To find the linear convolution a) in the time domain b) using fft(), ifft()

Code in Matlab

```
n = 0:3

x = gauspuls (n) - 2 * gauspuls (n - 2) + 4 * gauspuls (n - 5)
w = [1 1 2 -1]
CC = cconv ( x , w , 8 )
CL1 = conv( x, w )
m = length(x) + length(w) - 1
X = fft ( x, m )
W = fft ( w, m )
S = X.*W
CL2 = ifft (S)
```

Explanation of the commands used

Commands	Results	Comments
>> n = 0:3	n = 0 1 2 3	Defining the time interval defined by the signals $x[\,n\,]=\delta[\,n\,]-2\,\delta[\,n-2]+4\,\delta[\,n-5]\;\kappa\alpha\iota\;w[\,n\,]=[1,\,1{,}2,\,-1\,]$.

>> x = gauspuls (n) - 2 * gauspuls (n - 2) + 4 * gauspuls (n - 5)	x = 1 0 -2 0	Computation of the sequence $x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]$ in the field $0 \le n \le 3$.
>> w = [1 1 2 -1]	w = 1 1 2 -1	Compute the sequence $w[n] = [1, 1{,}2, -1, 0, 0, 0, 0]$ in the field $0 \le n \le 3$.
>> CC = cconv (x , w , 8)	CC = 1.0000 1.0000 0.0000 -3.0000 -4.0000 2.0000 -0.0000 0	Calculation of the circular convolution of the two sequences $x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]$ $\kappa\alpha\iota$ $w[n] = [1, 1{,}2, -1]$ in the time domain by calling the “ cconv ” function.
>> CL1 = conv(x, w)	CL1 = 1 1 0 -3 -4 2 0	Calculation of the linear convolution of the two sequences $x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]$ $\kappa\alpha\iota$ $w[n] = [1, 1{,}2, -1]$ in the time domain by calling the “ conv ” function in the time domain.
>> m = length(x) + length(w) - 1	m = 7	The number of points where the linear convolution will be calculated with DFT .
>> X = fft (x, m)	X = Columns 1 through 6 -1.0000 + 0.0000i 1.4450 + 1.9499i 2.8019 - 0.8678i -0.2470 - 1.5637i -0.2470 + 1.5637i 2.8019 + 0.8678i Column 7 1.4450 - 1.9499i	Fourier transform of the sequence $[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]$ by calling the “ fft ” function .
>> W = fft (w, m)	W = Columns 1 through 6 3.0000 + 0.0000i 2.0794 - 2.2978i - 1.6479 - 0.8890i 1.5685 + 2.1047i 1.5685 - 2.1047i -1.6479 + 0.8890i Column 7 2.0794 + 2.2978i	The fast Fourier transform of the sequence $w[n] = [1, 1{,}2, -1]$ by calling the “ fft ” function.
>> S = X.*W	S = Columns 1 through 6 -3.0000 + 0.0000i 7.4852 + 0.7341i - 5.3889 - 1.0609i 2.9037 - 2.9725i 2.9037 + 2.9725i -5.3889 + 1.0609i	Propagation of new DFTs.

	Column 7 7.4852 - 0.7341i	
>> CL2 = ifft (S)	CL2 = 1.0000 1.0000 0.0000 -3.0000 -4.0000 2.0000 -0.0000	Calculation of the linear convolution of the two sequences $x[n] = \delta[n] - 2\,\delta[n-2] + 4\,\delta[n-5]$ $\kappa\alpha\iota$ $w[n] = [1, 1,2, -1]$ by calling the “ ifft ” function.

EXERCISE 27

Calculate the power of the unit step sequence:

u[n], for -5000 ≤ n ≤ 5000.

Compare the result you found with the result from the theoretical solution.

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EXERCISE 28

Plot the magnitude, phase, real part, imaginary part of the DTFT of the sequence

$$x[n] = (0.5)\,n\,u[n],\ \ -30 \leq n \leq 30$$

in N=501 equidistant frequencies between 0 and π. Why is this observation interval (0 to π) sufficient?

Note, the calculation of DTFT should be based on the definition. In the diagram of the measure of the DTFT, the measure based on the theoretical value should be displayed (hold on) as well as the measure of the DTFT calculated by freqz() should also be displayed.

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EXERCISE 29

Plot the magnitude, phase, real part, imaginary part of the DTFT of the sequence

$$x[n] = \ (0.9 * \exp\left(\frac{jn}{3}\right))^3, \qquad 0 \leq n \leq 10$$

in N=501 equidistant frequencies. What observation interval is sufficient?

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Code in Matlab

```
n = 0:10

x1 = (j * pi) / 3;
x = (0.9 * exp (x1) ).^ n
k = 0:500
fk = k / 500
wk = 2 * fk * pi
X = x * exp(-j ).^ (n' * wk )
[F, w] = freqz ( [1 0], [1 0.9], 501)

figure
    stem( n, real(x))
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( 'x[n]' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
title ( 'x[n] = (0.9 * exp((j * pi) / 3)))^n' )
figure
subplots (121)
plot ( wk , abs (F))
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( 'abs (F)' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    title ( 'The measure of the DTFT of x[n]' )
    subplots (122)
plot ( wk , angle (F))
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( 'angle (F)' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    title ( 'The phase of the DTFT of x[n]' )
figure
subplots (121)
plot ( wk , real ( F ))
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( 'real (F)' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    title ( 'The real part of the DTFT of x[n]' )
    subplots (122)
plot ( wk , imag (F));
    xlabel ( 'n' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    ylabel ( ' imag (F)' , ' FontSize ' , 12, ' FontWeight ' , 'bold' )
    title ( 'The imaginary part of the DTFT of x[n]' )
```

Explanation of the commands used

Commands	Results	Comments
>> n = 0:10	n = 0 1 2 3 4 5 6 7 8 9 10	Define the time interval over which the sequence is defined $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$.
>> x1 = (j * pi) / 3; >> x = (0.9 * exp (x1)).^ n	x = Columns 1 through 6 1.0000 + 0.0000i 0.4500 + 0.7794i - 0.4050 + 0.7015i -	Computation of the sequence $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$ in the field $0 \leq n \leq 3$.

	0.7290 + 0.0000i -0.3281 - 0.5682i 0.2952 - 0.5114i Columns 7 through 11 0.5314 - 0.0000i 0.2391 + 0.4142i - 0.2152 + 0.3728i - 0.3874 + 0.0000i -0.1743 - 0.3020i	
>> k = 0:500 >> fk = k / 500 >> wk = 2 * fk * pi	k = Columns 1 through 22 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 Columns 23 through 44 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 … Columns 485 through 501 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 fk = Columns 1 through 13 0 0.0020 0.0040 0.0060 0.0080 0.0100 0.0120 0.0140 0.0160 0.0180 0.0200 0.0220 0.0240 Columns 14 through 26 0.0260 0.0280 0.0300 0.0320 0.0340 0.0360 0.0380 0.0400 0.0420 0.0440 0.0460 0.0480 0.0500 … Columns 495 through 501 0.9880 0.9900 0.9920 0.9940 0.9960 0.9980 1.0000	Calculation of the 501 corner equidistant frequencies.

	wk = Columns 1 through 13 0 0.0126 0.0251 0.0377 0.0503 0.0628 0.0754 0.0880 0.1005 0.1131 0.1257 0.1382 0.1508 Columns 14 through 26 0.1634 0.1759 0.1885 0.2011 0.2136 0.2262 0.2388 0.2513 0.2639 0.2765 0.2890 0.3016 0.3142 … Columns 495 through 501 6.2078 6.2204 6.2329 6.2455 6.2581 6.2706 6.2832	
>> X = x * exp(-j).^ (n' * wk)	X = Columns 1 through 6 0.2768 + 0.8864i 0.2825 + 0.9428i 0.2941 + 0.9991i 0.3117 + 1.0546i 0.3350 + 1.1085i 0.3640 + 1.1601i Columns 7 through 12 0.3982 + 1.2085i 0.4373 + 1.2532i 0.4807 + 1.2935i 0.5278 + 1.3289i 0.5780 + 1.3588i 0.6305 + 1.3828i Columns 13 through 18 0.6844 + 1.4008i 0.7390 + 1.4124i 0.7934 + 1.4176i 0.8467 + 1.4164i 0.8980 + 1.4089i 0.9463 + 1.3955i Columns 19 through 24 0.9909 + 1.3765i 1.0309 + 1.3525i 1.0656 + 1.3240i 1.0942 + 1.2917i 1.1163 + 1.2565i 1.1312 + 1.2194i … Columns 493 through 498	Multiplication of the sequence $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$ with $e^{j\omega}$.

	0.4184 + 0.5154i 0.3864 + 0.5485i 0.3574 + 0.5864i 0.3320 + 0.6286i 0.3108 + 0.6747i 0.2943 + 0.7241i Columns 499 through 501 0.2830 + 0.7763i 0.2770 + 0.8306i 0.2768 + 0.8864i	
>> [F, w] = freqz ([1 0], [1 0.9], 501)	F = 0.5263 + 0.0000i 0.5263 + 0.0016i 0.5263 + 0.0031i 0.5263 + 0.0047i 0.5263 + 0.0063i 0.5263 + 0.0078i 0.5263 + 0.0094i 0.5263 + 0.0109i 0.5263 + 0.0125i 0.5263 + 0.0141i 0.5263 + 0.0156i 0.5263 + 0.0172i 0.5264 + 0.0188i 0.5264 + 0.0203i 0.5264 + 0.0219i 0.5264 + 0.0235i 0.5264 + 0.0250i 0.5264 + 0.0266i 0.5264 + 0.0282i 0.5264 + 0.0297i 0.5264 + 0.0313i … 7.8840 + 3.9458i 8.2460 + 3.6797i 8.5963 + 3.3657i 8.9266 + 3.0028i 9.2279 + 2.5920i 9.4909 + 2.1362i 9.7068 + 1.6407i 9.8674 + 1.1129i 9.9665 + 0.5624i w = 0 0.0063 0.0125 0.0188 0.0251 0.0314 0.0376	The DTFT of the sequence $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$, by calling the " freqz " function.

	0.0439 0.0502 0.0564 0.0627 0.0690 0.0752 0.0815 0.0878 0.0941 0.1003 0.1066 0.1129 0.1191 0.1254 0.1317 0.1380 0.1442 0.1505 0.1568 0.1630 0.1693 0.1756 … 3.0789 3.0852 3.0914 3.0977 3.1040 3.1102 3.1165 3.1228 3.1291 3.1353	
>> figure >> stem(n, real(x)) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') > > ylabel ('x[n]', ' FontSize ', 12, ' FontWeight ', 'bold') > > title ('x[n] = (0.9 * exp((j * pi) / 3)))^n')	Figure 29.1	Designing the real part of the sequence $x[n] = (0.9 * \exp\left(\frac{j\pi}{3}\right))^3$, calling " stem " and " real ".
>> figure >> subplot (121) >> plot (wk , abs (F)) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('abs(F)', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('The measure of the DTFT of x[n]')	Figure 29.2	Plotting the measure of the DTFT of the sequence $x[n] = (0.9 * \exp\left(\frac{j\pi}{3}\right))^3$, by calling " plot " and " abs ".

>> subplot (122) >> plot (wk , angle (F)) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('angle (F)', ' FontSize ', 12, ' FontWeight ', 'bold') >> title ('The phase of the DTFT of x[n]')	Figure 29.2	Plotting the DTFT phase of the sequence $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$, by calling " plot " and " angle ".
>> figure >> subplot (121) >> plot (wk , real (F)) >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel ('real(F)', ' FontSize ', 12, ' FontWeight ', 'bold') >> title('The real part of the DTFT of x[n]')	Figure 29.3	Plotting the real part of the DTFT of the sequence $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$, calling " plot " and " real ".
>> subplot (122) >> plot (wk , imag (F)); >> xlabel ('n', ' FontSize ', 12, ' FontWeight ', 'bold') >> ylabel (' imag (F)', ' FontSize ', 12, ' FontWeight ', 'bold') >> title('The imaginary part of the DTFT of x[n]')	Figure 29.3	Plotting the imaginary part of the DTFT of the sequence $x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$, by calling `` plot " and `` imag ".

Graphic representations

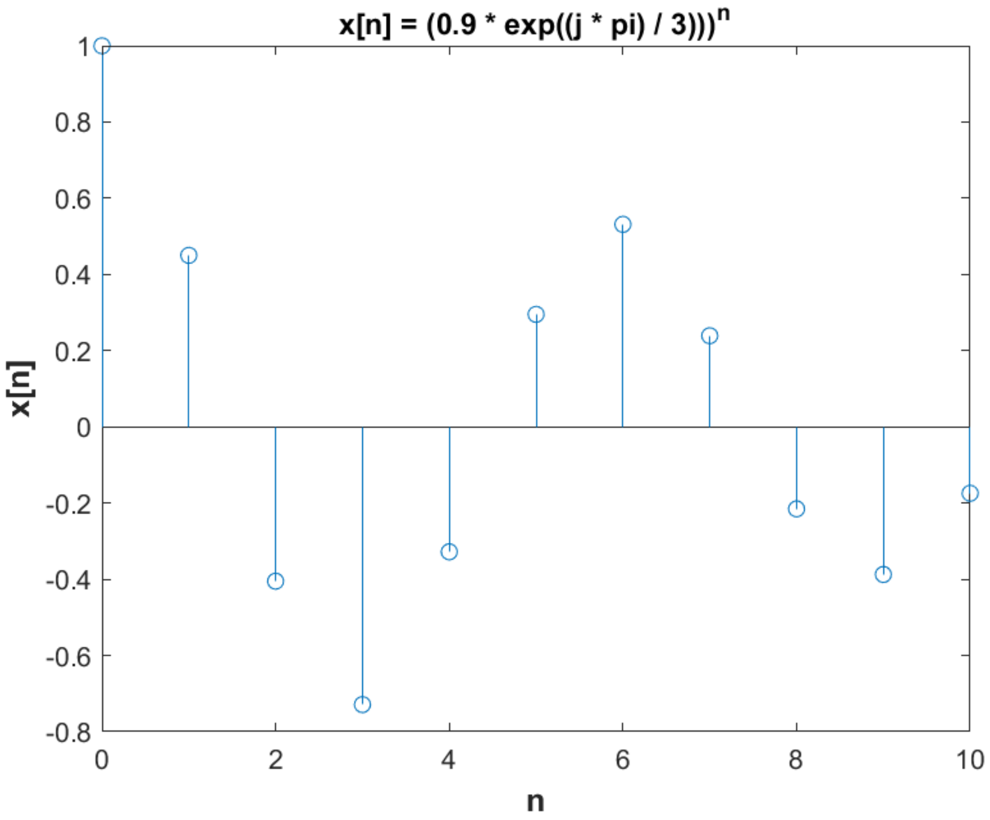

Figure 29.1 The real part of the sequence$x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$

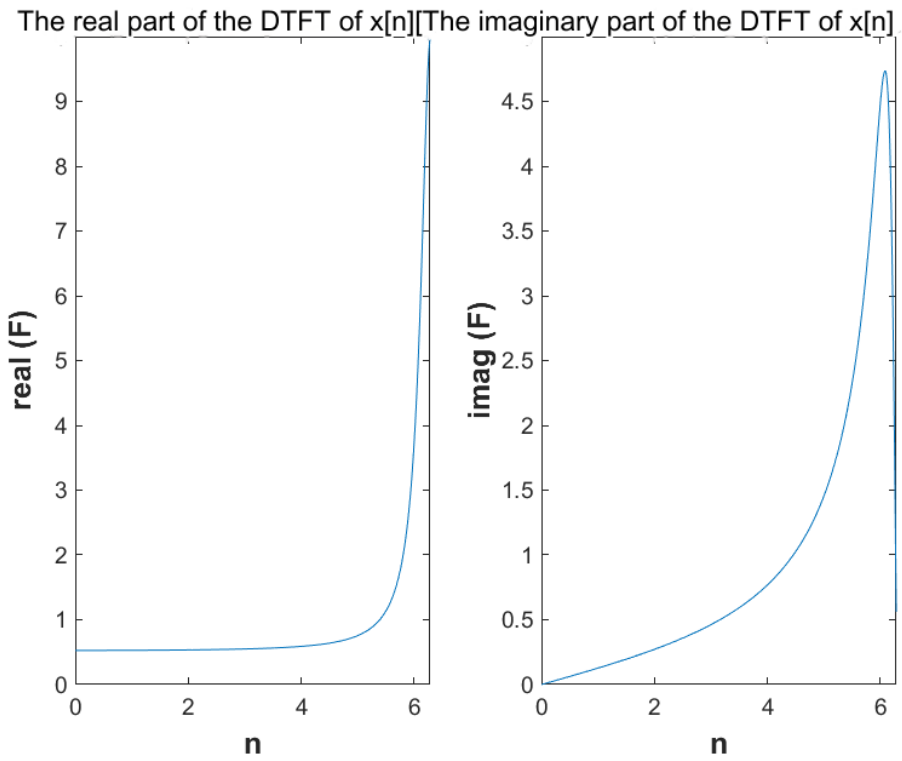

Figure 29.2 The magnitude and phase of the DTFT of the sequence$x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$

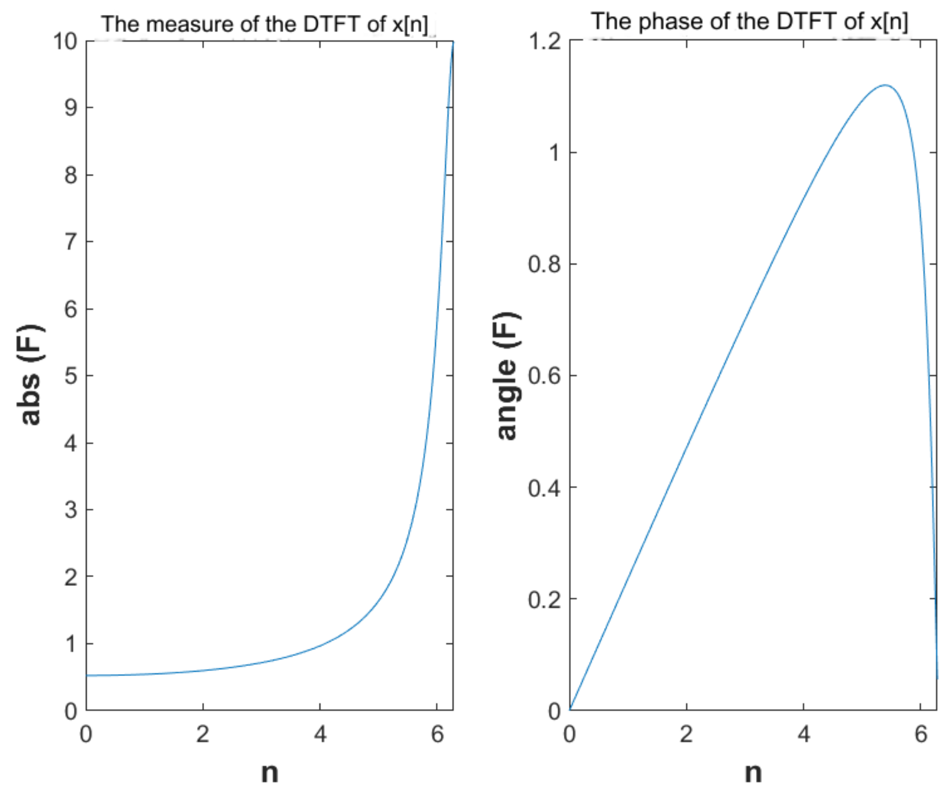

Figure 29.3 The real part and the imaginary part of the DTFT of the sequence$x[n] = (0.9 * \exp\left(\frac{jn}{3}\right))^3$

EXERCISE 30

Determine the linearity property for the DTFT with the following two discrete-time signals:

$$x1[n] = rand(1.11), \quad x2[n] = rand(1.11)$$

and α=2, β=3, in the interval $0 \leq n \leq 10$

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Thank you for your attention.